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Satellite Orbits and Trajectories

The study of orbits and trajectories of satellites and satellite launch vehicles is the most fundamental topic of the subject of satellite technology and perhaps also the most important one. It is important because it gives an insight into the operational aspects of this wonderful piece of technology. An understanding of the orbital dynamics would give a sound footing to address issues like types of orbits and their suitability for a given application, orbit stabilization, orbit correction and station keeping, launch requirements and typical launch trajectories for various orbits, Earth coverage and so on. This chapter and the one after this focus on all these issues and illustrate various concepts with the help of necessary mathematics and a large number of solved problems.

2.1 Definition of an Orbit and a Trajectory

While a trajectory is a path traced by a moving body, an orbit is a trajectory that is periodically repeated. While the path followed by the motion of an artificial satellite around Earth is an orbit, the path followed by a launch vehicle is a trajectory called the launch trajectory. The motion of different planets of the solar system around the sun and the motion of artificial satellites around Earth (Figure 2.1) are examples of orbital motion.

The term ‘trajectory’, on the other hand, is associated with a path that is not periodically revisited. The path followed by a rocket on its way to the right position for a satellite launch (Figure 2.2) or the path followed by orbiting satellites when they move from an intermediate orbit to their final destined orbit (Figure 2.3) are examples of trajectories.

2.2 Orbiting Satellites – Basic Principles

The motion of natural and artificial satellites around Earth is governed by two forces. One of them is the centripetal force directed towards the centre of the Earth due to the gravitational force of attraction of Earth and the other is the centrifugal force that acts outwards from the

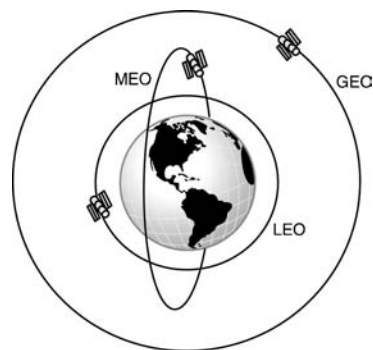


Figure 2.1 Example of orbital motion – satellites revolving around Earth

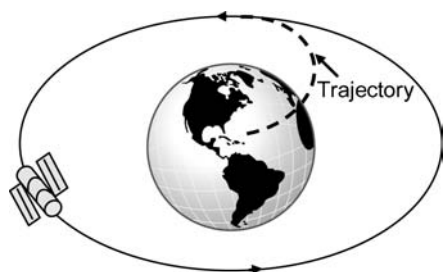


Figure 2.2 Example of trajectory – path followed by a rocket on its way during satellite launch

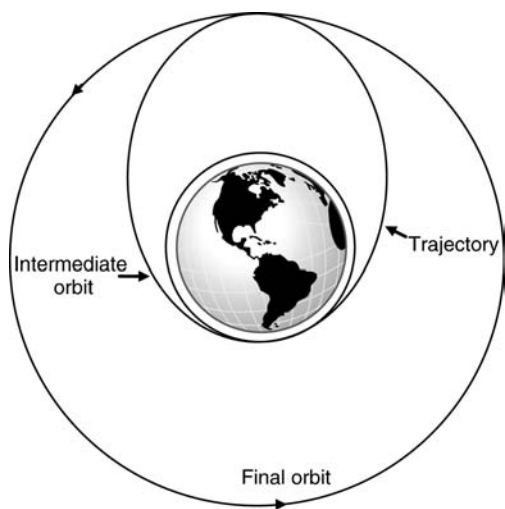


Figure 2.3 Example of trajectory – motion of a satellite from the intermediate orbit to the final orbit

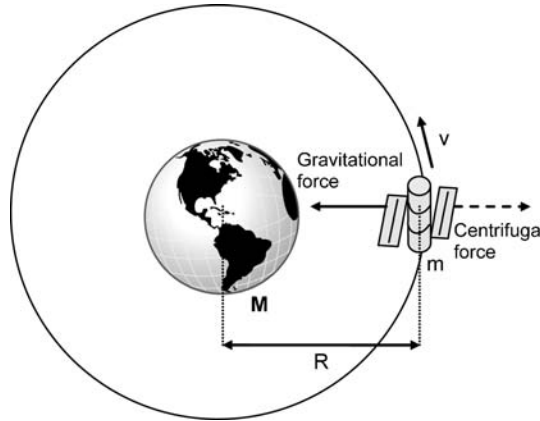


Figure 2.4 Gravitational force and the centrifugal force acting on bodies orbiting Earth

centre of the Earth (Figure 2.4). It may be mentioned here that the centrifugal force is the force exerted during circular motion, by the moving object upon the other object around which it is moving. In the case of a satellite orbiting Earth, the satellite exerts a centrifugal force. However, the force that is causing the circular motion is the centripetal force. In the absence of this centripetal force, the satellite would have continued to move in a straight line at a constant speed after injection. The centripetal force directed at right angles to the satellite's velocity towards the centre of the Earth transforms the straight line motion to the circular or elliptical one, depending upon the satellite velocity. Centripetal force further leads to a corresponding acceleration called centripetal acceleration as it causes a change in the direction of the satellite's velocity vector. The centrifugal force is simply the reaction force exerted by the satellite in a direction opposite to that of the centripetal force. This is in accordance with Newton's third law of motion, which states that for every action there is an equal and opposite reaction. This implies that there is a centrifugal acceleration acting outwards from the centre of the Earth due to the centripetal acceleration acting towards the centre of the Earth. The only radial force acting on the satellite orbiting Earth is the centripetal force. The centrifugal force is not acting on the satellite; it is only a reaction force exerted by the satellite.

The two forces can be explained from Newton's law of gravitation and Newton's second law of motion as outlined in the following paragraphs.

2.2.1 Newton's Law of Gravitation

According to Newton's law of gravitation, every particle irrespective of its mass attracts every other particle with a gravitational force whose magnitude is directly proportional to the product of the masses of the two particles and inversely proportional to the square of the distance between them and written as

$$F = \frac{Gm_1m_2}{r^2} \quad (2.1)$$

where

m_1, m_2 = masses of the two particles

r = distance between the two particles

G = gravitational constant = $6.67 \times 10^{-11} \text{ m}^3/\text{kg s}^2$

The force with which the particle with mass m_1 attracts the particle with mass m_2 equals the force with which particle with mass m_2 attracts the particle with mass m_1 . The forces are equal in magnitude but opposite in direction (Figure 2.5). The acceleration, which is force per unit mass, experienced by the two particles, however, would depend upon their masses. A larger mass experiences lesser acceleration. Newton also explained that although the law strictly applied to particles, it is applicable to real objects as long as their sizes are small compared to the distance between them. He also explained that a uniform spherical shell of matter would behave as if the entire mass of it were concentrated at its centre.

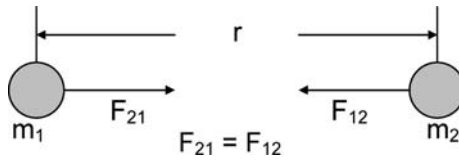


Figure 2.5 Newton's law of gravitation

2.2.2 Newton's Second Law of Motion

According to Newton's second law of motion, the force equals the product of mass and acceleration. In the case of a satellite orbiting Earth, if the orbiting velocity is v , then the acceleration, called centripetal acceleration, experienced by the satellite at a distance r from the centre of the Earth would be v^2/r . If the mass of satellite is m , it would experience a reaction force of mv^2/r . This is the centrifugal force directed outwards from the centre of the Earth and for a satellite is equal in magnitude to the gravitational force.

If the satellite orbited Earth with a uniform velocity v , which would be the case when the satellite orbit is a circular one, then equating the two forces mentioned above would lead to an expression for the orbital velocity v as follows:

$$\frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r} \quad (2.2)$$

$$v = \sqrt{\left(\frac{Gm_1}{r}\right)} = \sqrt{\left(\frac{\mu}{r}\right)} \quad (2.3)$$

where

m_1 = mass of Earth

m_2 = mass of the satellite

$\mu = Gm_1 = 3.986\,013 \times 10^5 \text{ km}^3/\text{s}^2 = 3.986\,013 \times 10^{14} \text{ N m}^2/\text{kg}$

The orbital period in such a case can be computed from

$$T = \frac{2\pi r^{3/2}}{\sqrt{\mu}} \quad (2.4)$$

In the case of an elliptical orbit, the forces governing the motion of the satellite are the same. The velocity at any point on an elliptical orbit at a distance d from the centre of the Earth is given by the formula

$$v = \sqrt{\left[\mu \left(\frac{2}{d} - \frac{1}{a} \right) \right]} \quad (2.5)$$

where

a = semi-major axis of the elliptical orbit

The orbital period in the case of an elliptical orbit is given by

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}} \quad (2.6)$$

The movement of a satellite in an orbit is governed by three Kepler's laws, explained below.

2.2.3 Kepler's Laws

Johannes Kepler, based on his lifetime study, gave a set of three empirical expressions that explained planetary motion. These laws were later vindicated when Newton gave the law of gravitation. Though given for planetary motion, these laws are equally valid for the motion of natural and artificial satellites around Earth or for any body revolving around another body. Here, these laws will be discussed with reference to the motion of artificial satellites around Earth.

2.2.3.1 Kepler's First Law

The orbit of a satellite around Earth is elliptical with the centre of the Earth lying at one of the foci of the ellipse (Figure 2.6). The elliptical orbit is characterized by its semi-major axis a and eccentricity e . Eccentricity is the ratio of the distance between the centre of the ellipse and either of its foci ($= ae$) to the semi-major axis of the ellipse a . A circular orbit is a special

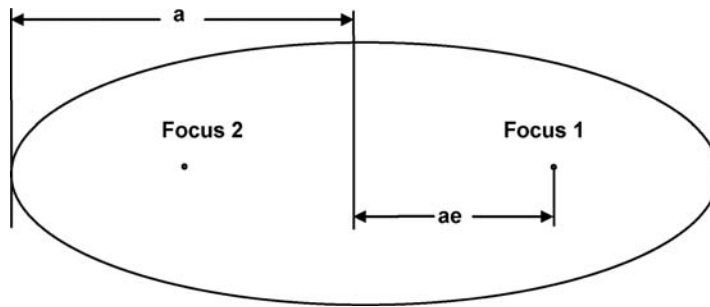


Figure 2.6 Kepler's first law

case of an elliptical orbit where the foci merge together to give a single central point and the eccentricity becomes zero. Other important parameters of an elliptical satellite orbit include its apogee (farthest point of the orbit from the Earth's centre) and perigee (nearest point of the orbit from the Earth's centre) distances. These are described in subsequent paragraphs.

For any elliptical motion, the law of conservation of energy is valid at all points on the orbit. The law of conservation of energy states that energy can neither be created nor destroyed; it can only be transformed from one form to another. In the context of satellites, it means that the sum of the kinetic and the potential energy of a satellite always remain constant. The value of this constant is equal to $-Gm_1m_2/(2a)$, where

m_1 = mass of Earth

m_2 = mass of the satellite

a = semi-major axis of the orbit

The kinetic and potential energies of a satellite at any point at a distance r from the centre of the Earth are given by

$$\text{Kinetic energy} = \frac{1}{2}(m_2v^2) \quad (2.7)$$

$$\text{Potential energy} = -\frac{Gm_1m_2}{r} \quad (2.8)$$

Therefore,

$$\frac{1}{2}(m_2v^2) - \frac{Gm_1m_2}{r} = -\frac{Gm_1m_2}{2a} \quad (2.9)$$

$$v^2 = Gm_1 \left(\frac{2}{r} - \frac{1}{a} \right) \quad (2.10)$$

$$v = \sqrt{\left[\mu \left(\frac{2}{r} - \frac{1}{a} \right) \right]} \quad (2.11)$$

2.2.3.2 Kepler's Second Law

The line joining the satellite and the centre of the Earth sweeps out equal areas in the plane of the orbit in equal time intervals (Figure 2.7); i.e. the rate (dA/dt) at which it sweeps area A is constant. The rate of change of the swept-out area is given by

$$\frac{dA}{dt} = \frac{\text{angular momentum of the satellite}}{2m} \quad (2.12)$$

where m is the mass of the satellite. Hence, Kepler's second law is also equivalent to the law of conservation of momentum, which implies that the angular momentum of the orbiting satellite given by the product of the radius vector and the component of linear momentum perpendicular to the radius vector is constant at all points on the orbit.

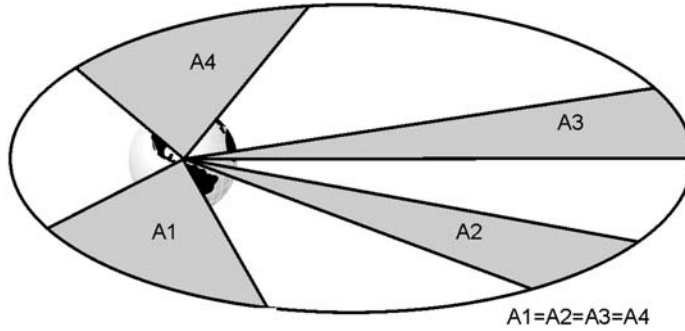


Figure 2.7 Kepler's second law

The angular momentum of the satellite of mass m is given by $mr^2\omega$, where ω is the angular velocity of the satellite. This further implies that the product $mr^2\omega = (m\omega r)(r) = mv'r$ remains constant. Here v' is the component of the satellite's velocity v in the direction perpendicular to the radius vector and is expressed as $v \cos \gamma$, where γ is the angle between the direction of motion of the satellite and the local horizontal, which is in the plane perpendicular to the radius vector r (Figure 2.8). This leads to the conclusion that the product $rv \cos \gamma$ is constant. The product reduces to rv in the case of circular orbits and also at apogee and perigee points in the case of elliptical orbits due to angle γ becoming zero. It is interesting to note here that the velocity component v' is inversely proportional to the distance r . Qualitatively, this implies that the satellite is at its lowest speed at the apogee point and the highest speed at the perigee point. In other words, for any satellite in an elliptical orbit, the dot product of its velocity vector and the radius vector at all points is constant. Hence,

$$v_p r_p = v_a r_a = vr \cos \gamma \quad (2.13)$$

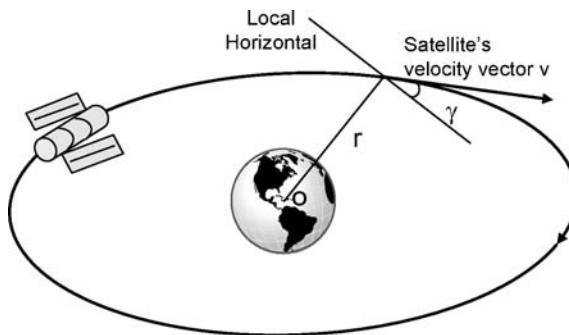


Figure 2.8 Satellite's position at any given time

where

v_p = velocity at the perigee point

r_p = perigee distance

v_a = velocity at the apogee point

r_a = apogee distance

v = satellite velocity at any point in the orbit

r = distance of the point

γ = angle between the direction of motion of the satellite and the local horizontal

2.2.3.3 Kepler's Third Law

According to the Kepler's third law, also known as the law of periods, the square of the time period of any satellite is proportional to the cube of the semi-major axis of its elliptical orbit. The expression for the time period can be derived as follows. A circular orbit with radius r is assumed. Remember that a circular orbit is only a special case of an elliptical orbit with both the semi-major axis and semi-minor axis equal to the radius. Equating the gravitational force with the centrifugal force gives

$$\frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r} \quad (2.14)$$

Replacing v by ωr in the above equation gives

$$\frac{Gm_1m_2}{r^2} = \frac{m_2\omega^2r^2}{r} = m_2\omega^2r \quad (2.15)$$

which gives $\omega^2 = Gm_1/r^3$. Substituting $\omega = 2\pi/T$ gives

$$T^2 = \left(\frac{4\pi^2}{Gm_1} \right) r^3 \quad (2.16)$$

This can also be written as

$$T = \left(\frac{2\pi}{\sqrt{\mu}} \right) r^{3/2} \quad (2.17)$$

The above equation holds good for elliptical orbits provided r is replaced by the semi-major axis a . This gives the expression for the time period of an elliptical orbit as

$$T = \left(\frac{2\pi}{\sqrt{\mu}} \right) a^{3/2} \quad (2.18)$$

2.3 Orbital Parameters

The satellite orbit, which in general is elliptical, is characterized by a number of parameters. These not only include the geometrical parameters of the orbit but also parameters that define its orientation with respect to Earth. The orbital elements and parameters will be discussed in

the following paragraphs:

1. Ascending and descending nodes
2. Equinoxes
3. Solstices
4. Apogee
5. Perigee
6. Eccentricity
7. Semi-major axis
8. Right ascension of the ascending node
9. Inclination
10. Argument of the perigee
11. True anomaly of the satellite
12. Angles defining the direction of the satellite

1. *Ascending and descending nodes.* The satellite orbit cuts the equatorial plane at two points: the first, called the descending node (N1), where the satellite passes from the northern hemisphere to the southern hemisphere, and the second, called the ascending node (N2), where the satellite passes from the southern hemisphere to the northern hemisphere (Figure 2.9).

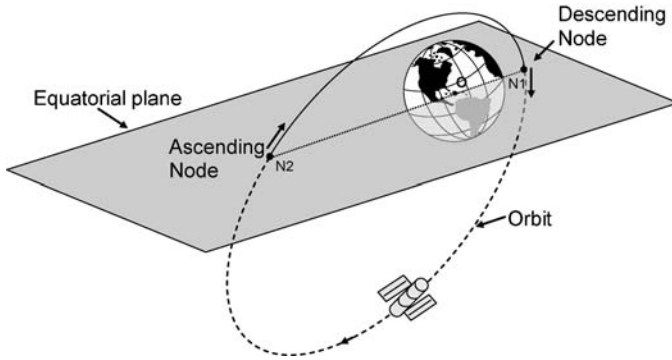


Figure 2.9 Ascending and descending nodes

2. *Equinoxes.* The inclination of the equatorial plane of Earth with respect to the direction of the sun, defined by the angle formed by the line joining the centre of the Earth and the sun with the Earth's equatorial plane follows a sinusoidal variation and completes one cycle of sinusoidal variation over a period of 365 days (Figure 2.10). The sinusoidal variation of the angle of inclination is defined by

$$\text{Inclination angle (in degrees)} = 23.4 \sin \left(\frac{2\pi t}{T} \right) \quad (2.19)$$

where T is 365 days. This expression indicates that the inclination angle is zero for $t = T/2$ and T . This is observed to occur on 20-21 March, called the spring equinox, and 22-23 September, called the autumn equinox. The two equinoxes are understandably spaced 6 months apart. During the equinoxes, it can be seen that the equatorial plane of Earth will be

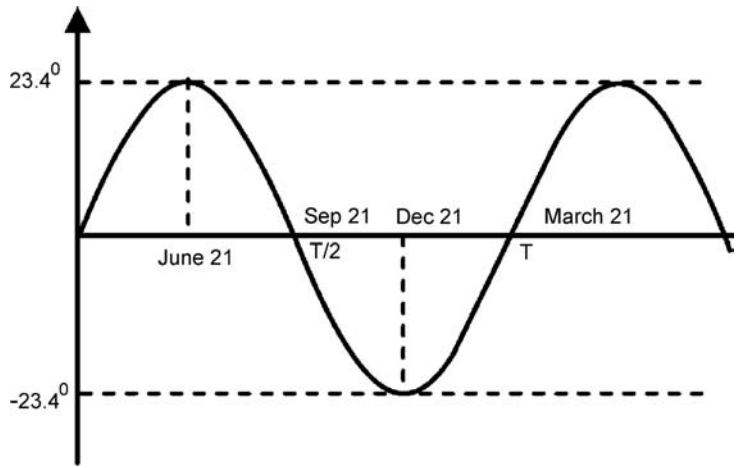


Figure 2.10 Yearly variation of angular inclination of Earth with the sun

aligned with the direction of the sun. Also, the line of intersection of the Earth's equatorial plane and the Earth's orbital plane that passes through the centre of the Earth is known as the line of equinoxes. The direction of this line with respect to the direction of the sun on 20-21 March determines a point at infinity called the vernal equinox (Y) (Figure 2.11).

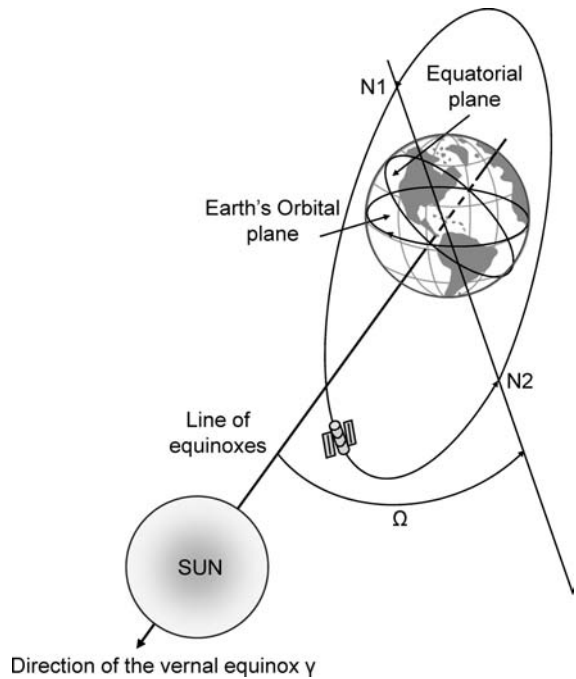


Figure 2.11 Vernal equinox

3. *Solstices*. Solstices are the times when the inclination angle is at its maximum, i.e. 23.4° . These also occur twice during a year on 20-21 June, called the summer solstice, and 21-22 December, called the winter solstice.
4. *Apogee*. Apogee is the point on the satellite orbit that is at the farthest distance from the centre of the Earth (Figure 2.12). The apogee distance can be computed from the known values of the orbit eccentricity e and the semi-major axis a from

$$\text{Apogee distance} = a(1 + e) \quad (2.20)$$

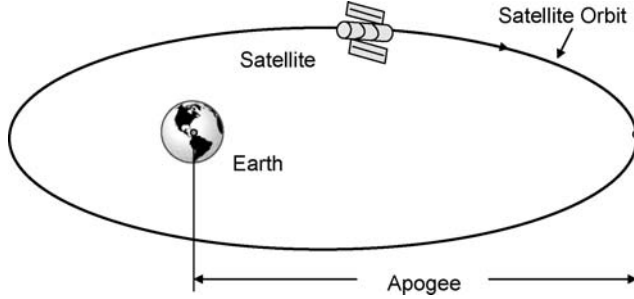


Figure 2.12 Apogee

The apogee distance can also be computed from the known values of the perigee distance and velocity at the perigee V_p from

$$V_p = \sqrt{\left(\frac{2\mu}{\text{Perigee distance}} - \frac{2\mu}{\text{Perigee distance} + \text{Apogee distance}} \right)} \quad (2.21)$$

where

$$V_p = V \left(\frac{d \cos \gamma}{\text{Perigee distance}} \right)$$

with V being the velocity of the satellite at a distance d from the centre of the Earth.

5. *Perigee*. Perigee is the point on the orbit that is nearest to the centre of the Earth (Figure 2.13). The perigee distance can be computed from the known values of orbit

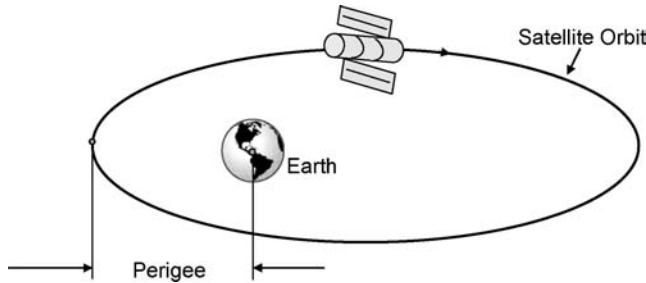


Figure 2.13 Perigee

eccentricity e and the semi-major axis a from

$$\text{Perigee distance} = a(1 - e) \quad (2.22)$$

6. *Eccentricity.* The orbit eccentricity e is the ratio of the distance between the centre of the ellipse and the centre of the Earth to the semi-major axis of the ellipse. It can be computed from any of the following expressions:

$$e = \frac{\text{apogee} - \text{perigee}}{\text{apogee} + \text{perigee}} \quad (2.23)$$

$$e = \frac{\text{apogee} - \text{perigee}}{2a} \quad (2.24)$$

Thus $e = \sqrt{(a^2 - b^2)}/a$, where a and b are semi-major and semi-minor axes respectively.

7. *Semi-major axis.* This is a geometrical parameter of an elliptical orbit. It can, however, be computed from known values of apogee and perigee distances as

$$a = \frac{\text{apogee} + \text{perigee}}{2} \quad (2.25)$$

8. *Right ascension of the ascending node.* The right ascension of the ascending node tells about the orientation of the line of nodes, which is the line joining the ascending and descending nodes, with respect to the direction of the vernal equinox. It is expressed as an angle Ω measured from the vernal equinox towards the line of nodes in the direction of rotation of Earth (Figure 2.14). The angle could be anywhere from 0° to 360° .

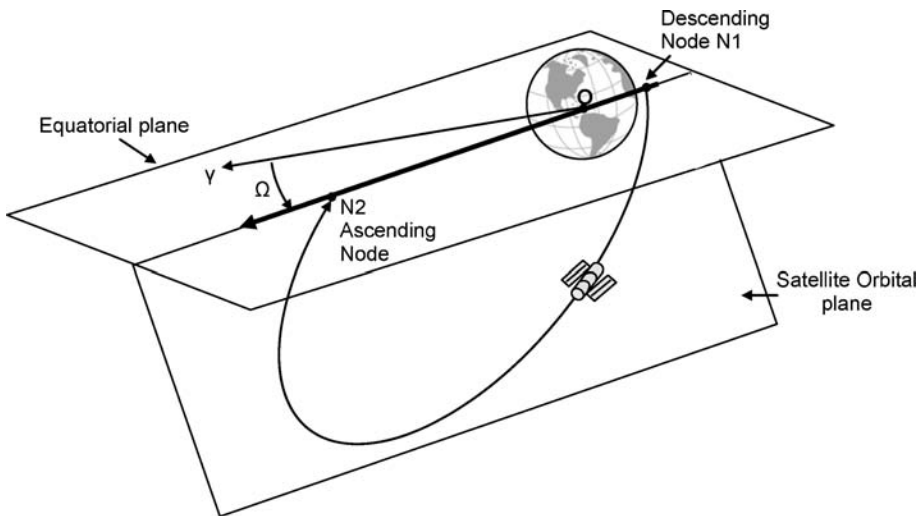


Figure 2.14 Right ascension of the ascending node

Acquisition of the correct angle of right ascension of the ascending node (Ω) is important to ensure that the satellite orbits in the given plane. This can be achieved by choosing an appropriate injection time depending upon the longitude. Angle Ω can be computed as the difference between two angles. One is the angle α between the direction of the vernal equinox and the longitude of the injection point and the other is the angle β between the line of nodes and the longitude of the injection point, as shown in Figure 2.15. Angle β can be computed from

$$\sin \beta = \frac{\cos i \sin l}{\cos l \sin i} \quad (2.26)$$

where $\angle i$ is the orbit inclination and l is the latitude at the injection point.

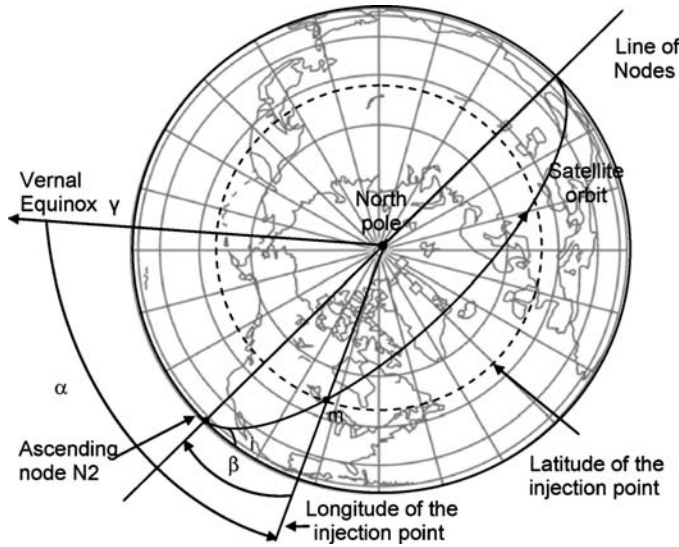


Figure 2.15 Computation of the right ascension of the ascending node

9. *Inclination.* Inclination is the angle that the orbital plane of the satellite makes with the Earth's equatorial plane. It is measured as follows. The line of nodes divides both the Earth's equatorial plane as well as the satellite's orbital plane into two halves. Inclination is measured as the angle between that half of the satellite's orbital plane containing the trajectory of the satellite from the descending node to the ascending node to that half of the Earth's equatorial plane containing the trajectory of a point on the equator from n_1 to n_2 , where n_1 and n_2 are respectively the points vertically below the descending and ascending nodes (Figure 2.16). The inclination angle can be determined from the latitude l at the injection point and the angle A_z between the projection of the satellite's velocity vector on the local horizontal and north. It is given by

$$\cos i = \sin A_z \cos l \quad (2.27)$$

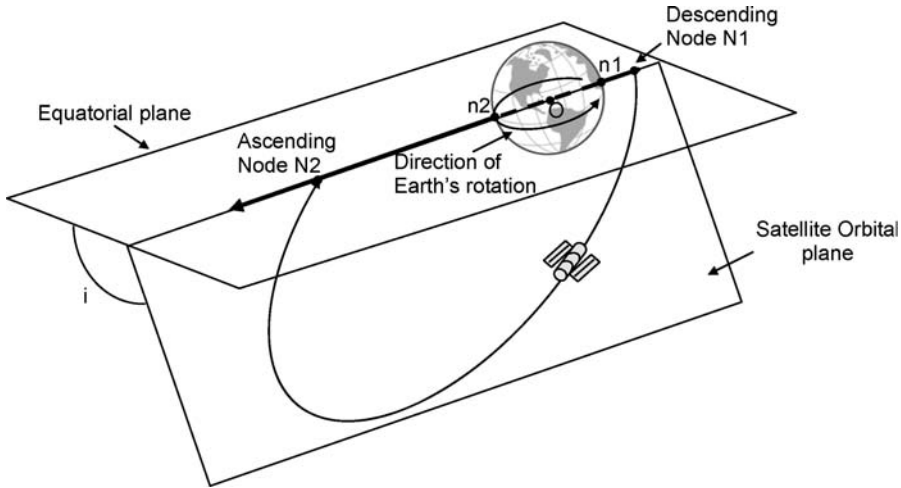


Figure 2.16 Angle of inclination

10. *Argument of the perigee.* This parameter defines the location of the major axis of the satellite orbit. It is measured as the angle ω between the line joining the perigee and the centre of the Earth and the line of nodes from the ascending node to the descending node in the same direction as that of the satellite orbit (Figure 2.17).

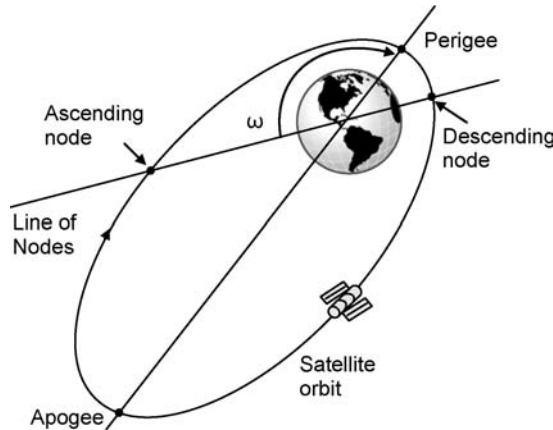


Figure 2.17 Argument of perigee

11. *True anomaly of the satellite.* This parameter is used to indicate the position of the satellite in its orbit. This is done by defining an angle θ , called the true anomaly of the satellite, formed by the line joining the perigee and the centre of the Earth with the line joining the satellite and the centre of the Earth (Figure 2.18).
12. *Angles defining the direction of the satellite.* The direction of the satellite is defined by two angles, the first by angle γ between the direction of the satellite's velocity

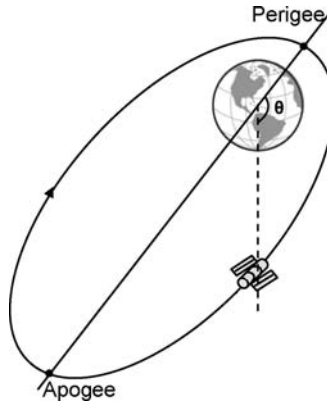


Figure 2.18 True anomaly of a satellite

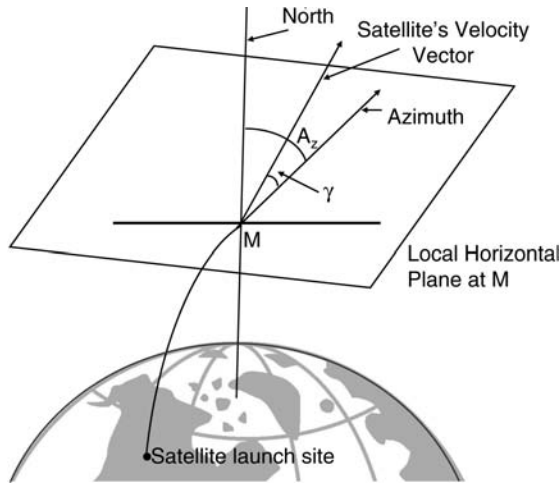


Figure 2.19 Angles defining the direction of the satellite

vector and its projection in the local horizontal and the second by angle A_z between the north and the projection of the satellite's velocity vector on the local horizontal (Figure 2.19).

Problem 2.1

A satellite is orbiting Earth in a uniform circular orbit at a height of 630 km from the surface of Earth. Assuming the radius of Earth and its mass to be 6370 km and 5.98×10^{24} kg respectively, determine the velocity of the satellite (Take the gravitational constant $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$).

Solution:

Orbit radius $R = 6370 + 630 = 7000 \text{ km} = 7\,000\,000 \text{ m}$

$$\begin{aligned}\text{Also, constant } \mu &= GM = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \\ &= 39.8 \times 10^{13} \text{ N m}^2/\text{kg} \\ &= 39.8 \times 10^{13} \text{ m}^3/\text{s}^2\end{aligned}$$

The velocity of the satellite can be computed from

$$V = \sqrt{\left(\frac{\mu}{R}\right)} = \sqrt{\left(\frac{39.8 \times 10^{13}}{7\,000\,000}\right)} = 7.54 \text{ km/s}$$

Problem 2.2

The apogee and perigee distances of a satellite orbiting in an elliptical orbit are respectively 45 000 km and 7000 km. Determine the following:

1. Semi-major axis of the elliptical orbit
2. Orbit eccentricity
3. Distance between the centre of the Earth and the centre of the elliptical orbit

Solution:

$$\begin{aligned}\text{1. Semi-major axis of the elliptical orbit } a &= \frac{\text{apogee} + \text{perigee}}{2} \\ &= \frac{45\,000 + 7000}{2} = 26\,000 \text{ km}\end{aligned}$$

$$\text{2. Eccentricity } e = \frac{\text{apogee} - \text{perigee}}{2a} = \frac{45\,000 - 7000}{2 \times 26\,000} = \frac{38\,000}{52\,000} = 0.73$$

$$\begin{aligned}\text{3. Distance between the centre of the Earth and the centre of the ellipse} &= ae \\ &= 26\,000 \times 0.73 \\ &= 18\,980 \text{ km}\end{aligned}$$

Problem 2.3

A satellite is moving in an elliptical orbit with the major axis equal to 42 000 km. If the perigee distance is 8000 km, find the apogee and the orbit eccentricity.

Solution:

Major axis = 42 000 km

Also, major axis = apogee + perigee = 42 000 km

Therefore apogee = 42 000 – 8000 = 34 000 km

$$\begin{aligned}\text{Also, eccentricity } e &= \frac{\text{apogee} - \text{perigee}}{\text{major axis}} \\ &= \frac{34\,000 - 8000}{42\,000} \\ &= \frac{26\,000}{42\,000} = 0.62\end{aligned}$$

Problem 2.4

Refer to the satellite orbit of Figure 2.20. Determine the apogee and perigee distances if the orbit eccentricity is 0.6.

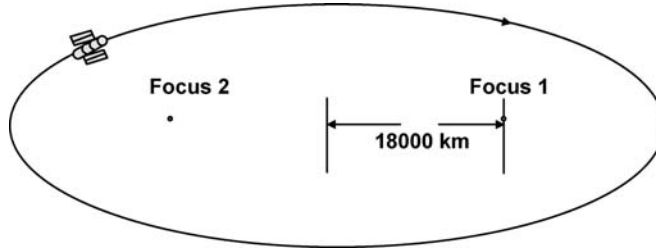


Figure 2.20 Figure for Problem 2.4

Solution: If e is the orbit eccentricity and a the semi-major axis of the elliptical orbit, then the distance between the centre of the Earth and the centre of the ellipse is equal to ae . Therefore $ae = 18\,000\text{ km}$

This gives $a = 18\,000/e = 18\,000/0.6 = 30\,000\text{ km}$

Apogee distance $= a(1 + e) = 30\,000 \times (1 + 0.6) = 48\,000\text{ km}$

Perigee distance $= a(1 - e) = 30\,000 \times (1 - 0.6) = 12\,000\text{ km}$

Problem 2.5

The difference between the furthest and the closest points in a satellite's elliptical orbit from the surface of the Earth is $30\,000\text{ km}$ and the sum of the distances is $50\,000\text{ km}$. If the mean radius of the Earth is considered to be 6400 km , determine orbit eccentricity.

Solution: Apogee – Perigee $= 30\,000\text{ km}$ as the radius of the Earth will cancel in this case

$$\text{Apogee} + \text{Perigee} = 50\,000 + 2 \times 6400 = 62\,800\text{ km}$$

$$\text{Orbit eccentricity} = (\text{Apogee} - \text{Perigee})/(\text{Apogee} + \text{Perigee}) = 30\,000/62\,800 = 0.478$$

Problem 2.6

Refer to Figure 2.21. Satellite A is orbiting Earth in a near-Earth circular orbit of radius 7000 km . Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of $47\,000\text{ km}$ and 7000 km respectively. Determine the velocities of the two satellites at point X. (Take $\mu = 39.8 \times 10^{13}\text{ m}^3/\text{s}^2$.)

Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$V = \sqrt{\left(\frac{\mu}{R}\right)}$$

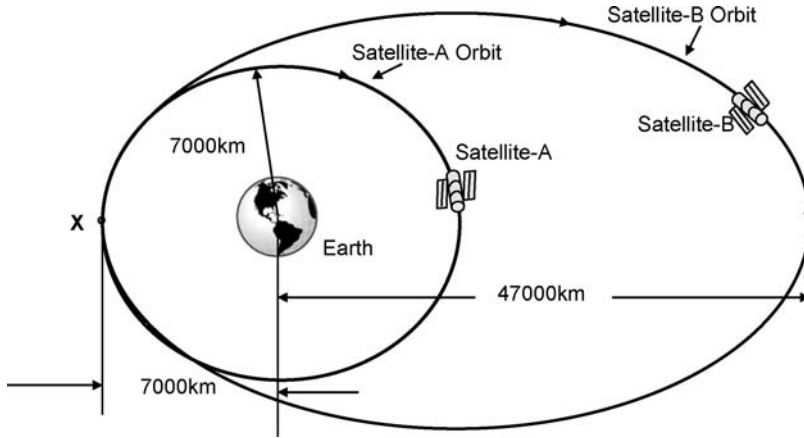


Figure 2.21 Figure for Problem 2.6

Therefore the velocity of satellite A at point X = $\sqrt{\left(\frac{39.8 \times 10^{13}}{7\,000\,000}\right)}$

$$= 7.54 \text{ km/s}$$

The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. Here $R = 7000 \text{ km}$ and $a = (47\,000 + 7000)/2 = 27\,000 \text{ km}$. Therefore, velocity of satellite B at point X is given by

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{7\,000\,000} - \frac{1}{27\,000\,000}\right)\right]} = 9.946 \text{ km/s}$$

Problem 2.7

Refer to Figure 2.22. Satellite A is orbiting Earth in an equatorial circular orbit of radius 42 000 km. Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of 42 000 km and 7000 km respectively. Determine the velocities of the two satellites at point X. (Take $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$v = \sqrt{\left(\frac{\mu}{R}\right)}$$

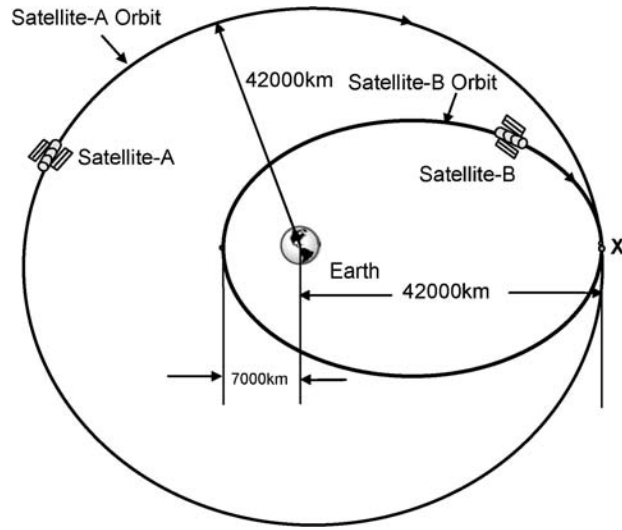


Figure 2.22 Figure for Problem 2.7

Therefore, the velocity of satellite A at point X =
$$\sqrt{\left(\frac{39.8 \times 10^{13}}{42\,000\,000}\right)}$$

$$= 3.078 \text{ km/s}$$

The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. Here $R = 42\,000 \text{ km}$ and $a = (42\,000 + 7\,000)/2 = 24\,500 \text{ km}$. Therefore, velocity of satellite B at point X is given by

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{42\,000\,000} - \frac{1}{24\,500\,000}\right)\right]} = 1.645 \text{ km/s}$$

Problem 2.8

Refer to Figure 2.23. Satellite A is orbiting Earth in a circular orbit of radius 25 000 km. Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of 43 000 km and 7 000 km respectively. Determine the velocities of the two satellites at the indicated points X and Y. (Take $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$v = \sqrt{\left(\frac{\mu}{R}\right)}$$

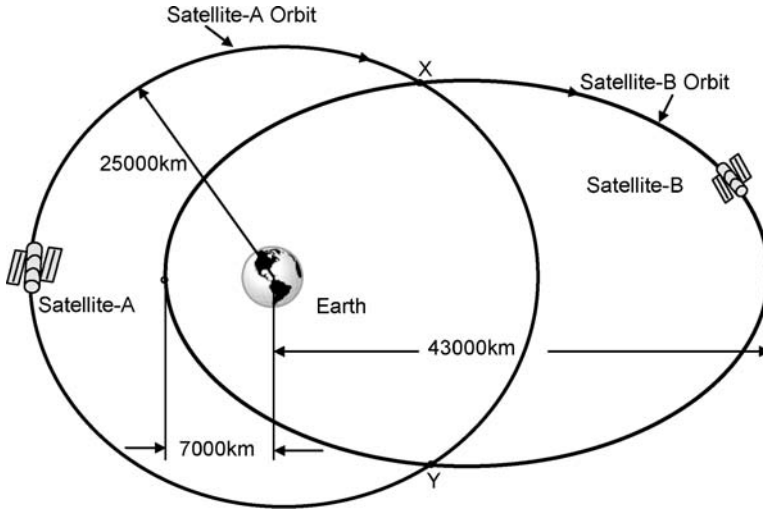


Figure 2.23 Figure for Problem 2.8

Therefore the velocity of satellite A at points X and Y = $\sqrt{\left(\frac{39.8 \times 10^{13}}{25\,000\,000}\right)} = 3.989 \text{ km/s}$
 The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. For satellite B, at points X and Y, $R = 25\,000 \text{ km}$ and $a = (43\,000 + 7\,000)/2 = 25\,000 \text{ km}$. Therefore, velocity of satellite B at points X and Y

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{25\,000\,000} - \frac{1}{25\,000\,000}\right)\right]} = 3.989 \text{ m/s}$$

Please note: In Problems 2.6, 2.7 and 2.8, three different cases of satellites moving in circular and elliptical orbits have been considered. Note that the velocity of the satellite at a point in the elliptical orbit is more than the velocity of the satellite in a circular orbit if the distance of the satellite from the centre of the Earth happens to be less than the radius of the circular orbit. It will be less than the velocity in the circular orbit if the distance of the satellite in the elliptical orbit happens to be more than the radius of the circular orbit. The two velocities are the same at a point where the distance of the satellite from the centre of the Earth in the elliptical orbit equals the radius of the circular orbit provided that the semi-major axis of the ellipse also equals the radius of the circular orbit.

Problem 2.9

Calculate the orbital period of a satellite moving in an elliptical orbit having a major axis of 50 000 km. (Take $\mu = 39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$).

Solution: Semi-major axis, $a = 50\,000/2 = 25\,000 \text{ km}$

Orbital time period, $T = 2\pi \times \sqrt{(a^3/\mu)} = 2 \times 3.14 \times \sqrt{(25\,000\,000)^3/(39.8 \times 10^{13})}$
 $= 39\,344 \text{ seconds} = 10 \text{ hours } 55 \text{ minutes } 44 \text{ seconds}$

Problem 2.10

Refer to Figure 2.24. The semi-major axes of the two satellites shown in the figure are 18 000 km (satellite 1) and 24 000 km (satellite 2). Determine the relationship between their orbital periods.

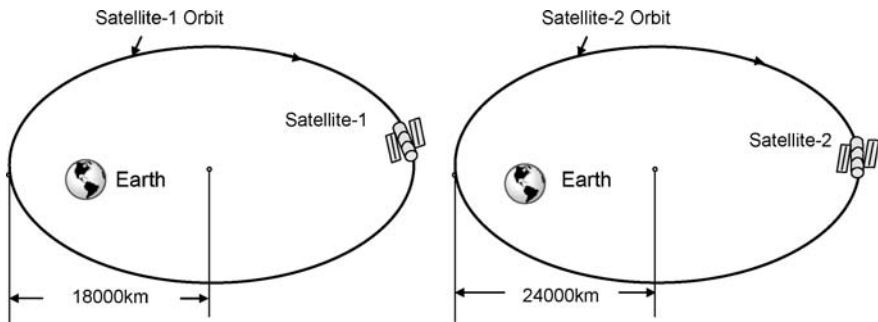


Figure 2.24 Figure for Problem 2.10

Solution: The orbital time period $T = 2\pi \times \sqrt{(a^3/\mu)}$ where a is the semi-major axis. If a_1 and a_2 are the semi-major axes in the two cases and T_1 and T_2 are their time periods, then

$$T_1 = 2\pi \times \sqrt{\left(\frac{a_1^3}{\mu}\right)}$$

and

$$T_2 = 2\pi \times \sqrt{\left(\frac{a_2^3}{\mu}\right)}$$

This gives

$$\frac{T_2}{T_1} = \left(\frac{a_2}{a_1}\right)^{3/2} = \left(\frac{24\,000}{18\,000}\right)^{3/2} = 1.54$$

The orbital period of satellite 2 is 1.54 times that of satellite 1.

Problem 2.11

The elliptical eccentric orbit of a satellite has its semi-major and semi-minor axes as 25 000 km and 18 330 km respectively. Determine the apogee and perigee distances.

Solution: If r_a and r_p are the apogee and perigee distances respectively, a and b are the semi-major and semi-minor axis respectively and e is the eccentricity, then

$$a = \frac{r_a + r_p}{2}$$

and

$$b = \sqrt{r_a r_p}$$

Now $(r_a + r_p)/2 = 25\,000$, which gives $r_a + r_p = 50\,000$

and $\sqrt{r_a r_p} = 18\,330$, which gives $r_a r_p = 335\,988\,900$

Substituting the value of r_p from these equations gives

$$r_a(50\,000 - r_a) = 335\,988\,900$$

which gives

$$r_a^2 - 50\,000r_a + 335\,988\,900 = 0$$

Solving this quadratic equation gives two solutions:

$$r_a = 42\,000 \text{ km}, 8000 \text{ km}$$

If the semi-major axis is 25 000 km, r_a cannot be 8000 km. Therefore

$$\text{Apogee distance } r_a = 42\,000 \text{ km}$$

and

$$\text{Perigee distance } r_p = 50\,000 - 42\,000 = 8000 \text{ km}$$

Alternative solution:

$$e = \frac{\sqrt{a^2 - b^2}}{a} = e = \frac{\sqrt{[(25\,000)^2 - (18\,330)^2]}}{25\,000} = 0.68$$

This gives

$$\text{Apogee distance} = a(1 + e) = 25\,000 \times 1.68 = 42\,000 \text{ km}$$

$$\text{and Perigee distance} = a(1 - e) = 25\,000 \times 0.32 = 8000 \text{ km}$$

Therefore, from both the methods,

$$\text{Apogee distance} = 42\,000 \text{ km and Perigee distance} = 8000 \text{ km}$$

Problem 2.12

Refer to Figure 2.25. The satellite is moving in an elliptical orbit with its semi-major and semi-minor axes as a and b respectively and an eccentricity of 0.6. The satellite takes

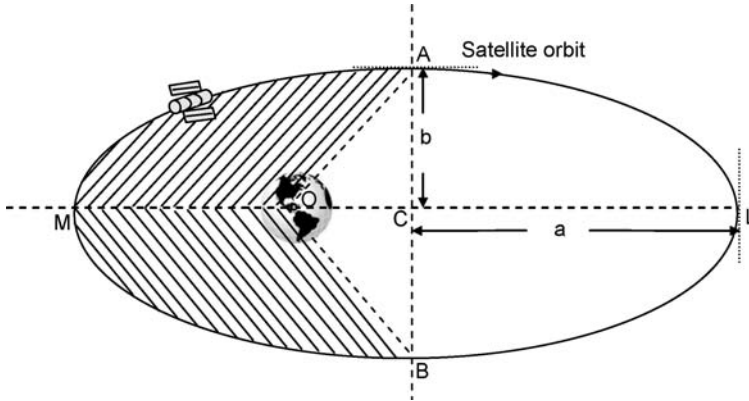


Figure 2.25 Figure for Problem 2.12

3 hours to move from point B to point A. How much time will it take to move from point A to point B?

Solution: This problem can be solved by applying Kepler's law of areas, which says that the line joining the satellite and centre of the Earth spans equal areas of the ellipse in equal times. Now, the area spanned when the satellite moves from point B to point A (area of the shaded region) is given by

$$\begin{aligned}
 \frac{1}{2} \times (\text{area of ellipse}) - (\text{area of } \triangle AOB) &= \frac{\pi ab}{2} - b \cdot OC \\
 &= \frac{\pi ab}{2} - bae \text{ (where } OC = ae) \\
 &= 1.57ab - 0.6ab \\
 &= 0.97ab
 \end{aligned}$$

The area spanned when the satellite moves from point A to point B (area of the unshaded region) is given by

$$\begin{aligned}
 \frac{1}{2} \times (\text{area of ellipse}) + (\text{area of } \triangle AOB) &= 1.57ab + 0.6ab \\
 &= 2.17ab
 \end{aligned}$$

The ratio of the two areas is given by $2.17/0.97 = 2.237$. The time taken by the satellite to move from point A to point B should therefore be 2.237 times the time taken by the satellite to move from point B to point A. Therefore

$$\text{Time taken} = 3 \times 2.237 = 6.711 \text{ hours}$$

Problem 2.13

The apogee and perigee distances of a certain elliptical satellite orbit are 42 000 km and 8000 km respectively. If the velocity at the perigee point is 9.142 km/s, what would be the velocity at the apogee point?

Solution: The product of the distance of a satellite from the centre of the Earth and the component of velocity vector in the direction of local horizontal at that point is constant at all possible points on the orbit; i.e. $dv \cos \gamma = \text{constant}$, where γ is the angle between the direction of motion of the satellite and the local horizontal. This has been explained earlier. The angle γ is zero at the apogee and perigee points. The expression thus reduces to $dv = \text{constant}$ for the apogee and perigee points. Therefore

$$\text{Velocity at the apogee} = \frac{8000 \times 9.142}{42\,000} = 1.741 \text{ km/s}$$

Problem 2.14

In the satellite orbit of Problem 2.13, at a certain point in the orbit at a distance of 16 000 km from the centre of the Earth, the direction of the satellite makes an angle of 56.245° with the local horizontal at that point (Figure 2.26). Determine the satellite velocity at that point.

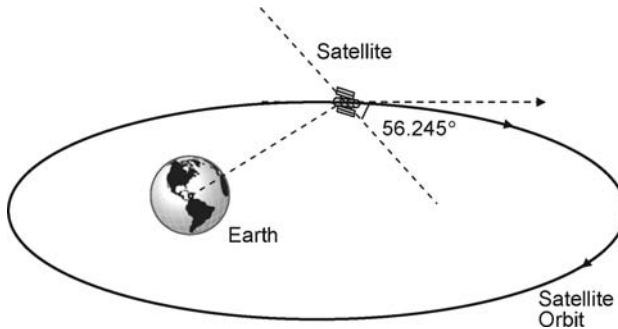


Figure 2.26 Figure for Problem 2.14

Solution: The product $dv \cos \gamma = 8000 \times 9.142$, which gives

$$\begin{aligned} v &= \frac{8000 \times 9.142}{16\,000 \times \cos 56.245^\circ} \\ &= \frac{73\,136}{16\,000 \times 0.555} = \frac{73\,136}{8880} = 8.236 \text{ km/s} \end{aligned}$$

Therefore

$$v = 8.236 \text{ km/s}$$

2.4 Injection Velocity and Resulting Satellite Trajectories

The horizontal velocity with which a satellite is injected into space by the launch vehicle with the intention of imparting a specific trajectory to the satellite has a direct bearing on the satellite trajectory. The phenomenon is best explained in terms of the three cosmic velocities. The general expression for the velocity of a satellite at the perigee point (V_P), assuming an elliptical orbit, is given by

$$V_P = \sqrt{\left[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)\right]} \quad (2.28)$$

where

R = apogee distance

r = perigee distance

$\mu = GM = \text{constant}$

The first cosmic velocity V_1 is the one at which apogee and perigee distances are equal, i.e. $R = r$, and the orbit is circular. The above expression then reduces to

$$V_1 = \sqrt{\left(\frac{\mu}{r}\right)} \quad (2.29)$$

Thus, irrespective of the distance r of the satellite from the centre of the Earth, if the injection velocity is equal to the first cosmic velocity, also sometimes called the first orbital velocity, the satellite follows a circular orbit (Figure 2.27) and moves with a uniform velocity equal to $\sqrt{(\mu/r)}$. A simple calculation shows that for a satellite at 35 786 km above the surface of the Earth, the first cosmic velocity turns out to be 3.075 km/s and the orbital period is 23 hours 56 minutes, which is equal to the time period of one sidereal day – the time taken by Earth to complete one full rotation around its axis with reference to distant stars. This confirms why a geostationary satellite needs to be at a height of 35 786 km above the surface of the Earth. Different types of orbits are discussed at length in the following pages.

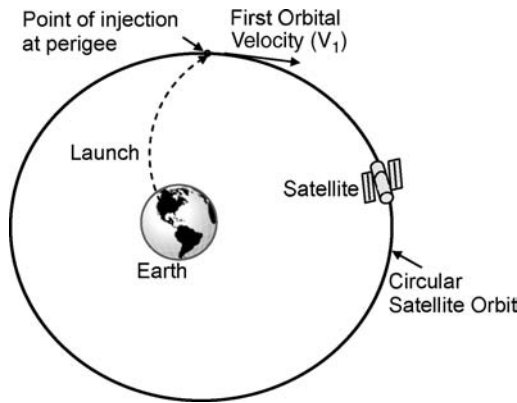


Figure 2.27 Satellite's path where the injection velocity is equal to the first orbital velocity

If the injection velocity happens to be less than the first cosmic velocity, the satellite follows a ballistic trajectory and falls back to Earth. In fact, in this case, the orbit is elliptical and the injection point is at the apogee and not the perigee. If the perigee lies in the atmosphere or exists only virtually below the surface of the Earth, the satellite accomplishes a ballistic flight and falls back to Earth (Figure 2.28).

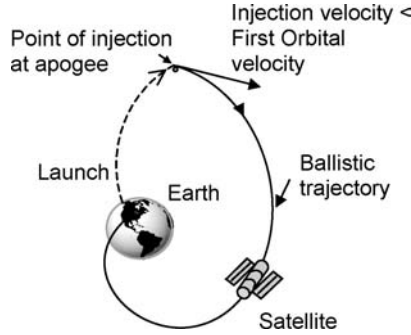


Figure 2.28 Satellite's path where the injection velocity is less than the first orbital velocity

For injection velocity greater than the first cosmic velocity and less than the second cosmic velocity, i.e. $V > \sqrt{\mu/r}$ and $V < \sqrt{2\mu/r}$, the orbit is elliptical and eccentric. The orbit eccentricity is between 0 and 1. The injection point in this case is the perigee and the apogee distance attained in the resultant elliptical orbit depends upon the injection velocity. The higher the injection velocity, the greater is the apogee distance. The apogee distance can also be computed from the known value of injection velocity, which is also the velocity at the perigee point as the perigee coincides with the injection point, and the velocity v at any other point in the orbit distant d from the centre of the Earth using

$$V_p = \sqrt{\left[(2\mu/r) - \left(\frac{2\mu}{R+r} \right) \right]} = \frac{vd \cos \gamma}{r} \quad (2.30)$$

When the injection velocity equals $\sqrt{2\mu/r}$, the apogee distance R becomes infinite and the orbit takes the shape of a parabola (Figure 2.29) and the orbit eccentricity is 1. This is the second cosmic velocity v_2 . At this velocity, the satellite escapes Earth's gravitational pull. For an injection velocity greater than the second cosmic velocity, the trajectory is hyperbolic within the solar system and the orbit eccentricity is greater than 1.

If the injection velocity is increased further, a stage is reached where the satellite succeeds in escaping from the solar system. This is known as the third cosmic velocity and is related to the motion of planet Earth around the sun. The third cosmic velocity (V_3) is mathematically expressed as

$$V_3 = \sqrt{\left[\frac{2\mu}{r} - V_t^2(3 - 2\sqrt{2}) \right]} \quad (2.31)$$

where V_t is the speed of Earth's revolution around the sun.

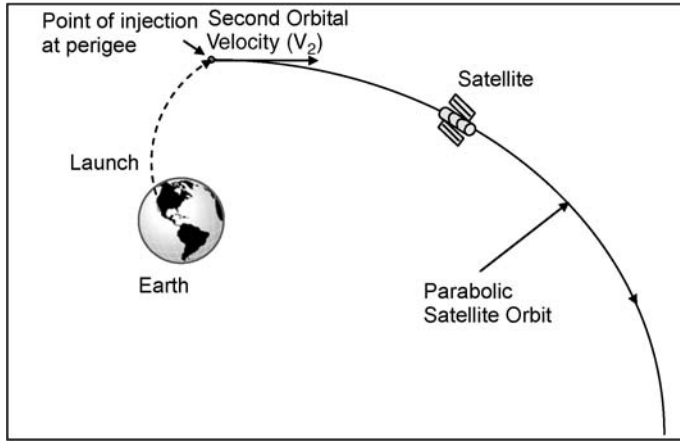


Figure 2.29 Satellite's path where the injection velocity is equal to the second orbital velocity

Beyond the third cosmic velocity, there is a region of hyperbolic flights outside the solar system. Coming back to elliptical orbits, the greater the injection velocity from the first cosmic velocity, the greater is the apogee distance. This is evident from the generalized expression for the velocity of the satellite in elliptical orbits according to which

V_p = velocity at the perigee point

$$= \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R+r} \right) \right]} \quad (2.32)$$

For a given perigee distance r , a higher velocity at the perigee point, which is also the injection velocity, necessitates that the apogee distance R is greater. Figure 2.30 shows a family of curves that can be used to find out the attained apogee height for a given value of injection velocity at the perigee point. The dashed line shows the relationship between the injection velocity and altitude for a circular orbit. For example, for a perigee height of 1000 km above the surface of the Earth, the injection velocity for a circular orbit as seen from the curve is about 7.3 km/s. If the injection velocity is increased to 8 km/s for this perigee height, the orbit attains an apogee height of about 4200 km. If it is increased further to 9 km/s, the apogee height goes up to 16 000 km. It can also be seen that the second cosmic velocity for this perigee height is about 10.3 km/s, as given by the vertical line to which the 1000 km perigee height curve approaches asymptotically.

In fact, for a given perigee distance r , it can be proved that the injection velocities and corresponding apogee distances are related by

$$\left(\frac{v_2}{v_1} \right)^2 = \frac{1 + r/R_1}{1 + r/R_2} \quad (2.33)$$

Refer to Problem 2.15 for the derivation of this expression.

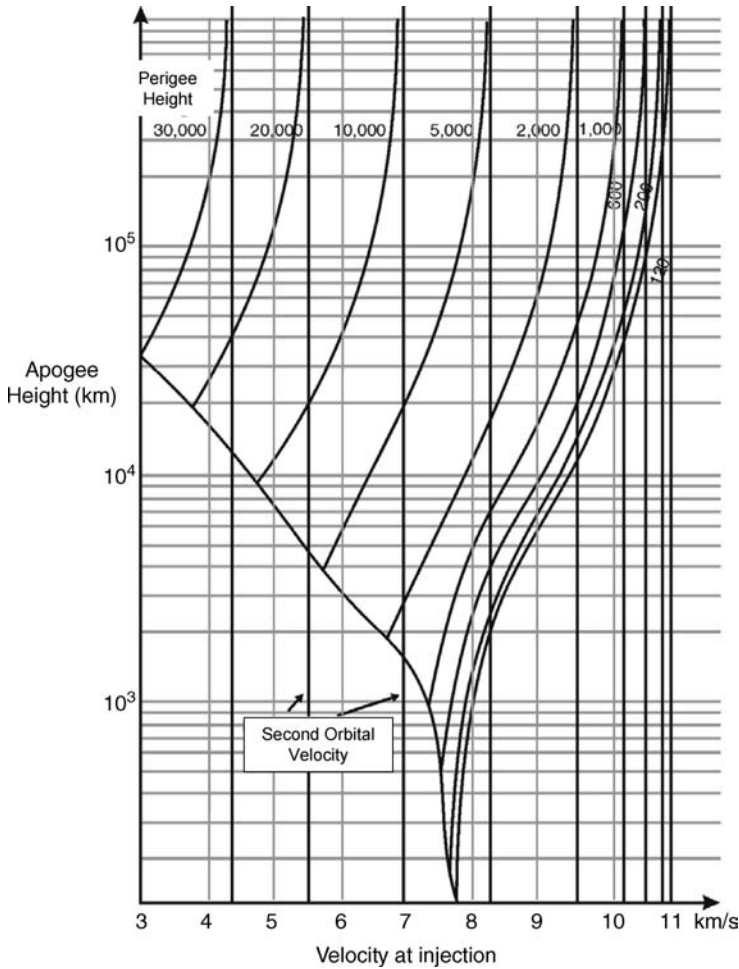


Figure 2.30 Family of curves showing the apogee height for various values of the injection velocity

Problem 2.15

A satellite launched with an injection velocity of v_1 from a point above the surface of the Earth at a distance r from the centre of the Earth attains an elliptical orbit with an apogee distance of R_1 . The same satellite when launched with an injection velocity of v_2 from the same perigee distance attains an elliptical orbit with an apogee distance of R_2 . Derive the relationship between v_1 and v_2 in terms of r , R_1 and R_2 .

Solution: Refer to Figure 2.31. Expressions for the velocity at perigee points can be written as

$$v_1 = \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R_1 + r} \right) \right]}$$

$$v_2 = \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R_2 + r} \right) \right]}$$

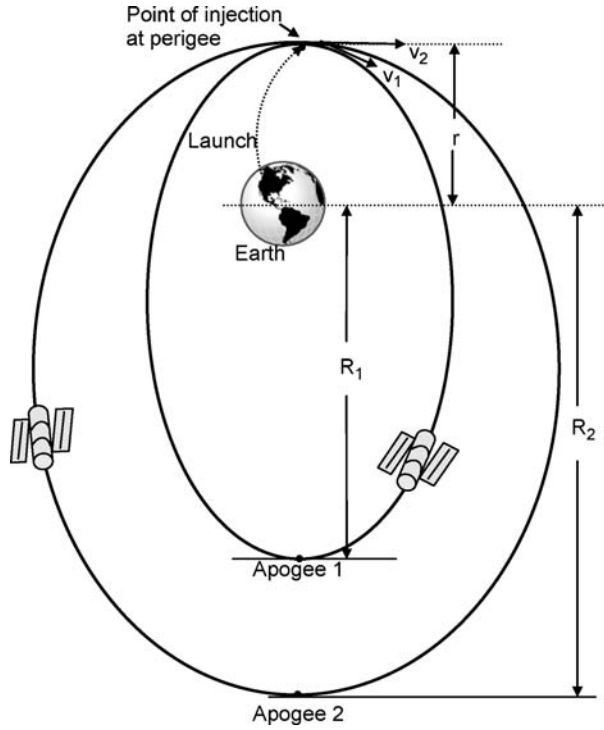


Figure 2.31 Figure for Problem 2.15

Squaring the two expressions and then taking the ratio of the two yields

$$\begin{aligned}
 \left(\frac{v_2}{v_1} \right)^2 &= \frac{(1/r) - 1/(R_2 + r)}{(1/r) - 1/(R_1 + r)} \\
 &= \frac{R_2/r(R_2 + r)}{R_1/r(R_1 + r)} \\
 &= \left(\frac{R_2}{R_1} \right) \left(\frac{R_1 + r}{R_2 + r} \right) \\
 &= \frac{1 + r/R_1}{1 + r/R_2}
 \end{aligned}$$

Therefore,

$$\left(\frac{v_2}{v_1} \right)^2 = \frac{1 + r/R_1}{1 + r/R_2}$$

Problem 2.16

A satellite launched with an injection velocity of (v_1) from a point above the surface of the Earth at a distance of (r) of 8000 km from the centre of the Earth attains an elliptical orbit

with an apogee distance of 12 000 km. The same satellite when launched with an injection velocity of (v_2) that is 20 % higher than (v_1) from the same perigee distance attains an elliptical orbit with an apogee distance of (R_2). Determine the new apogee distance.

Solution:

We know that: $(v_2/v_1)^2 = [(1 + r/R_1)/(1 + r/R_2)]$

Substituting $v_2 = 1.2v_1$, we get $1.44 = [(1 + 8000/12\,000)/(1 + 8000/R_2)]$

or $1.44 \times 3 \times (1 + 8000/R_2) = 5$ or $(1 + 8000/R_2) = 1.1574$

$R_2 = 8000/0.1574 = 50\,826$ km

Problem 2.17

A rocket injects a satellite with a horizontal velocity of 8 km/s from a height of 1620 km from the surface of the Earth. What will be the velocity of the satellite at a point distant 10 000 km from the centre of the Earth, if the direction of the satellite makes an angle of 30° with the local horizontal at that point? Assume radius of the Earth to be 6380 km.

Solution:

We know that $V_2 = \sqrt{[2\mu/r] - [2\mu/(R+r)]} = (v \cdot d \cdot \cos \gamma)/r$

Here $V_2 = 8$ km/s, $r = 1620 + 6380 = 8000$ km, $d = 10\,000$ km and $\gamma = 30^\circ$

Therefore, $[v \times 10\,000 \times \cos 30^\circ / 8000] = 8$

or $v = (8 \times 8000)/(10\,000 \times 0.866) = 7.39$ km/s

Problem 2.18

A rocket injects a satellite with a certain horizontal velocity from a height of 620 km from the surface of the Earth. The velocity of the satellite at a point distant 9000 km from the centre of the Earth is observed to be 8 km/s. If the direction of the satellite makes an angle of 30° with the local horizontal at that point, determine the apogee distance of the satellite orbit. (Assume that the radius of the Earth is 6380 km and $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution:

It is known that $V_p = \sqrt{\left[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)\right]} = \frac{vd \cos \gamma}{r}$

Here,

$r = 620 + 6380 = 7000$ km

$d = 9000$ km

$v = 8$ km/s

$\gamma = 30^\circ$

$R = \text{apogee distance} = ?$

Therefore,

$$\sqrt{\left[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)\right]} = 8 \times 9000 \times \frac{0.866}{7000} = 8.90 \text{ km/s} = 8.90 \times 10^3 \text{ m/s}$$

or

$$\frac{2\mu}{r} - \frac{2\mu}{R+r} = (8.90 \times 10^3)^2 = 79.34 \times 10^6$$

or

$$\frac{2\mu}{R+r} = \frac{2 \times 39.8 \times 10^{13}}{7\,000\,000} - (79.34 \times 10^6) = 34.36 \times 10^6$$

This gives

$$R+r = \frac{2 \times 39.8 \times 10^{13}}{34.36 \times 10^6} = 2.317 \times 10^7 \text{ m}$$

$$\text{or } R = (2.317 \times 10^7 - 7\,000\,000) \text{ m} = 16\,170 \text{ km}$$

2.5 Types of Satellite Orbits

As described at length in the earlier pages of this chapter, satellites travel around Earth along predetermined repetitive paths called orbits. The orbit is characterized by its elements or parameters, which have been covered at length in earlier sections. The orbital elements of a particular satellite depend upon its intended application. The satellite orbits can be classified on the basis of:

1. Orientation of the orbital plane
2. Eccentricity
3. Distance from Earth

2.5.1 Orientation of the Orbital Plane

The orbital plane of the satellite can have various orientations with respect to the equatorial plane of Earth. The angle between the two planes is called the angle of inclination of the satellite. On this basis, the orbits can be classified as equatorial orbits, polar orbits and inclined orbits.

In the case of an equatorial orbit, the angle of inclination is zero, i.e. the orbital plane of the satellite coincides with the Earth's equatorial plane (Figure 2.32). A satellite in the equatorial orbit has a latitude of 0° . For an angle of inclination equal to 90° , the satellite is said to be in the polar orbit (Figure 2.33). For an angle of inclination between 0° and 180° , the orbit is said to be an inclined orbit.

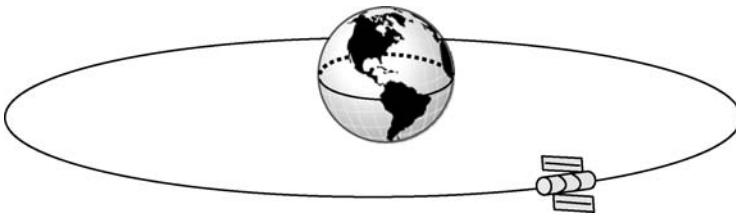


Figure 2.32 Equatorial orbit

For inclinations between 0° and 90° , the satellite travels in the same direction as the direction of rotation of the Earth. The orbit in this case is referred to as a direct or prograde orbit



Figure 2.33 Polar orbit

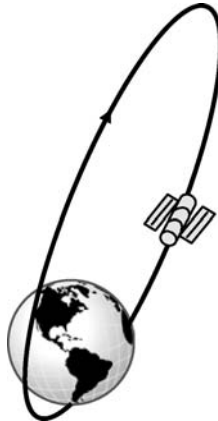


Figure 2.34 Prograde orbit

(Figure 2.34). For inclinations between 90° and 180° , the satellite orbits in a direction opposite to the direction of rotation of the Earth and the orbit in this case is called a retrograde orbit (Figure 2.35).

2.5.2 *Eccentricity of the Orbit*

On the basis of eccentricity, the orbits are classified as elliptical (Figure 2.36 (a)) and circular (Figure 2.36 (b)) orbits. Needless to say, when the orbit eccentricity lies between 0 and 1, the orbit is elliptical with the centre of the Earth lying at one of the foci of the ellipse. When the eccentricity is zero, the orbit becomes circular. It may be mentioned here that all circular orbits are eccentric to some extent. As an example, the eccentricity of orbit of geostationary



Figure 2.35 Retrograde orbit

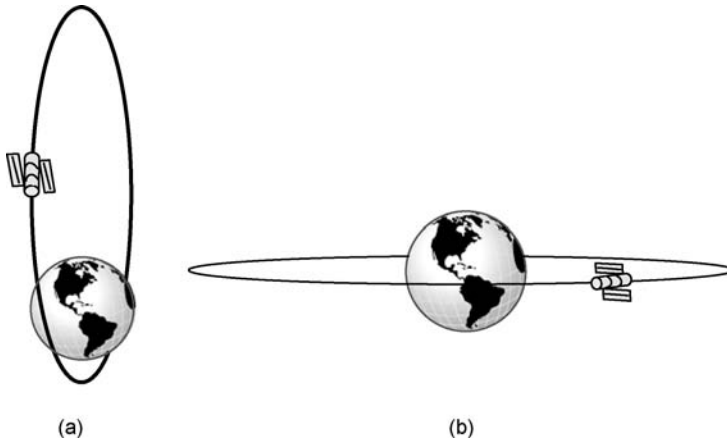


Figure 2.36 (a) Elliptical orbit and (b) circular orbit

satellite INSAT-3B, an Indian satellite in the INSAT series providing communication and meteorological services, is 0.000 252 6. Eccentricity figures for orbits of GOES-9 and Meteosat-7 geostationary satellites, both offering weather forecasting services, are 0.000 423 3 and 0.000 252 6 respectively.

2.5.2.1 Molniya Orbit

Highly eccentric, inclined and elliptical orbits are used to cover higher latitudes, which are otherwise not covered by geostationary orbits. A practical example of this type of orbit is the Molniya orbit (Figure 2.37). It is widely used by Russia and other countries of the former Soviet Union to provide communication services. Typical eccentricity and orbit inclination

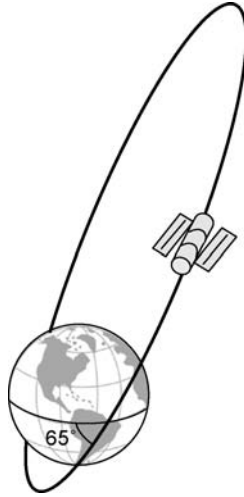


Figure 2.37 Molniya orbit

figures for the Molniya orbit are 0.75 and 65° respectively. The apogee and perigee points are about 40 000 km and 400 km respectively from the surface of the Earth.

The Molniya orbit serves the purpose of a geosynchronous orbit for high latitude regions. It is a 12 hour orbit and a satellite in this orbit spends about 8 hours above a particular high latitude station before diving down to a low level perigee at an equally high southern latitude. Usually, three satellites at different phases of the same Molniya orbit are capable of providing an uninterrupted service.

2.5.3 Distance from Earth

Again depending upon the intended mission, satellites may be placed in orbits at varying distances from the surface of the Earth. Depending upon the distance, these are classified as low Earth orbits (LEOs), medium Earth orbits (MEOs) and geostationary Earth orbits (GEOs), as shown in Figure 2.38.

Satellites in the low Earth orbit (LEO) circle Earth at a height of around 160 to 500 km above the surface of the Earth. These satellites, being closer to the surface of the Earth, have much shorter orbital periods and smaller signal propagation delays. A lower propagation delay makes them highly suitable for communication applications. Due to lower propagation paths, the power required for signal transmission is also less, with the result that the satellites are of small physical size and are inexpensive to build. However, due to a shorter orbital period, of the order of an hour and a half or so, these satellites remain over a particular ground station for a short time. Hence, several of these satellites are needed for 24 hour coverage. One important application of LEO satellites for communication is the project Iridium, which is a global communication system conceived by Motorola (Figure 2.39). A total of 66 satellites are arranged in a distributed architecture, with each satellite carrying 1/66 of the total system capacity. The system is intended to provide a variety of telecommunication

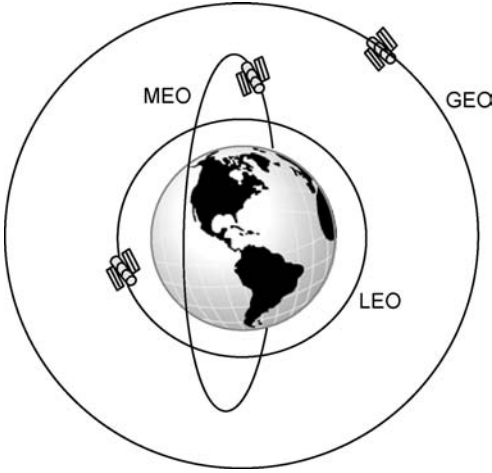


Figure 2.38 LEO, MEO and GEO



Figure 2.39 Iridium constellation of satellites

services at the global level. The project is named ‘Iridium’ as earlier the constellation was proposed to have 77 satellites and the atomic number of iridium is 77. Other applications where LEO satellites can be put to use are surveillance, weather forecasting, remote sensing and scientific studies.

Medium Earth orbit (MEO) satellites orbit at a distance of approximately 10 000 to 20 000 km above the surface of the Earth. They have an orbital period of 6 to 12 hours. These satellites stay in sight over a particular region of Earth for a longer time. The transmission distance and propagation delays are greater than those for LEO satellites. These orbits are generally polar in nature and are mainly used for communication and navigation applications.

A geosynchronous Earth orbit is a prograde orbit whose orbital period is equal to Earth's rotational period. If such an orbit were in the plane of the equator and circular, it would remain stationary with respect to a given point on the Earth. These orbits are referred to as the geostationary Earth orbits (GEOs). For the satellite to have such an orbital velocity, it needs to be at a height of about 36 000 km, 35 786 km to be precise, above the surface of the Earth.

To be more precise and technical, in order to remain above the same point on the Earth's surface, a satellite must fulfil the following conditions:

1. It must have a constant latitude, which is possible only at 0° latitude.
2. The orbit inclination should be zero.
3. It should have a constant longitude and thus have a uniform angular velocity, which is possible when the orbit is circular.
4. The orbital period should be equal to 23 hours 56 minutes, which implies that the satellite must orbit at a height of 35 786 km above the surface of the Earth.
5. The satellite motion must be from west to east.

In the case where these conditions are fulfilled, then as the satellite moves from a position O_1 to O_2 in its orbit, a point vertically below on the equator moves with the same angular velocity and moves from E_1 to E_2 , as shown in Figure 2.40. Satellites in geostationary orbits play an essential role in relaying communication and TV broadcast signals around the globe. They also perform meteorological and military surveillance functions very effectively.

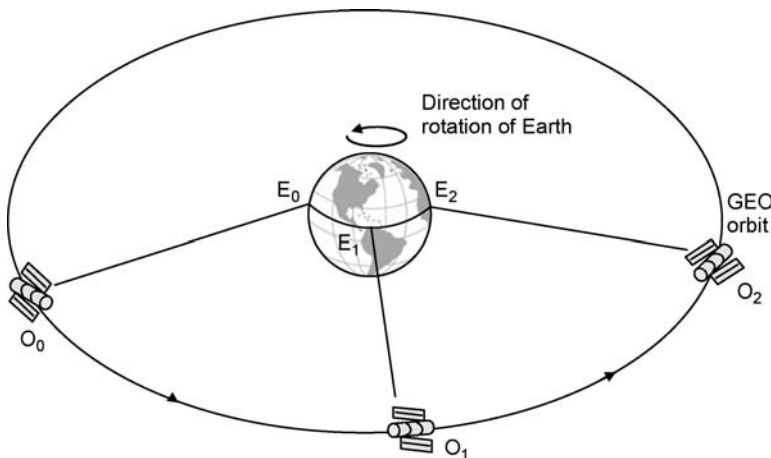


Figure 2.40 GEO satellites appear stationary wrt to a point on Earth

2.5.4 Sun-synchronous Orbit

Another type of satellite orbit, which could have been categorized as an LEO on the basis of distance from the surface of the Earth, needs a special mention and treatment because of its particular importance to satellites intended for remote sensing and military reconnaissance applications. A sun-synchronous orbit, also known as a helio-synchronous orbit, is one that lies in a plane that maintains a fixed angle with respect to the Earth–sun direction. In other words, the orbital plane has a fixed orientation with respect to the Earth–sun direction and the angle between the orbital plane and the Earth–sun line remains constant throughout the year, as shown in Figure 2.41. Satellites in sun-synchronous orbits are particularly suited to applications like passive remote sensing, meteorological, military reconnaissance and atmospheric studies.

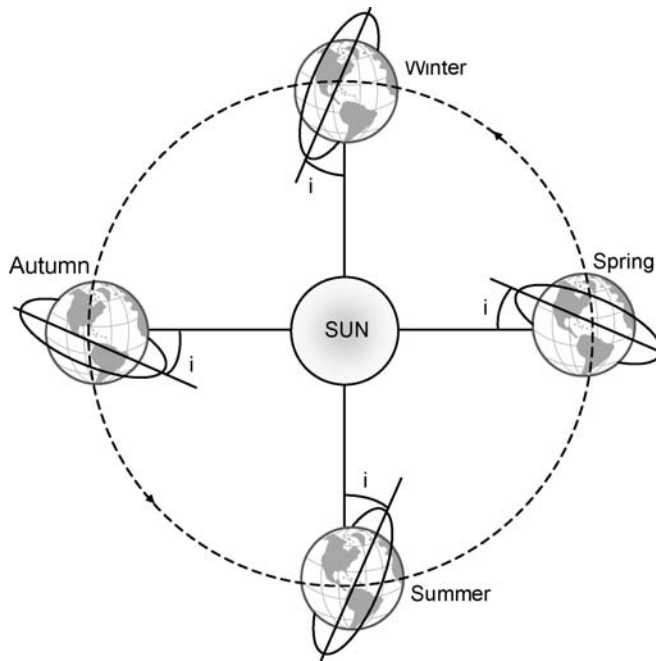


Figure 2.41 Sun-synchronous orbit

As a result of this property, sun-synchronous orbits ensure that:

1. The satellite passes over a given location on Earth every time at the same local solar time, thereby guaranteeing almost the same illumination conditions, varying only with seasons.
2. The satellite ensures coverage of the whole surface of the Earth, being quasi-polar in nature.

Every time a sun-synchronous satellite completes one revolution around Earth, it traverses a thin strip on the surface of the Earth. During the next revolution it traverses another strip

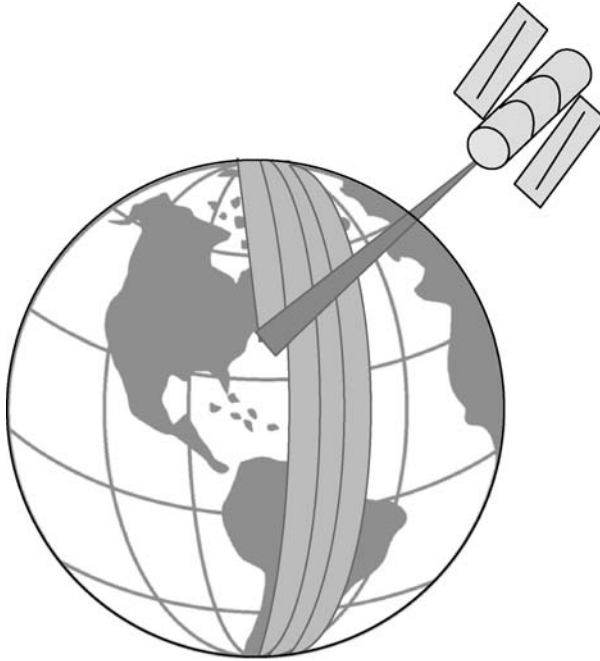


Figure 2.42 Earth coverage of sun-synchronous satellites

shifted westwards and the process of shift continues with successive revolutions, as shown in Figure 2.42. Depending upon the orbital parameters and speed of rotation of Earth, after making a certain number of revolutions around Earth, it comes back close to the first strip that it had traversed. It may not exactly overlap the first strip, as the mean distance between the two strips, called the tracking interval, may not be an integral multiple of the equatorial perimeter. However, the number of revolutions required before the satellite repeats the same strip sequence can certainly be calculated. This is called one complete orbital cycle, which is basically the time that elapses before the satellite revisits a given location in the same direction. To be more precise, orbital cycle means the whole number of orbital revolutions that a satellite must describe in order to be once again flying in the same direction over the same point on the Earth's surface. Landsat-1, -2 and -3 satellites, for instance, have an orbital cycle of 18 days. IRS-1A and IRS-P2 satellites of India have orbital cycles of 22 days and 24 days respectively.

The concept of an orbital cycle will be further illustrated with the help of an example. Take the case of Landsat-1, -2 and -3 satellites each having an orbital cycle of 18 days. During the course of a given revolution, say revolution n , the satellite crosses the equatorial plane at the descending node above a certain point on the surface of the Earth. During the next adjacent pass or revolution, the track shifts to the west by a distance known as the tracking interval. The tracking interval depends upon the speed of the Earth's rotation and the nodal precession of the orbit concerned for one revolution. Fourteen revolutions later, which take about 24 hours and a few minutes, the track is found to be located slightly to the west of the first revolution, designated revolution n . Due to the cumulative effect of these small drifts,

after every 14 revolutions it is found that the satellite flies over the same point on Earth in the same direction once every 251 revolutions. The figure of 251 comes from $18 \times 14 - 1$, thus explaining an 18 day orbital cycle for Landsat-1, -2 and -3 satellites.

Another parameter of relevance to satellites in a sun-synchronous orbit is the crossing time for descending and ascending nodes. Satellites of the Landsat series, the SPOT series and the IRS series move along their descending trajectory above the sunward face and their ascending trajectory at night. Descending node-crossing times for Landsat-1, -2 and -3 satellites is 9 h 30 min and for Landsat-4 and -5 it is 9 h 37 min. In the case of IRS series and SPOT series satellites, it is 10 h 30 min.

Problem 2.19

Verify that a geostationary satellite needs to be at a height of about 35 780 km above the surface of the Earth. Assume the radius of the Earth to be 6380 km and $\mu = 39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$

Solution:

The orbital period of a satellite in circular orbit is given by $T = 2\pi r^{3/2} / \sqrt{\mu}$

Orbital period of a geostationary satellite is equal to 23 hours, 56 minutes, which is equal to 86 160 seconds.

Therefore, $2 \times 3.14 \times (r)^{3/2} / \sqrt{(39.8 \times 10^{13})} = 86\,160$

$6.28 \times (r)^{3/2} = 86\,160 \times \sqrt{(39.8 \times 10^{13})} = 1\,718\,886 \times 10^6$

or $(r)^{3/2} = 273\,708 \times 10^6$

or $r = (273\,708 \times 10^6)^{2/3} = 4215.5 \times 10^4 \text{ m} = 42\,155 \text{ km}$

Hence, height of satellite orbit above the surface of the Earth = $42\,155 - 6380 = 35\,775 \text{ km}$

Problem 2.20

A typical Molniya orbit has perigee and apogee heights above the surface of Earth as 400 km and 40 000 km respectively. Verify that the orbit has a 12 hour time period assuming the radius of the Earth to be 6380 km and $\mu = 39.8 \times 10^{13} \text{ N m}^2/\text{kg}$.

Solution: The orbital period of a satellite in elliptical orbit is given by

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$

where a = semi-major axis of the elliptical orbit. Now $a = (40\,000 + 400 + 6380 + 6380)/2 = 26\,580 \text{ km}$. Therefore

$$\begin{aligned} T &= 6.28 \times \frac{(26\,580 \times 10^3)^{3/2}}{\sqrt{(39.8 \times 10^{13})}} \\ &= 43\,137 \text{ seconds} = 11.98 \text{ hours} \approx 12 \text{ hours} \end{aligned}$$

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6. http://www.thetech.org/exhibits_events/online/satellite/home.html

Glossary

Apogee: Point on the satellite orbit farthest from the centre of the Earth. The apogee distance is the distance of the apogee point from the centre of the Earth

Argument of perigee: This parameter defines the location of the major axis of the satellite orbit. It is measured as the angle between the line joining the perigee and the centre of the Earth and the line of nodes from the ascending to the descending node in the same direction as that of the satellite orbit

Ascending node: The point where the satellite orbit cuts the Earth's equatorial plane, when it passes from the southern hemisphere to the northern hemisphere

Centrifugal force: The force acting outwards from the centre of the Earth on any body orbiting it

Centripetal force: A force that is directed towards the centre of the Earth due to the gravitational force of attraction of Earth

Descending node: The point where the satellite orbit cuts the Earth's equatorial plane, when it passes from the northern hemisphere to the southern hemisphere

Eccentricity: Referring to an elliptical orbit, it is the ratio of the distance between the centre of the Earth and the centre of the ellipse to the semi-major axis of the ellipse. It is zero for a circular orbit and between 0 and 1 for an elliptical orbit

Equatorial orbit: An orbit in which the satellite's orbital plane coincides with the Earth's equatorial plane

Equinox: An equinox is said to occur when the angle of inclination of the Earth's equatorial plane with respect to the direction of the sun is zero. Such a situation occurs twice a year, one on 21 March called the spring equinox and the other on 21 September called the autumn equinox

First cosmic velocity: This is the injection velocity at which the apogee and perigee distances are equal, with the result that the satellite orbit is circular

Geostationary Earth orbit (GEO): A satellite orbit with an orbit height at 35 786 km above the surface of the Earth. This height makes the satellite's orbital velocity equal to the speed of rotation of Earth, thus making the satellite look stationary from a given point on the surface of the Earth

Inclination: Inclination is the angle that the orbital plane of the satellite makes with the Earth's equatorial plane

Inclined orbit: An orbit having an angle of inclination between 0° and 180°

Injection velocity: This is the horizontal velocity with which a satellite is injected into space by the launch vehicle with the intention of imparting a specific trajectory to the satellite

Kepler's first law: The orbit of an artificial satellite around Earth is elliptical with the centre of the Earth lying at one of its foci

Kepler's second law: The line joining the satellite and the centre of the Earth sweeps out equal areas in the plane of the orbit in equal times

Kepler's third law: The square of the time period of any satellite is proportional to the cube of the semi-major axis of its elliptical orbit

Low Earth orbit (LEO): A satellite orbit with an orbital height of around 150 km to 500 km above the surface of Earth. These orbits have lower orbital periods, shorter propagation delays and lower propagation losses

Medium Earth orbit (MEO): A satellite orbit with an orbital height around 10 000 km to 20 000 km above the surface of the Earth

Molniya orbit: A highly inclined and eccentric orbit used by Russia and other countries of the erstwhile Soviet Union for providing communication services

Orbit: A trajectory that is periodically repeated

Perigee: A point on a satellite orbit closest to the centre of the Earth. The perigee distance is the distance of the perigee point from the centre of the Earth

Polar orbit: An orbit having an angle of inclination equal to 90°

Prograde orbit: Also called a direct orbit, an orbit where the satellite travels in the same direction as the direction of rotation of Earth. This orbit has an angle of inclination between 0° and 90°

Project Iridium: Project Iridium is a global communication system conceived by Motorola that makes use of satellites in low Earth orbits. A total of 66 satellites are arranged in a distributed architecture with each satellite carrying 1/66 of the total system capacity

Retrograde orbit: An orbit where the satellite travels in a direction opposite to the direction of rotation of Earth. This orbit has an angle of inclination between 90° and 180°

Right ascension of the ascending node: The right ascension of the ascending node indicates the orientation of the line of nodes, which is the line joining the ascending and descending nodes, with respect to the direction of the vernal equinox. It is expressed as an angle (Ω) measured from the vernal equinox towards the line of nodes in the direction of rotation of Earth. The angle could be anywhere from 0° to 360°

Second cosmic velocity: This is the injection velocity at which the apogee distance becomes infinite and the orbit takes the shape of a parabola. It equals $\sqrt{2}$ times the first cosmic velocity

Solstices: Solstices are said to occur when the angle of inclination of the Earth's equatorial plane with respect to the direction of the sun is at its maximum, i.e. 23.4° . These are like equinoxes and also occur twice during the year, one on 21 June called the summer solstice and the other on 21 December called the winter solstice

Sun-synchronous orbit: A sun-synchronous orbit, also known as a helio-synchronous orbit, is one that lies in a plane that maintains a fixed angle with respect to the Earth–sun direction

Third cosmic velocity: This is the injection velocity at which the satellite succeeds in escaping from the solar system. It is related to the motion of Earth around the sun. For injection velocities beyond the third cosmic velocity, there is a region of hyperbolic flights outside the solar system

Trajectory: A path traced by a moving body

True anomaly of a satellite: This parameter is used to indicate the position of the satellite in its orbit. This is done by defining an angle (θ), called the true anomaly of the satellite, formed by the line joining the perigee and the centre of the Earth with the line joining the satellite and the centre of the Earth

3

Satellite Launch and In-orbit Operations

Fundamental issues such as laws governing motion of artificial satellites around Earth, different orbital parameters, types of orbits and their suitability for a given application, etc., related to orbital dynamics were addressed in the previous chapter. The next obvious step is to understand the launch requirements to acquire the desired orbit. This should then lead to various in-orbit operations such as orbit stabilization, orbit correction and station keeping that are necessary for keeping the satellite in the desired orbit. The present chapter will address all these issues and also other related issues like Earth coverage, eclipses and pointing towards a given satellite from Earth. Again the chapter is amply illustrated with mathematics wherever necessary and a large number of solved problems.

3.1 Acquiring the Desired Orbit

In order to ensure that the satellite acquires the desired orbit, i.e. the orbit with desired values of orbital elements/parameters such as orbital plane, apogee and perigee distances, etc., it is important that correct conditions are established at the satellite injection point. For instance, in order to ensure that the satellite orbits within a given plane, the satellite must be injected at a certain specific time, depending upon the longitude of the injection point, at which the line of nodes makes the required angle with the direction of the vernal equinox. Put in simple words, for a given orbital plane, the line of nodes will have a specific angle with the vernal equinox. To acquire this angle, the satellite must be injected at the desired time depending upon the longitude of the injection point.

3.1.1 *Parameters Defining the Satellite Orbit*

The satellite orbit is completely defined or specified by the following parameters:

1. Right ascension of the ascending node
2. Inclination angle

3. Position of the major axis of the orbit
4. Shape of the elliptical orbit
5. Position of the satellite in its orbit

1. *Right ascension of the ascending node.* The angle Ω defining the right ascension of the ascending node is basically the difference between two angles, θ_1 and θ_2 , where θ_1 is the angle made by the longitude of the injection point at the time of launch with the direction of vernal equinox and θ_2 is the angle made by the longitude of the injection point at the time of launch with the line of nodes, as shown in Figure 3.1. Angle θ_2 can be computed from

$$\sin \theta_2 = \frac{\cos i \sin L}{\cos L \sin i} \quad (3.1)$$

where i = angle of inclination and L = longitude of the injection point. Thus, for a known angle of inclination, the time of launch and hence the longitude of the injection point can be so chosen as to get the desired angle (Ω).

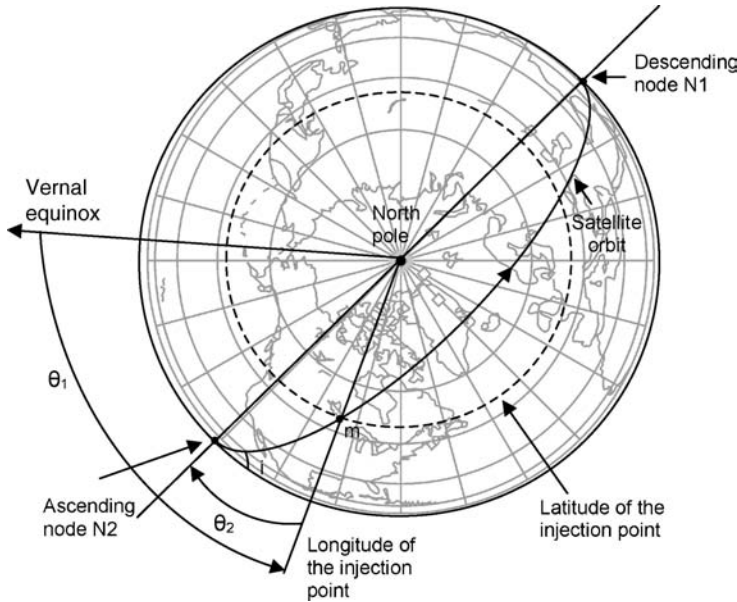


Figure 3.1 Right ascension of the ascending node

2. *Inclination angle.* The angle of inclination (i) of the orbital plane can be determined from the known values of the angle of azimuth A_z and the latitude of the injection point l using the expression

$$\cos i = \sin A_z \cos l \quad (3.2)$$

The azimuth angle at a given point in a satellite orbit was defined in the previous chapter as the angle made by the projection of the satellite velocity vector at that point in the

local horizontal with the north. It is obvious from the above expression that for the angle of inclination to be zero, the right-hand side of the expression must equal unity. This is possible only for $A_z = 90^\circ$ and $l = 0^\circ$. Other inferences that can be drawn from the above expression are:

- (a) For $A_z = 90^\circ$, $i = l$.
- (b) For $A_z < 90^\circ$, $\sin A_z < 1$ and therefore $i > l$.

Thus it can be concluded that the satellite will tend to orbit in a plane which will be inclined to the equatorial plane at an angle equal to or greater than the latitude of the injection point. This will be evident if some of the prominent satellite launch sites and the corresponding latitudes of those locations are looked at.

The launch site at Kourou in French Guyana is at a latitude of 5.2°N . When the typical launch sequences are discussed in the latter part of this chapter, it will be seen that the orbital plane acquired by the satellite launched from Kourou immediately after injection is 7° , which is later corrected to get an inclination of zero or near zero through manoeuvres. The latitude of another prominent launch site at Baikonur in Russia is 45.9° . A study of the launch sequence for satellites launched from this base shows that the satellite acquires an initial orbital inclination of 51° . Both these examples confirm the axiom or the concept outlined in an earlier section. A similar result will be found if other prominent launch sites at Cape Canaveral, Sriharikota, Vandenberg and Xichang are examined.

3. *Position of the major axis of the orbit.* The position of the major axis of the satellite orbit is defined by argument of the perigee ω . Refer to Figure 3.2. If the injection point happens to be different from the perigee point, then ω is the difference of two angles, ϕ and θ , as shown in Figure 3.2. Angle ϕ can be computed from

$$\sin \phi = \frac{\sin l}{\sin i} \quad (3.3)$$

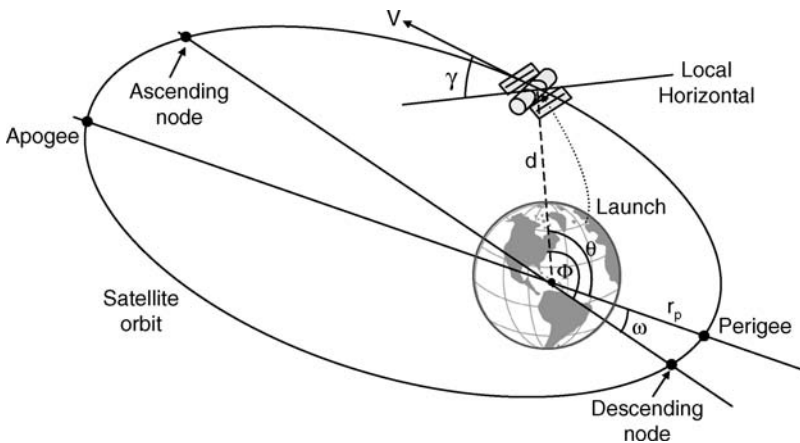


Figure 3.2 Argument of the perigee

and θ can be computed from

$$\cos \theta = \frac{dV^2 \cos^2 \gamma - \mu}{e\mu} \quad (3.4)$$

The terms have their usual meaning and are indicated in Figure 3.2.

In the case where the injection point is the same as the perigee point (Figure 3.3), $\gamma = 0^\circ$ and $\theta = 0^\circ$. This gives

$$\sin \omega = \sin \phi = \frac{\sin l}{\sin i} \quad (3.5)$$

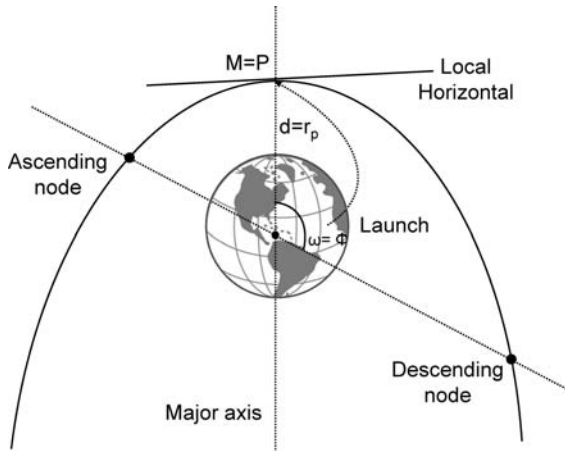


Figure 3.3 Argument of the perigee when the injection point is the same as the perigee point

4. *Shape of the elliptical orbit.* The shape of the orbit is defined by the orbit eccentricity e , the semi-major axis a , the apogee distance r_a and the perigee distance r_p . The elliptical orbit can be completely defined by either a and e or by r_a and r_p . The orbit is usually defined by the apogee and perigee distances. The perigee distance can be computed from known values of the distance d from the focus (which in this case is the centre of the Earth) at any point in the orbit and the angle γ that the velocity vector V makes with the local horizontal. The relevant expression to compute r_p is

$$r_p = \frac{V^2 d^2 \cos^2 \gamma}{\mu(1 + e)} \quad (3.6)$$

The apogee distance r_a can be computed with the help of the following expression:

$$r_a = \frac{V^2 d^2 \cos^2 \gamma}{\mu(1 - e)} \quad (3.7)$$

5. *Position of the satellite in its orbit.* The position of the satellite in its orbit can be defined by a time parameter t , which is the time that elapsed after a time instant t_0 when the satellite last passed through a reference point. The reference point is usually the perigee. The time

t that has elapsed after the satellite last passed through the perigee point can be computed from

$$t = \left(\frac{T}{2\pi} \right) (u - e \sin u) \quad (3.8)$$

where T is the orbital period and the angle u is the eccentric anomaly of the current location of the satellite (Figure 3.4).

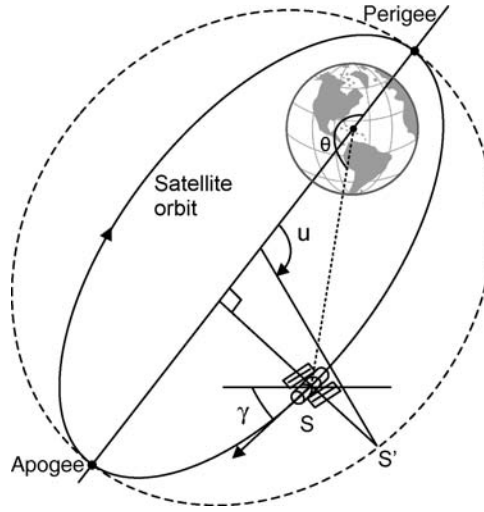


Figure 3.4 Position of the satellite in its orbit

In this section, the parameters that define an orbit as well as the factors upon which each one of these parameters is dependent, have been discussed, with the help of relevant mathematical expressions. However, when it comes to acquiring a desired orbit, this is perhaps not enough. The discussion would be complete only if it was known how each one of these parameters could be modified if needed, with the help of commands transmitted to the satellite from the ground control station. In the section that follows, this process will be discussed.

3.1.2 Modifying the Orbital Parameters

1. *Right ascension of the ascending node.* This parameter, as mentioned before, defines the orientation of the orbital plane of the satellite. Due to the Earth's equatorial bulge, there is nodal precession, i.e. a change in the orientation of the orbital plane with time as Earth revolves around the sun. It will be seen that this perturbation to the satellite orbital plane depends upon the orbit inclination angle i and also the apogee and perigee distances. The rotational perturbation experienced by the satellite orbital plane (in degrees) during one orbit

period is given by

$$\Delta\Omega = -0.58 \left(\frac{D}{r_a + r_p} \right)^2 \left(\frac{1}{1 - e^2} \right)^2 \cos i \quad (3.9)$$

where D = diameter of the Earth. It is clear from the above expression that the perturbation is zero only in the case of the inclination angle being 90° . Also, more the deviation of the inclination angle from 90° and the shorter the apogee and perigee distances, the larger is the perturbation. Any experimental manoeuvre that changes the orientation is a very expensive affair. It is always preferable to depend upon the natural perturbation for this purpose.

When the orbit inclination angle $i < 90^\circ$, $\Delta\Omega$ is negative; i.e. the satellite orbital plane rotates in a direction opposite to the direction of the rotation of Earth (Figure 3.5). When the orbit inclination angle $i > 90^\circ$, $\Delta\Omega$ is positive; i.e. the satellite orbital plane rotates in the same direction as the direction of the rotation of Earth (Figure 3.6).

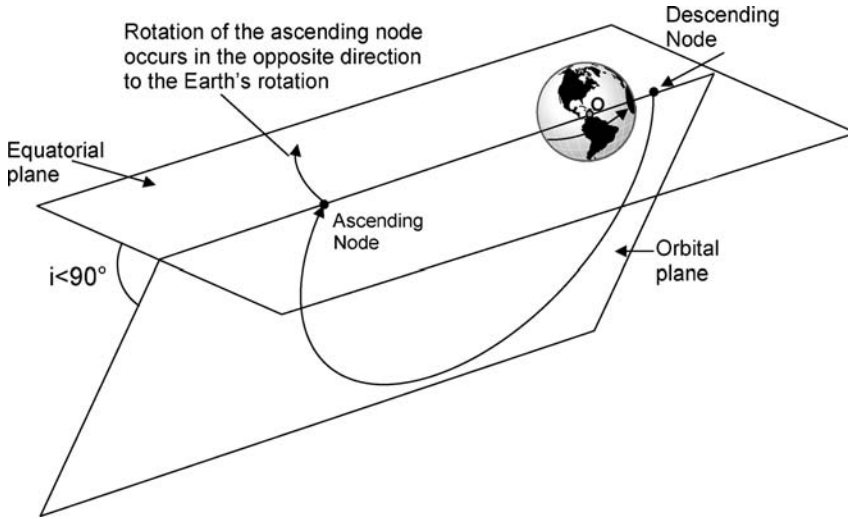


Figure 3.5 Rotation of the orbital plane when the inclination angle is less than 90°

2. *Inclination angle.* Though there is a natural perturbation experienced by the satellite orbital plane inclination w.r.t. the Earth's equatorial plane due to the sun-moon attraction phenomenon, this change is small enough to be considered negligible. A small change Δi to the inclination angle i can be affected externally by applying a thrust Δv to the velocity vector V at an angle of $90^\circ + \Delta i/2$ to the direction of the satellite, as shown in Figure 3.7. The thrust Δv is given by

$$\Delta v = 2V \sin \left(\frac{\Delta i}{2} \right) \quad (3.10)$$

The thrust is applied at either of the two nodes.

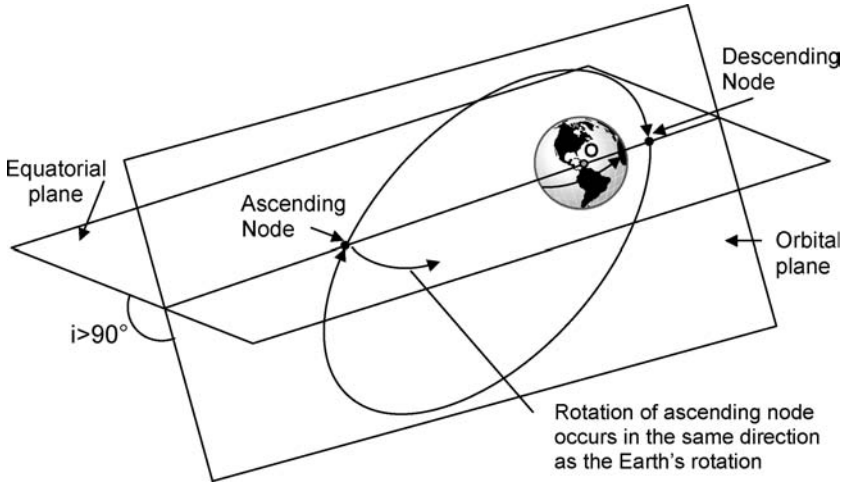


Figure 3.6 Rotation of the orbital plane when the inclination angle is greater than 90°

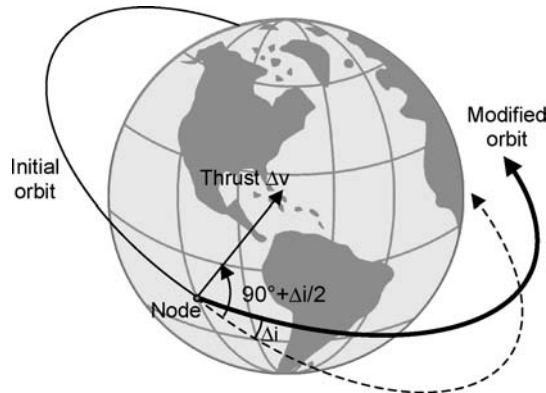


Figure 3.7 Changing the inclination angle

3. *Position of the major axis of the orbit.* The position of the major axis, as mentioned in the previous section, is defined by the parameter called the argument of the perigee ω . This parameter, like the right ascension of the ascending node, also undergoes natural perturbations due to the equatorial bulge of Earth. The phenomenon is known as apsidal precession. The further the angle of inclination from $63^\circ 26'$, the larger is the rotation of the perigee. Also, closer the satellite to the centre of the Earth, the larger is the rotation. Moreover, the rotation of the perigee occurs in the direction opposite to the satellite motion if the inclination angle is greater than $63^\circ 26'$ and in the same direction as the satellite motion if the inclination angle is less than $63^\circ 26'$. This is illustrated in Figure 3.8 (a) and (b).

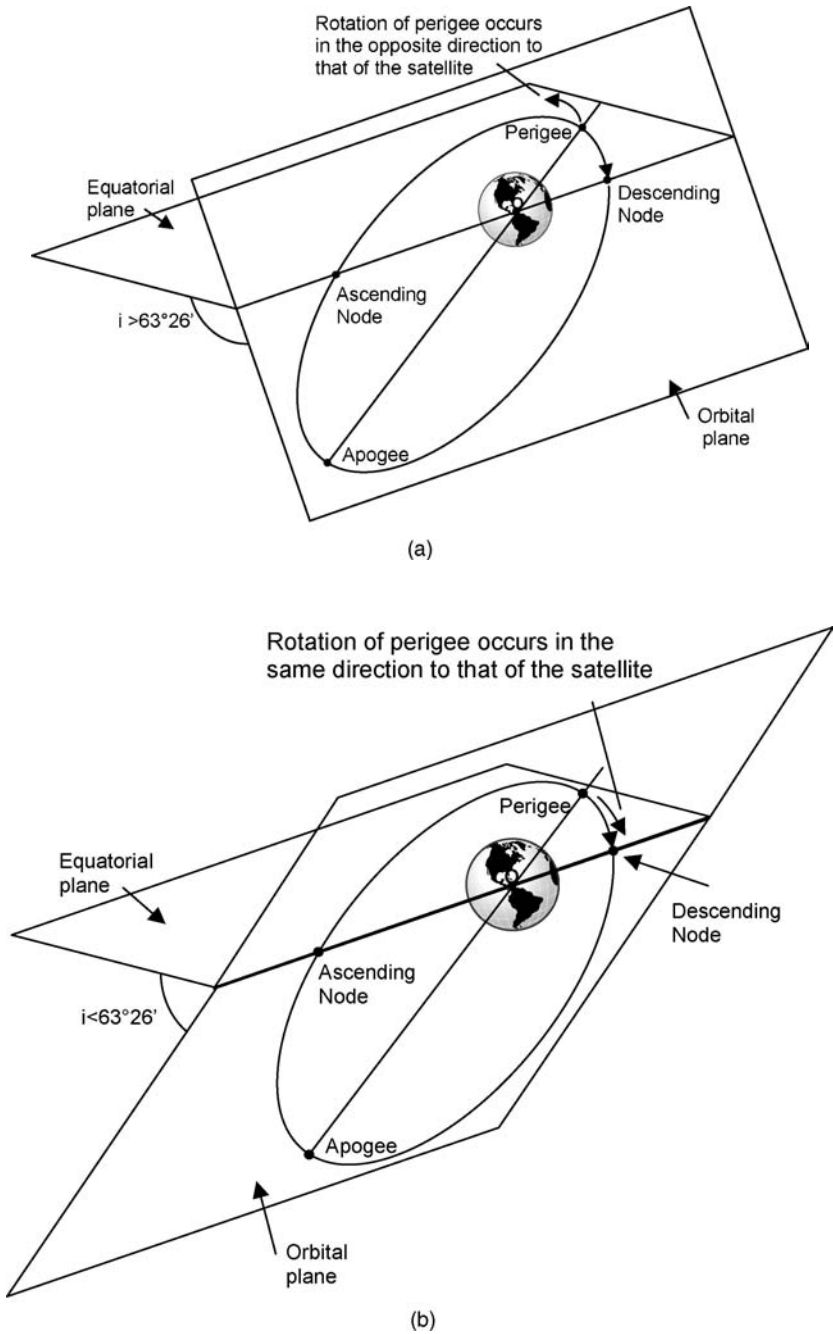


Figure 3.8 (a) Rotation of the perigee when the inclination angle is greater than $63^\circ 26'$. (b) Rotation of the perigee when the inclination angle is less than $63^\circ 26'$

The rotation experienced by ω in degrees due to natural perturbation in one orbit is given by

$$\Delta\omega = 0.29 \left[\frac{4 - 5 \sin^2 i}{(1 - e^2)^2} \right] \left[\left(\frac{D}{r_a + r_p} \right)^2 \right] \quad (3.11)$$

In the case of a circular orbit, $r_a = r_p$ and $e = 0$. In order to affect an intentional change ($\Delta\omega$), a thrust Δv is applied at a point in the orbit where the line joining this point and the centre of the Earth makes an angle $(\Delta\omega/2)$ with the major axis. The thrust is applied in a direction towards the centre of the Earth, as shown in Figure 3.9. The thrust Δv can be computed from

$$\Delta v = 2 \sqrt{\left(\frac{\mu}{r_p} \right)} \left[\frac{e}{\sqrt{1+e}} \right] \sin \left(\frac{\omega}{2} \right) \quad (3.12)$$

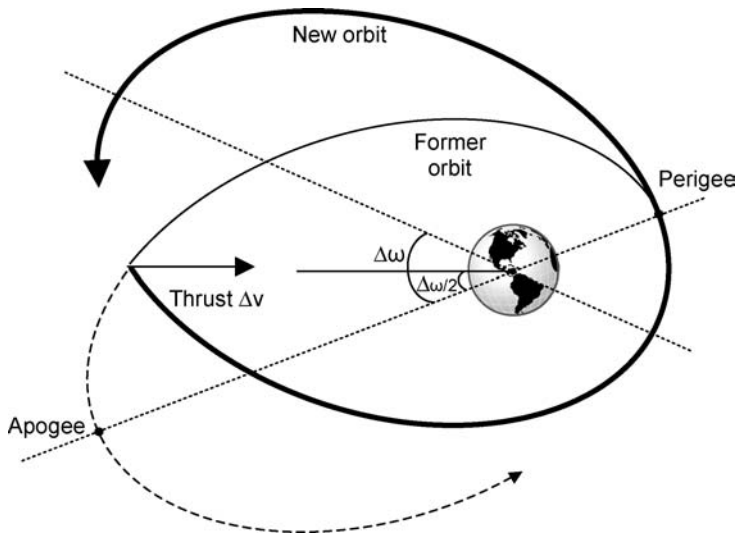


Figure 3.9 Changing the argument of the perigee

4. *Shape of the elliptical orbit.* The apogee and perigee distances fully specify the shape of the orbit as the other parameters such as eccentricity e and the semi-major axis a can be computed from this pair of distances. The apogee distance becomes affected; in fact, it reduces due to atmospheric drag and to a lesser extent by the solar radiation pressure. In fact, every elliptical orbit tends to become circular with time, with a radius equal to the perigee distance. It is interesting to note that the probable lifetime of a satellite in a circular orbit at a height of 200 km is only a few days and that of a satellite at a height of 800 km is about 100 years. For a geostationary satellite, it would be more than a million years.

Returning to modifying the apogee distance, it can be intentionally increased or decreased by applying a thrust Δv at the perigee point in the direction of motion of the satellite or opposite to the direction of motion of the satellite respectively (Figure 3.10). Thus Δr_a can

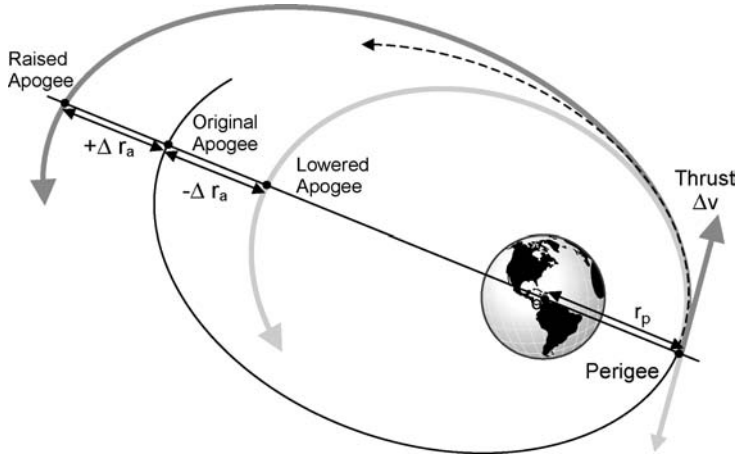


Figure 3.10 Changing the apogee distance

be computed from

$$\Delta v = \Delta r_a \frac{\mu}{V_p(r_a + r_p)^2} \quad (3.13)$$

where V_p is the velocity at the perigee point. The perigee distance can similarly be increased or decreased by applying a thrust Δv at the apogee point in the direction of motion of the satellite or opposite to the direction of motion of the satellite respectively (Figure 3.11).

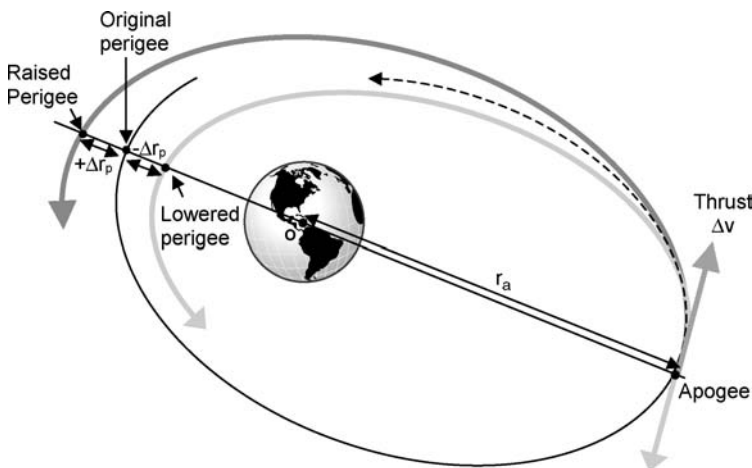


Figure 3.11 Changing the perigee distance

Thus Δr_p can be computed from

$$\Delta v = \Delta r_p \frac{\mu}{V_a(r_a + r_p)^2} \quad (3.14)$$

where V_a is the velocity at the apogee point.

So far modifying the apogee and perigee distances has been discussed. A very economical manoeuvre that can be used to change the radius of a circular orbit is the Homann transfer, which uses an elliptical trajectory tangential to both the current and modified circular orbits. The process requires application of two thrusts, Δv and $\Delta v'$. The thrust is applied in the direction of motion of the satellite if the orbit radius is to be increased (Figure 3.12) and opposite to the direction of motion of the satellite if the orbit radius is to be decreased (Figure 3.13). The two thrusts can be computed from

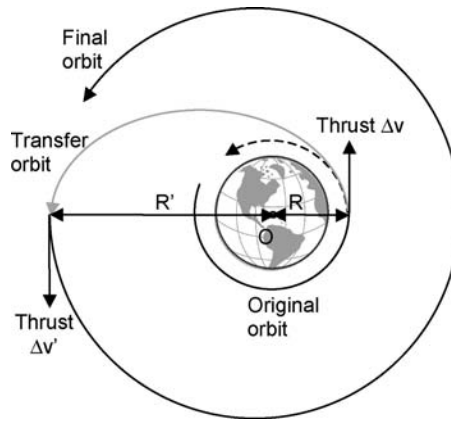


Figure 3.12 Increasing the radius of the circular orbit

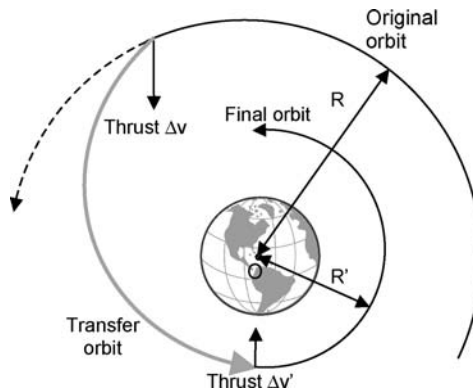


Figure 3.13 Decreasing the radius of the circular orbit

$$\Delta v = \sqrt{\left[\frac{2\mu R'}{R(R+R')} \right]} - \sqrt{\left(\frac{\mu}{R} \right)} \quad (3.15)$$

$$\Delta v' = \sqrt{\left(\frac{\mu}{R'} \right)} - \sqrt{\left[\frac{2\mu R}{R'(R+R')} \right]} \quad (3.16)$$

where R = radius of the initial orbit and R' = radius of the final orbit.

5. *Position of the satellite in its orbit.* The position of the satellite was defined earlier in terms of its passage time at the perigee point. Modifying the apogee and perigee distances also changes the time of a satellite's passage at the perigee point due to the change in velocity of the satellite. A similar manoeuvre can be used to modify the longitudinal location of a geostationary satellite.

Problem 3.1

A geostationary satellite was launched from Kourou launch site in French Guiana. The satellite was injected into a highly eccentric elliptical transfer orbit from a height of 200 km. The projection of the injection velocity vector in the local horizontal plane made an angle of 85° with the north. Determine the inclination angle attained by this transfer orbit given that latitude and longitude of the Kourou site are 5.2°N and 52.7°W respectively.

Solution: The inclination angle can be computed from

$$\cos i = \sin A_z \cos l$$

where A_z = azimuth angle of the injection point = 85° and l = latitude of the launch site = 5.2° . This gives

$$\begin{aligned} \cos i &= \sin 85^\circ \cos 5.2^\circ = 0.9962 \times 0.9959 = 0.9921 \\ i &= 7.2^\circ \end{aligned}$$

Problem 3.2

A geostationary satellite was launched into an eccentric elliptical transfer orbit having an orbital plane inclination of 7° and its apogee at the geostationary height. This orbit was then circularized by an appropriate thrust manoeuvre without affecting any change to the inclination. If the velocity of the satellite in the circularized orbit is 3 km/s, determine the velocity thrust required to make the inclination 0° . Also show the relevant diagram indicating the direction of thrust to be applied.

Solution: The required thrust can be computed from

$$\Delta v = 2V \sin \left(\frac{\Delta i}{2} \right)$$

where $V = 3000 \text{ m/s}$ and $\Delta i = 7^\circ$. Therefore

$$\Delta v = 2V \sin\left(\frac{\Delta i}{2}\right) = 2 \times 3000 \times \sin 3.5^\circ = 6000 \times 0.061 = 366 \text{ m/s}$$

The thrust is applied when the satellite is at either of the nodes and is applied at an angle of $(90^\circ + \Delta i/2) = 93.5^\circ$ to the direction of motion of the satellite, as shown in Figure 3.14.

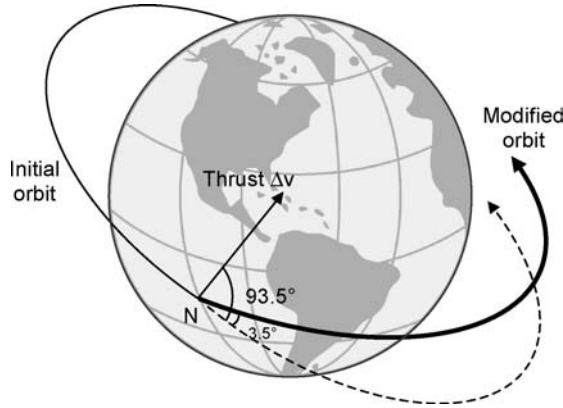


Figure 3.14 Figure for Problem 3.2

Problem 3.3

A satellite is launched into an eccentric ($e = 0.69$) elliptical orbit. The line joining the perigee and centre of the Earth makes an angle of 60° with the line of nodes. It is intended to rotate the perigee of the orbit so as to make this angle equal to 50° . If the perigee distance is 7000 km, determine the velocity thrust required to bring about the change. Also illustrate the application of thrust *vis-à-vis* the point of application and its direction. (Assume that $\mu = 39.8 \times 10^{13} \text{ N m}^2/\text{kg}$.)

Solution: The velocity thrust required to rotate the perigee point by the desired amount can be computed from

$$\Delta v = 2\sqrt{\left(\frac{\mu}{r_p}\right)} \left[\frac{e}{\sqrt{1+e}} \right] \sin\left(\frac{\omega}{2}\right)$$

Now $r_p = 7000 \text{ km} = 7\,000\,000 \text{ m}$, $e = 0.69$ and $\omega = 60^\circ$. Substituting these values gives

$$\begin{aligned} \Delta v &= 2\sqrt{\left(\frac{39.8 \times 10^{13}}{7\,000\,000}\right)} \left(\frac{0.69}{\sqrt{1.69}}\right) \sin 30^\circ \\ &= 15080.74 \times 0.53 \times 0.5 = 3996.4 \text{ m/s} = 3.996 \text{ km/s} \end{aligned}$$

The point of application of this thrust and its direction are shown in Figure 3.15.

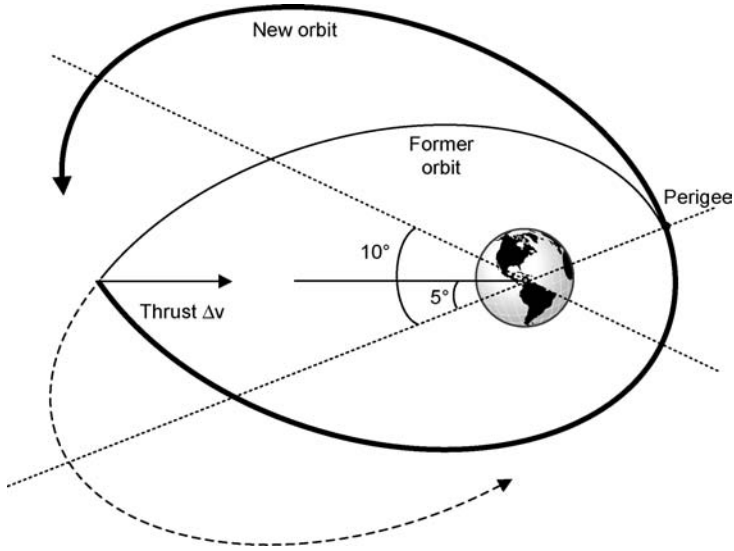


Figure 3.15 Figure for Problem 3.3

Problem 3.4

A certain satellite is moving in an elliptical orbit with apogee and perigee distances of 15 000 km and 7000 km respectively. It is intended to circularize the orbit with a radius equal to the perigee distance. Determine the magnitude and direction of the velocity thrust required to achieve the objective. Also indicate the point in the orbit where the thrust would be applied. Take the value of the constant μ as $39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$.

Solution: Stated in other words, the objective is to reduce the apogee distance to 7000 km from 15 000 km. The required velocity thrust Δv can be computed from

$$\Delta v = \Delta r_a \frac{\mu}{V_p(r_a + r_p)^2}$$

where Δr_a = required change in the apogee distance = 8000 km. Also, the velocity at the perigee point can be computed from

$$\begin{aligned} V_p &= \sqrt{\left(\frac{2\mu}{\text{perigee distance}} - \frac{2\mu}{\text{perigee distance} + \text{apogee distance}} \right)} \\ &= \sqrt{\left(\frac{2 \times 39.8 \times 10^{13}}{7 \times 10^6} - \frac{2 \times 39.8 \times 10^{13}}{7 \times 10^6 + 15 \times 10^6} \right)} \\ &= 10^3 \sqrt{\left(\frac{796}{7} - \frac{796}{22} \right)} \\ &= 8.805 \text{ km/s} \end{aligned}$$

Therefore,

$$\Delta v = 8\,000\,000 \times \frac{39.8 \times 10^{13}}{8805 \times (15\,000\,000 + 7\,000\,000)^2} = 747.1 \text{ m/s}$$

The thrust will be applied at the perigee point and in the direction opposite to the motion of the satellite, as illustrated in Figure 3.16.

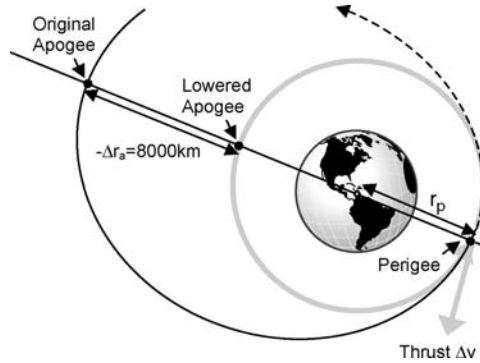


Figure 3.16 Figure for Problem 3.4

Problem 3.5

A certain satellite is moving in an elliptical orbit with apogee and perigee distances of 15 000 km and 7000 km respectively. It is intended to increase the perigee distance to 8000 km. Determine the magnitude and direction of the velocity thrust required to achieve the objective. Also indicate the point in orbit where the thrust would be applied. Take the value of constant (μ) as $39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$ and radius of the Earth as 6378 km.

Solution:

The objective is to increase the perigee distance to 8000 km.

The required velocity thrust Δv can be computed from $\Delta v = \frac{\Delta r_p \mu}{[v_a(r_a + r_p)^2]}$ where $\Delta r_p =$

Required change in perigee distance = 1000 km

Also velocity at the perigee point can be computed from:

$$\begin{aligned} V_p &= \sqrt{[2\mu/(\text{Perigee distance})] - \{2\mu/(\text{Perigee distance} + \text{Apogee distance})\}} \\ &= \sqrt{[2\mu/(7\,000\,000)] - \{2\mu/(7\,000\,000 + 15\,000\,000)\}} \\ &= \sqrt{(77\,532\,469.6)} = 8.805 \text{ km/s} \end{aligned}$$

The velocity at the apogee $V_a = (V_p \times \text{Distance of perigee from center of the Earth})/(\text{Distance of apogee from center of the Earth})$

$$V_a = 8.805 \times (7\,000\,000)/(15\,000\,000) \text{ km/s} = 4.109 \text{ km/s}$$

$$\text{Therefore, } \delta v = 1\,000\,000 \times 39.8 \times 10^{13} / [4109 \times (15\,000\,000 + 7\,000\,000)^2] = 200.1 \text{ m/s}$$

The thrust will be applied at the apogee point and in the direction of motion of the satellite.

Problem 3.6

A certain satellite is moving in a circular orbit with an altitude of 500 km. The orbit altitude is intended to be increased to 800 km. Suggest a suitable manoeuvre to do this and calculate the magnitudes of the thrusts required. Show the point/s of application of the thrust/s. Take the value of the constant μ as $39.8 \times 10^{13} \text{ N m}^2/\text{kg}$.

Solution: The objective is to increase the orbit altitude to 800 km. This can be achieved by making use of the Homann transfer method by first applying a thrust Δv to transform the current circular orbit to an elliptical one, with its perigee at the current orbit radius and apogee at the desired orbit radius. Since the altitude of desired orbit is greater than the current one, the thrust is applied in the direction of motion of the satellite. A second thrust $\Delta v'$ is then applied at the apogee of the elliptical orbit in the direction of the motion of the satellite in order to circularize the orbit with the desired radius. The two thrusts can be computed from

$$\Delta v = \sqrt{\left[\frac{2\mu R'}{R(R + R')} \right]} - \sqrt{\left(\frac{\mu}{R} \right)}$$

$$\Delta v' = \sqrt{\left(\frac{\mu}{R'} \right)} - \sqrt{\left[\frac{2\mu R}{R'(R + R')} \right]}$$

where R = radius of the initial orbit and R' = radius of the final orbit.

Now $R = 6878 \text{ km}$ and $R' = 7178 \text{ km}$. The two thrusts are computed as

$$\begin{aligned} \Delta v &= \sqrt{\left[\frac{2\mu R'}{R(R + R')} \right]} - \sqrt{\left(\frac{\mu}{R} \right)} \\ &= \sqrt{\left[\frac{2 \times 39.8 \times 10^{13} \times 7\,178\,000}{6\,878\,000 \times (6\,878\,000 + 7\,178\,000)} \right]} - \sqrt{\left(\frac{39.8 \times 10^{13}}{6\,878\,000} \right)} \\ &= 7687.70 - 7606.95 = 80.75 \text{ m/s} \\ \Delta v' &= \sqrt{\left(\frac{\mu}{R'} \right)} - \sqrt{\left[\frac{2\mu R}{R'(R + R')} \right]} \\ &= \sqrt{\left(\frac{39.8 \times 10^{13}}{7\,178\,000} \right)} - \sqrt{\left[\frac{2 \times 39.8 \times 10^{13} \times 6\,878\,000}{7\,178\,000 \times (6\,878\,000 + 7\,178\,000)} \right]} \\ &= 7446.29 - 7366.40 = 79.89 \text{ m/s} \end{aligned}$$

Thus, the two thrusts to be applied are $\Delta v = 80.75 \text{ m/s}$ and $\Delta v' = 79.89 \text{ m/s}$.

Figure 3.17 shows the point and direction of application of the two thrusts.

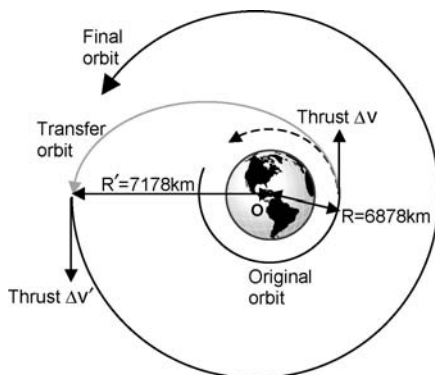


Figure 3.17 Figure for Problem 3.6

3.2 Launch Sequence

In the last chapter, the various types of satellite orbits were discussed, the prominent among them being the low Earth circular orbits, elliptical eccentric orbits and geostationary orbits. The suitability of each type of orbit for a given application was also discussed. In the present chapter, we have continued with a basic discussion on orbits and trajectories. The various factors important for acquiring the desired orbit, as well as various natural causes that cause perturbations to certain orbital parameters and external manoeuvres used to bring about intended changes in them have been discussed. In the present section, the discussion will centre on typical launch sequences employed worldwide for putting satellites in the geostationary orbit, as putting a satellite in a lower circular or elliptical orbit is only a step towards achieving the geostationary orbit.

3.2.1 Types of Launch Sequence

There are two broad categories of launch sequence, one that is employed by expendable launch vehicles such as Ariane of the European Space Agency and Atlas Centaur and Thor Delta of the United States and the other that is employed by a re-usable launch vehicle such as the Space Shuttle of the United States and the recent development Buran of Russia.

Irrespective of whether a satellite is launched by a re-usable launch vehicle or an expendable vehicle, the satellite heading for a geostationary orbit is first placed in a transfer orbit. The transfer orbit is elliptical in shape with its perigee at an altitude between 200 km and 300 km and its apogee at the geostationary altitude.

In some cases, the launch vehicle injects the satellite directly into a transfer orbit of this type. Following this, an apogee manoeuvre circularizes the orbit at the geostationary altitude. The last step is then to correct the orbit for its inclination. This type of launch sequence is illustrated

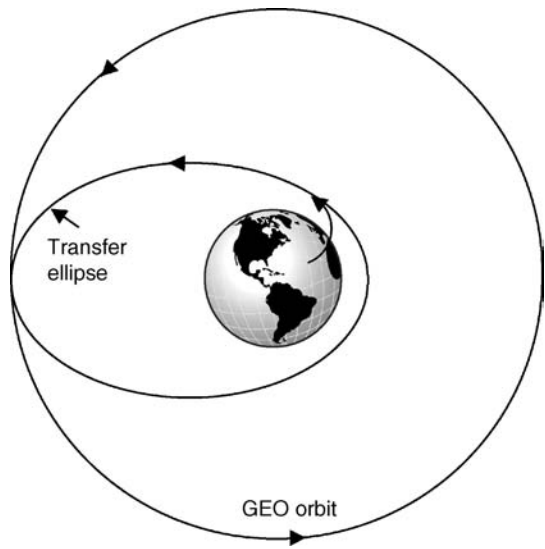


Figure 3.18 Possible geostationary satellite launch sequence

in Figure 3.18. In the second case, the satellite is first injected into a low Earth circular orbit. In the second step, the low Earth circular orbit is transformed into an elliptical transfer orbit with a perigee manoeuvre. Circularization of the transfer orbit and then correction of the orbit inclination follow this. This type of sequence is illustrated in Figure 3.19.

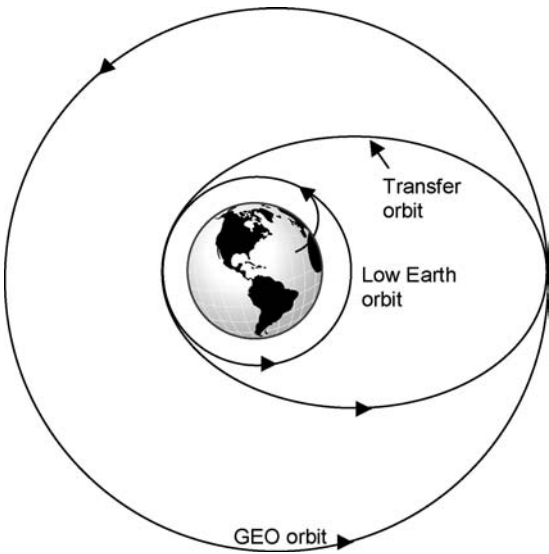


Figure 3.19 Another possible geostationary satellite launch sequence

In the following paragraphs, we will discuss some typical launch sequences observed in the case of various launch vehicles used to deploy geostationary satellites from some of the prominent launch sites all over the world. The cases presented here include the launch of geostationary satellites from Kourou in French Guiana and Cape Canaveral in the United States, both situated towards the eastern coast of America, and also Baikonur in Russia. A typical Space Shuttle launch will also be discussed. The two types of launch sequence briefly described above will be amply evident in the real-life illustrations that follow.

3.2.1.1 Launch from Kourou

A typical Ariane launch of a geostationary satellite from Kourou in French Guiana will be illustrated. Different steps involved in the entire process are:

1. The launch vehicle takes the satellite to a point that is intended to be the perigee of the transfer orbit, at a height of about 200 km above the surface of the Earth. The satellite along with its apogee boost motor is injected before the launch vehicle crosses the equatorial plane, as shown in Figure 3.20. The injection velocity is such that the injected satellite attains an eccentric elliptical orbit with an apogee altitude at about 36 000 km. The orbit is inclined at about 7° , which is expected, as the latitude of the launch site is 5.2° .

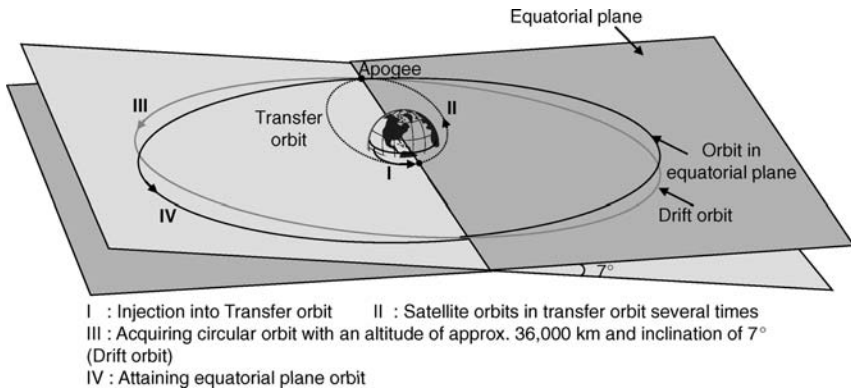


Figure 3.20 Typical launch of a geostationary satellite from Kourou

2. In the second step, after the satellite has completed several revolutions in the transfer orbit, the apogee boost motor is fired during the passage of the satellite at the apogee point. The resulting thrust gradually circularizes the orbit. The orbit now is a circular orbit with an altitude of 36 000 km.
3. Further thrust is applied at the apogee point to bring the inclination to 0° , thus making the orbit a true circular and equatorial orbit.
4. The last step is to attain the correct longitude and attitude. This is also achieved by applying thrust either tangential or normal to the orbit.

3.2.1.2 Launch from Cape Canaveral

Different steps involved in the process of the launch of a geostationary satellite from Cape Canaveral are:

1. The launch vehicle takes the satellite to a point that is intended to be the perigee of the transfer orbit, at a height of about 300 km above the surface of the Earth, and injects the satellite first into a circular orbit called the parking orbit. The orbit is inclined at an angle of 28.5° with the equatorial plane, as shown in Figure 3.21. Reasons for this inclination angle have been explained earlier.

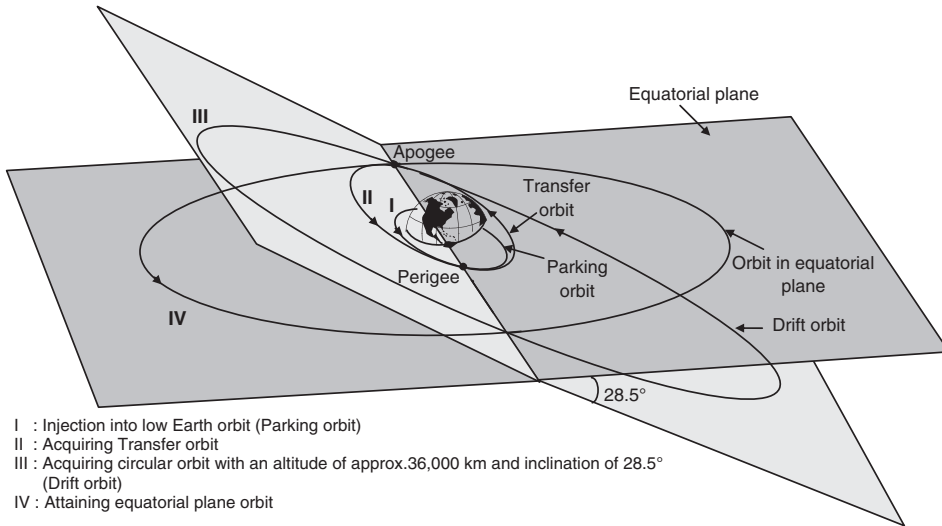


Figure 3.21 Typical launch of a geostationary satellite from Cape Canaveral

2. In the second step, a perigee manoeuvre associated with the firing of a perigee boost motor transforms the circular parking orbit to an eccentric elliptical transfer orbit with perigee and apogee altitudes of 300 km and 36 000 km respectively.
3. In the third step, an apogee manoeuvre similar to the one used in the case of the Kourou launch circularizes the transfer orbit. Till now, the orbit inclination is 28.5° . In another apogee manoeuvre, the orbit inclination is brought to 0° . The thrust required in this manoeuvre is obviously much larger than that required in the case of inclination correction in the Kourou launch.
4. In the fourth and last step, several small manoeuvres are used to put the satellite in the desired longitudinal position.

3.2.1.3 Launch from Baikonur

The launch procedure for a geostationary satellite from Baikonur is similar to the one described in case of the launch from Cape Canaveral. Different steps involved in the process of the launch of a geostationary satellite from Baikonur are:

1. The launch vehicle injects the satellite in a circular orbit with an altitude of 200 km and an inclination of 51° .
2. In the second step, during the first passage of the satellite through the intended perigee, a manoeuvre puts the satellite in the transfer orbit with an apogee altitude of approximately 36 000 km. The orbit inclination now is 47° .
3. In the third step, the transfer orbit is circularized and the inclination corrected at the descending node.
4. In the fourth and last step, the satellite drifts to its final longitudinal position. Different steps are depicted in Figure 3.22.

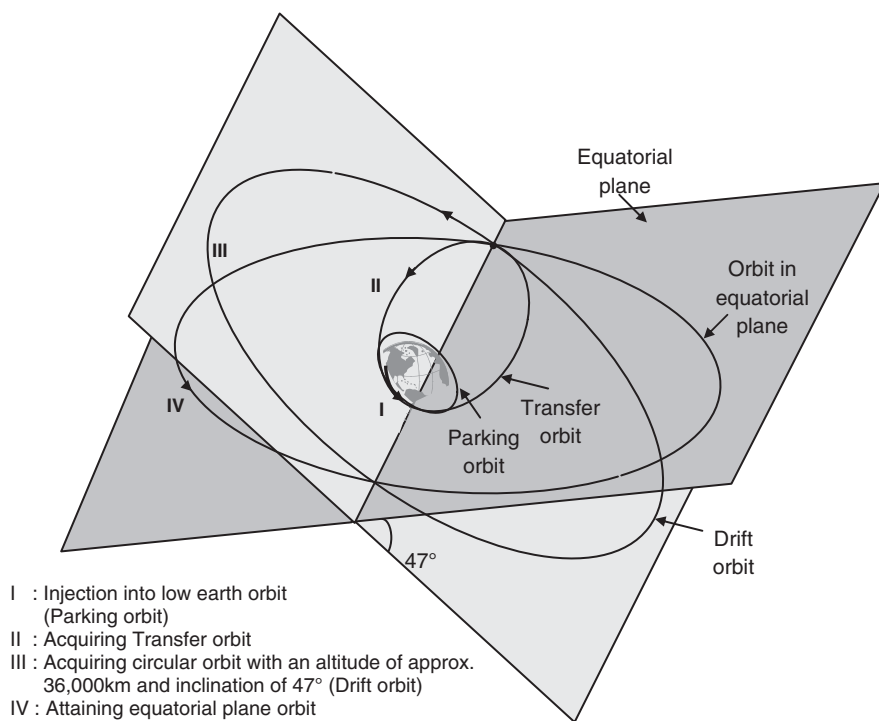


Figure 3.22 Typical launch of a geostationary satellite from Baikonour

3.2.1.4 Space Shuttle Launch

The Space Shuttle launch is also similar to the launch procedure described in the case of a launch from Cape Canaveral and Baikonur. The Space Shuttle injects the satellite along with its perigee motor in a circular parking orbit at an altitude of 100 km. After several revolutions, the perigee motor is fired by signals from an onboard timing mechanism and the thrust generated thereby puts the satellite in a transfer orbit with an apogee at the geostationary height. The perigee motor is then jettisoned. The orbit is then circularized with an apogee manoeuvre.

3.2.2 Launch Vehicles

Launch vehicles are used to launch the satellites from the Earth into their desired orbits. In order to launch the satellite into its desired orbit, two parameters that are very important are the velocity vector and the orbital height. Launch vehicles can be further classified as Expendable Launch Vehicles (ELV) and Reusable Launch Vehicles (RLV). ELVs are designed to be used only once and their components are not recovered after launch. They mostly comprise of multi stage rockets and the job of each stage is to provide the desired orbital manoeuvre. As the job of the stage is completed, it is expended. The process goes on till the satellite is placed into the desired trajectory. The number of rocket stages can be as many as five. It may be mentioned here that some launch vehicles used to put the satellite into small LEO orbits can comprise of a single rocket stage. In addition to the rocket stages, launch vehicles also comprise of boosters that are used to aid the rockets during main orbital manoeuvres or to provide small orbital corrections. Figure 3.23 shows the launch sequence of GSLV-FO4 launch vehicle developed by Indian Space Research Organization (ISRO). It is a three stage vehicle comprising of a solid rocket engine based first stage (GS-1) and liquid rocket engine based second and third stages (GS-2 and GS-3).

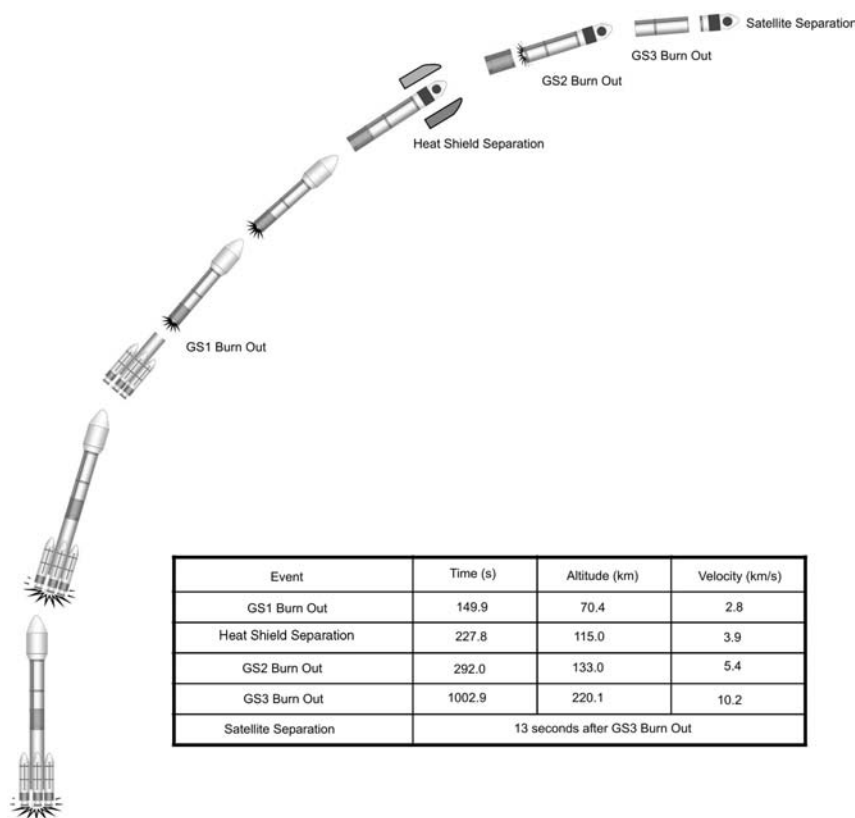


Figure 3.23 Launch sequence of GSLV-FO4 launch vehicle

Ariane (Europe), Atlas (USA), Delta (USA), GSLV (India), PSLV (India), Long March (China) and Proton (Russia) are some of the launch vehicles being used internationally to launch satellites. The Ariane launch vehicle from the European Space Agency (ESA) has entered the fifth generation with ARIANE-5 series. The ARIANE-5 ECA (Enhanced Capability-A) (Figure 3.24) launch vehicle of the Ariane-5 series has the capacity of launching 12 tonnes of payload to geostationary transfer orbit. The GSLV launch vehicle (Figure 3.25), developed by Indian Space Research Organization (ISRO) can launch a 2 to 2.5 tonne satellite into GTO ($200 \text{ km} \times 36\,000 \text{ km}$).



Figure 3.24 Ariane 5 ECA launch vehicle (Reproduced by permission of © ESA-D.DUCROS)

Reusable launch vehicles (RLV) are designed to be recovered intact and used again for subsequent launches. The Space Shuttle (Figure 3.26) from the USA is one example. It is generally used for human spaceflight missions.

Launch vehicles are also classified according to the mass they carry into orbit. As an example the PSLV (Polar Satellite Launch Vehicle), developed by ISRO, India has the capability of launching a payload of 1000 to 1200 kg into polar sun-synchronous orbit. Another launch vehicle developed by ISRO, GSLV (Geostationary Satellite Launch Vehicle) has the capability to launch a 2 to 2.5 tonne payload to GTO ($200 \text{ km} \times 36\,000 \text{ km}$).

3.3 Orbital Perturbations

The satellite, once placed in its orbit, experiences various perturbing torques that cause variations in its orbital parameters with time. These include gravitational forces from other bodies



Figure 3.25 GSLV launch vehicle (Courtesy: ISRO)



Figure 3.26 Space shuttle (Courtesy: NASA)

like solar and lunar attraction, magnetic field interaction, solar radiation pressure, asymmetry of Earth's gravitational field etc. Due to these factors, the satellite orbit tends to drift and its orientation also changes and hence the true orbit of the satellite is different from that defined using Kepler's laws. The satellite's position thus needs to be controlled both in the east–west as well as the north–south directions. The east–west location needs to be maintained to prevent radio frequency (RF) interference from neighbouring satellites. It may be mentioned here that in the case of a geostationary satellite, a 1° drift in the east or west direction is equivalent to a drift of about 735 km along the orbit (Figure 3.27). The north–south orientation has to be maintained to have proper satellite inclination.

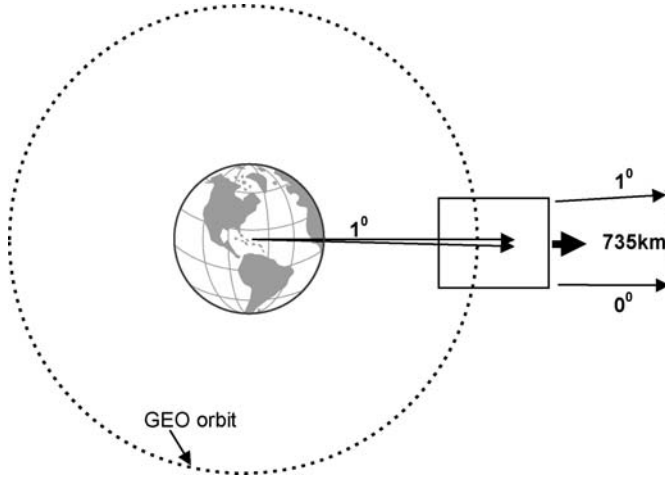


Figure 3.27 Drift of a geostationary satellite

The Earth is not a perfect sphere and is flattened at the poles. The equatorial diameter is about 20–40 km more than the average polar diameter. Also, the equatorial radius of the Earth is not constant. In addition, the average density of Earth is not uniform. All of this results in a non-uniform gravitational field around the Earth which in turn results in variation in gravitational force acting on the satellite due to the Earth. The effect of variation in the gravitational field of the Earth on the satellite is more predominant for geostationary satellites than for satellites orbiting in low Earth orbits as in the case of these satellites the rapid change in the position of the satellite with respect to the Earth's surface will lead to the averaging out of the perturbing forces. In the case of a geostationary satellite, these forces result in an acceleration or deceleration component that varies with the longitudinal location of the satellite.

In addition to the variation in the gravitational field of the Earth, the satellite is also subjected to the gravitational pulls of the sun and the moon. The Earth's orbit around the sun is an ellipse whose plane is inclined at an angle of 7° with respect to the equatorial plane of the sun. The Earth is tilted around 23° away from the normal to the ecliptic. The moon revolves around the Earth with an inclination of around 5° to the equatorial plane of the Earth. Hence, the satellite in orbit is subjected to a variety of out-of-plane forces which change the inclination on the satellite's orbit. The gravitational pulls of Earth, sun and moon have negligible effect of the satellites orbiting in LEO orbits, where the effect of atmospheric drag is more predominant.

As the perturbed orbit is not an ellipse anymore, the satellite does not return to the same point in space after one revolution. The time elapsed between the successive perigee passages is referred to as anomalistic period. The anomalistic period (T_A) is given by equation 3.17.

$$t_A = \frac{2\pi}{\omega_{\text{mod}}} \quad (3.17)$$

Where,

$$\omega_{\text{mod}} = \omega_0 \left[1 + \frac{K(1 - 1.5 \sin^2 i)}{a^2(1 - e^2)^{3/2}} \right]$$

ω_0 is the angular velocity for spherical Earth, $K = 66\,063.1704 \text{ km}^2$, a is the semi-major axis, e is the eccentricity and $i = \cos^{-1} W_Z$, W_Z is the Z axis component of the orbit normal.

The attitude and orbit control system maintains the satellite's position and its orientation and keeps the antenna pointed correctly in the desired direction (bore-sighted to the center of the coverage area of the satellite). The orbit control is performed by firing thrusters in the desired direction or by releasing jets of gas. It is also referred to as station keeping. Thrusters and gas jets are used to correct the longitudinal drifts (in-plane changes) and the inclination changes (out-of-plane changes). It may be mentioned that the manoeuvres required for correcting longitudinal drifts (referred to as the north-south manoeuvre) require a much larger velocity increment as compared to the manoeuvres required for correcting the inclination changes (referred to as the east-west manoeuvre). Hence, generally a different set of thrusters or gas jets is used for north-south and east-west manoeuvres.

3.4 Satellite Stabilization

Commonly employed techniques for satellite attitude control include:

1. Spin stabilization
2. Three-axis or body stabilization

3.4.1 Spin Stabilization

In a spin-stabilized satellite, the satellite body is spun at a rate between 30 and 100 rpm about an axis perpendicular to the orbital plane (Figure 3.28). Like a spinning top, the rotating body offers inertial stiffness, which prevents the satellite from drifting from its desired orientation. Spin-stabilized satellites are generally cylindrical in shape. For stability, the satellite should be spun about its major axis, having a maximum moment of inertia. To maintain stability, the moment of inertia about the desired spin axis should at least be 10 % greater than the moment of inertia about the transverse axis.

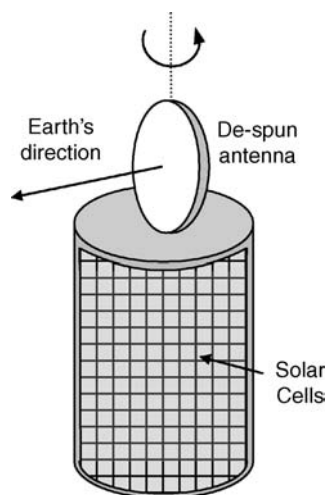


Figure 3.28 Spin stabilized satellite

There are two types of spinning configurations employed in spin-stabilized satellites. These include the simple spinner configuration and the dual spinner configuration. In the simple spinner configuration, the satellite payload and other subsystems are placed in the spinning section, while the antenna and the feed are placed in the de-spun platform. The de-spun platform is spun in a direction opposite to that of the spinning satellite body. In the dual spinner configuration, the entire payload along with the antenna and the feed is placed on the de-spun platform and the other subsystems are located on the spinning body. Modern spin-stabilized satellites almost invariably employ the dual spinner configuration. It may be mentioned here that mounting of the antennae system on the de-spun platform in both the configurations ensures a constant pointing direction of the antennae. In both configurations, solar cells are mounted on the cylindrical body of the satellite. Intelsat-1 to Intelsat-4, Intelsat-6 and TIROS-1 are some of the popular spin-stabilized satellites. Figure 3.29 shows the photograph of Intelsat-4 satellite.

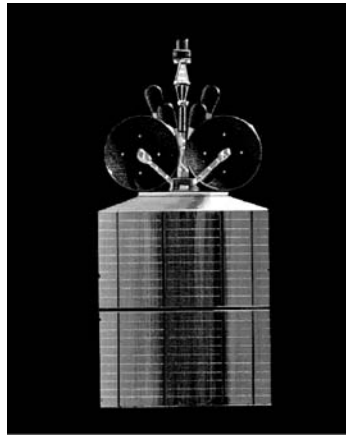


Figure 3.29 Spin-stabilized satellite (Intelsat-4) (Reproduced by permission of © Intelsat)

3.4.2 Three-axis or Body Stabilization

In the case of three-axis stabilization, also known as body stabilization, the stabilization is achieved by controlling the movement of the satellite along the three axes, i.e. yaw, pitch and roll, with respect to a reference (Figure 3.30). The system uses reaction wheels or momentum

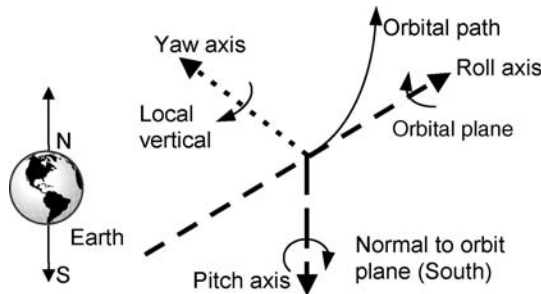


Figure 3.30 Three-axis stabilization

wheels to correct orbit perturbations. The stability of the three-axis system is provided by the active control system, which applies small corrective forces on the wheels to correct the undesirable changes in the satellite orbit.

Most three-axis stabilized satellites use momentum wheels. The basic control technique used here is to speed up or slow down the momentum wheel depending upon the direction in which the satellite is perturbed. The satellite rotates in a direction opposite to that of speed change of the wheel. For example, an increase in speed of the wheel in the clockwise direction will make the satellite to rotate in a counterclockwise direction. The momentum wheels rotate in one direction and can be twisted by a gimbal motor to provide the required dynamic force on the satellite.

An alternative approach is to use reaction wheels. Three reaction wheels are used, one for each axis. They can be rotated in either direction depending upon the active correction force. The satellite body is generally box shaped for three-axis stabilized satellites. Antennae are mounted on the Earth-facing side and on the lateral sides adjacent to it. These satellites use flat solar panels mounted above and below the satellite body in such a way that they always point towards the sun, which is an obvious requirement.

Some popular satellites belonging to the category of three-axis stabilized satellites include Intelsat-5, Intelsat-7, Intelsat-8, GOES-8, GOES-9, TIROS-N and the INSAT series of satellites. Figure 3.31 is a photograph of the Intelsat-5 satellite.



Figure 3.31 Three-axis stabilized satellite (Intelsat-5) (Reproduced by permission of © Intelsat)

3.4.3 Comparison between Spin-stabilized and Three-axis Stabilized Satellites

1. In comparison to spin-stabilized satellites, three-axis stabilized satellites have more power generation capability and more additional mounting area available for complex antennae structures.
2. Spin-stabilized satellites are simpler in design and less expensive than three-axis stabilized satellites.
3. Three-axis stabilized satellites have the disadvantage that the extendible solar array used in these satellites are unable to provide power when the satellite is in the transfer orbit, as the array is still stored inside the satellite during this time.

3.4.4 Station Keeping

Station keeping is the process of maintenance of the satellite's orbit against different factors that cause temporal drift. Satellites need to have their orbits adjusted from time to time because the satellite, even though initially placed in the correct orbit, can undergo a progressive drift due to some natural forces such as minor gravitational perturbations due to the sun and moon, solar radiation pressure, Earth being an imperfect sphere, etc. The orbital adjustments are usually made by releasing jets of gas or by firing small rockets tied to the body of the satellite.

In the case of spin-stabilized satellites, station keeping in the north–south direction is maintained by firing thrusters parallel to the spin axis in a continuous mode. The east–west station keeping is obtained by firing thrusters perpendicular to the spin axis. In the case of three-axis stabilization, station keeping is achieved by firing thrusters in the east–west or the north–south directions in a continuous mode. The amplitude and direction of thrusts required to carry out a variety of orbital adjustments have already been discussed at length in Section 3.1.

3.5 Orbital Effects on Satellite's Performance

As we know the satellite is revolving constantly around the Earth. The motion of the satellite has significant effects on its performance. These include the Doppler shift, effect due to variation in the orbital distance, effect of solar eclipse and sun's transit outage.

3.5.1 Doppler Shift

The geostationary satellites appear stationary with respect to an Earth station terminal whereas in the case of satellites orbiting in low Earth orbits, the satellite is in relative motion with respect to the terminal. However, in the case of geostationary satellites also there are some variations between the satellite and the Earth station terminal. As the satellite is moving with respect to the Earth station terminal, the frequency of the satellite transmitter also varies with respect to the receiver on the Earth station terminal. If the frequency transmitted by the satellite is f_T , then the received frequency f_R is given by equation 3.18.

$$\left(\frac{f_R - f_T}{f_T} \right) = \left(\frac{\Delta f}{f_T} \right) = \left(\frac{v_T}{v_P} \right) \quad (3.18)$$

Where,

v_T is the component of the satellite transmitter velocity vector directed towards the Earth station receiver

v_P is the phase velocity of light in free space (3×10^8 m/s)

3.5.2 Variation in the Orbital Distance

Variation in the orbital distance results in variation in the range between the satellite and the Earth station terminal. If a Time Division Multiple Access (TDMA) scheme is employed by the satellite, the timing of the frames within the TDMA bursts should be worked out carefully so that the user terminals receive the correct data at the correct time. Range variations are more predominant in low and medium Earth orbiting satellites as compared to the geostationary satellites.

3.5.3 Solar Eclipse

There are times when the satellites do not receive solar radiation due to obstruction from a celestial body. During these periods the satellites operate using onboard batteries. The design of the battery is such so as to provide continuous power during the period of the eclipse. Ground control stations perform battery conditioning routines prior to the occurrence of an eclipse to ensure best performance during the eclipse. These include discharging the batteries close to their maximum depth of discharge and then fully recharging them just before the eclipse occurs. Also, the rapidity with which the satellite enters and exits the shadow of the celestial body creates sudden temperature stress situations. The satellite is designed in such a manner so as to cope with these thermal stresses.

3.5.4 Sun Transit Outrage

There are times when the satellite passes directly between the sun and the Earth as shown in Figure 3.32. The Earth station antenna will receive signals from the satellite as well as the microwave radiation emitted by the sun (the sun is a source of radiation with an equivalent temperature varying between 6000 K to 11000 K depending upon the time of the 11-year sunspot cycle). This might cause temporary outage if the magnitude of the solar radiation exceeds the fade margin of the receiver. The traffic of the satellite may be shifted to other satellites during such periods.

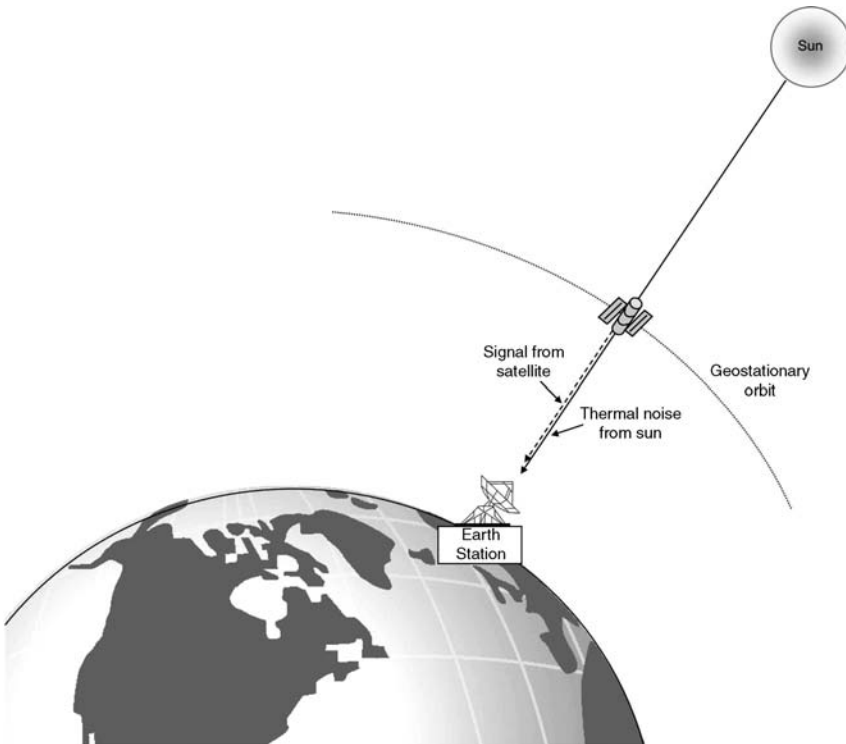


Figure 3.32 Sun outage conditions

3.6 Eclipses

With reference to satellites, an eclipse is said to occur when the sunlight fails to reach the satellite's solar panel due to an obstruction from a celestial body. The major and most frequent source of an eclipse is due to the satellite coming in the shadow of the Earth (Figure 3.33). This is known as a solar eclipse. The eclipse is total; i.e. the satellite fails to receive any light whatsoever if it passes through the umbra, which is the dark central region of the shadow, and receives very little light if it passes through the penumbra, which is the less dark region surrounding the umbra (Figure 3.34). The eclipse occurs as the Earth's equatorial plane is inclined at a constant angle of about 23.5° to its ecliptic plane, which is the plane of the Earth's orbit extended to infinity. The eclipse is seen on 42 nights during the spring and an equal number of nights during the autumn by the geostationary satellite. The effect is the worst during the equinoxes and lasts for about 72 minutes. The equinox, as explained earlier, is the point in time when the sun crosses the equator, making the day and night equal in length. The spring and autumn equinoxes respectively occur on 20–21 March and 22–23 September. During the equinoxes in March and September, the satellite, the Earth and the sun are aligned at midnight local time and the satellite spends about 72 minutes in total darkness. From 21 days before and 21 days after the equinoxes, the satellite crosses the umbral cone each day for some time, thereby receiving only a part of solar light for that time. During the rest of the year, the geostationary satellite orbit passes either above or below the umbral cone. It is at the maximum distance at the time of the solstices, above the umbral cone at the time of the summer solstice (20–21 June) and below it at the time of the winter solstice (21–22 December). Figure 3.35 further illustrates the phenomenon.

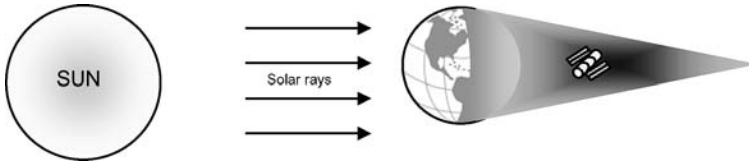


Figure 3.33 Solar eclipse



Figure 3.34 Umbra and penumbra

Hence, the duration of an eclipse increases from zero to about 72 minutes starting 21 days before the equinox and then decreases from 72 minutes to zero during 21 days following the equinox. The duration of an eclipse on a given day around the equinox can be seen from the graph in Figure 3.36. Another type of eclipse known as the lunar eclipse occurs when

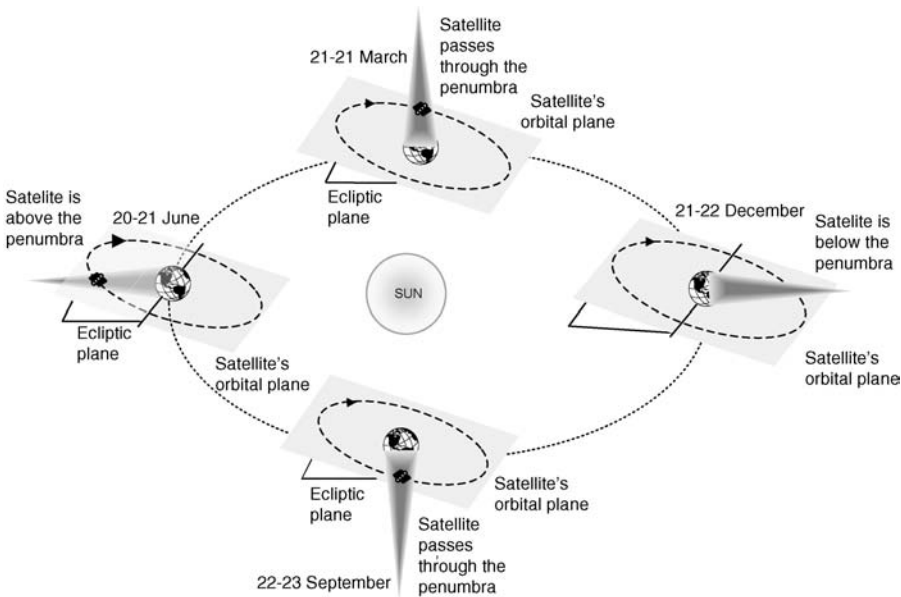


Figure 3.35 Positions of the geostationary satellite during the equinoxes and solstices

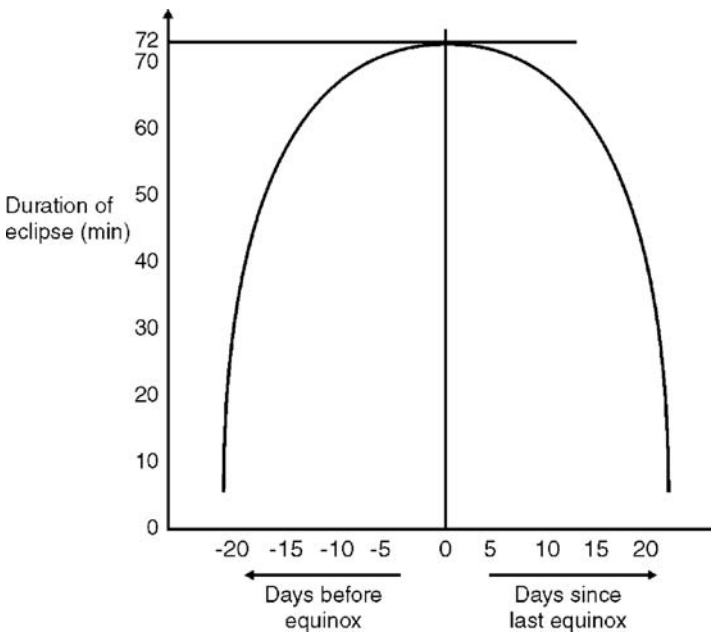


Figure 3.36 Duration of the eclipse before and after the equinox

the moon's shadow passes across the satellite (Figure 3.37). This is much less common and occurs once in 29 years. In fact, for all practical purposes, when an eclipse is mentioned with respect to satellites, it is a solar eclipse that is referred to.

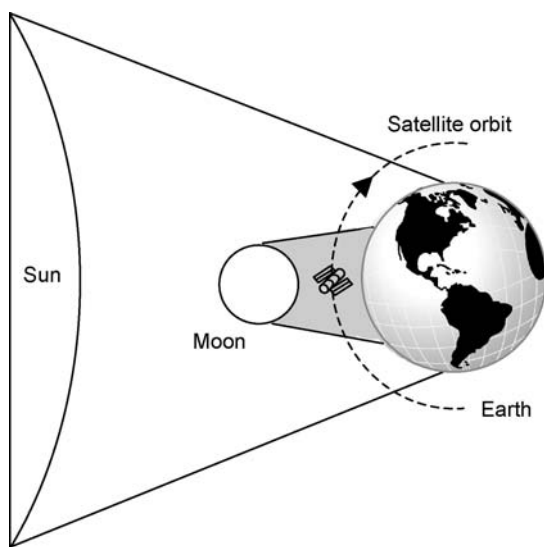


Figure 3.37 Lunar eclipse

While a solar eclipse takes place, the failure of sunlight to reach the satellite interrupts the battery recharging process. The satellite is depleted of its electrical power capacity. It does not significantly affect low power satellites, which can usually continue their operation with back-up power. The high power satellites, however, shut down for all but essential services.

3.7 Look Angles of a Satellite

The look angles of a satellite refer to the coordinates to which an Earth station must be pointed in order to communicate with the satellite and are expressed in terms of azimuth and elevation angles. In the case where an Earth station is within the footprint or coverage area of a geostationary satellite, it can communicate with the satellite by simply pointing its antenna towards it. The process of pointing the Earth station antenna accurately towards the satellite can be accomplished if the azimuth and elevation angles of the Earth station location are known. Also, the elevation angle, as we shall see in the following paragraphs, affects the slant range, i.e. line of sight distance between the Earth station and the satellite.

In order to determine the look angles of a satellite, its precise location should be known. The location of a satellite is very often determined by the position of the sub-satellite point. The sub-satellite point is the location on the surface of the Earth that lies directly between the satellite and the centre of the Earth. To an observer on the sub-satellite point, the satellite will appear to be directly overhead (Figure 3.38).

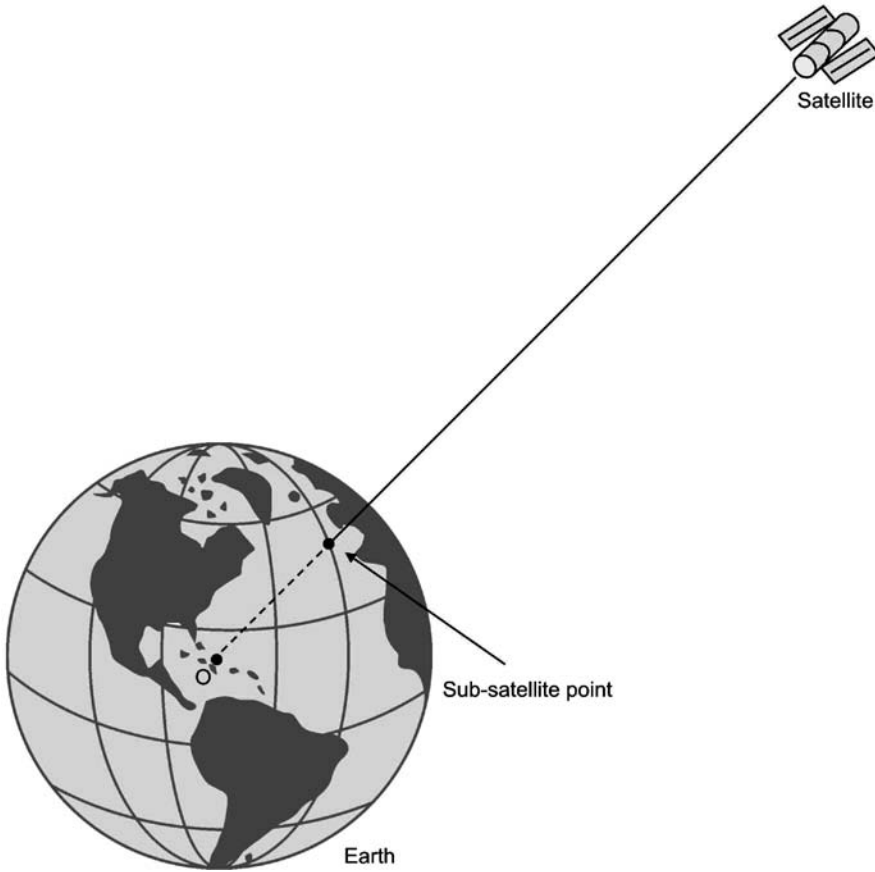


Figure 3.38 Sub-satellite point

3.7.1 Azimuth Angle

The azimuth angle A of an Earth station is defined as the angle produced by the line of intersection of the local horizontal plane and the plane passing through the Earth station, the satellite and the centre of the Earth with the true north (Figure 3.39). We can visualize that this line of intersection between the two above-mentioned planes would be one of the many possible tangents that can be drawn at the point of location of the Earth station. Depending upon the location of the Earth station and the sub-satellite point, the azimuth angle can be computed as follows:

Earth station in the northern hemisphere:

$$A = 180^\circ - A' \quad \text{when the Earth station is to the west of the satellite} \quad (3.19)$$

$$A = 180^\circ + A' \quad \text{when the Earth station is to the east of the satellite} \quad (3.20)$$

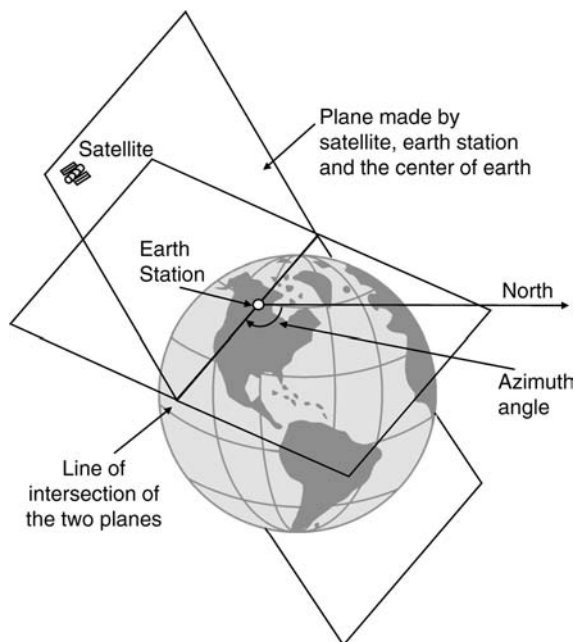


Figure 3.39 Azimuth angle

Earth station in the southern hemisphere:

$$A = A' \dots \quad \text{when the Earth station is to the west of the satellite} \quad (3.21)$$

$$A = 360^\circ - A' \dots \quad \text{when the Earth station is to the east of the satellite} \quad (3.22)$$

where A' can be computed from

$$A' = \tan^{-1} \left(\frac{\tan |\theta_s - \theta_L|}{\sin \theta_1} \right) \quad (3.23)$$

where

θ_s = satellite longitude

θ_L = Earth station longitude

θ_1 = Earth station latitude

3.7.2 Elevation Angle

The Earth station elevation angle E is the angle between the line of intersection of the local horizontal plane and the plane passing through the Earth station, the satellite and the centre of the Earth with the line joining the Earth station and the satellite. Figures 3.40 (a) and (b) show the elevation angles for two different satellite and Earth station positions. It can be computed from

$$E = \tan^{-1} \left[\frac{r - R \cos \theta_1 \cos |\theta_s - \theta_L|}{R \sin \{\cos^{-1}(\cos \theta_1 \cos |\theta_s - \theta_L|)\}} \right] - \cos^{-1}(\cos \theta_1 \cos |\theta_s - \theta_L|) \quad (3.24)$$

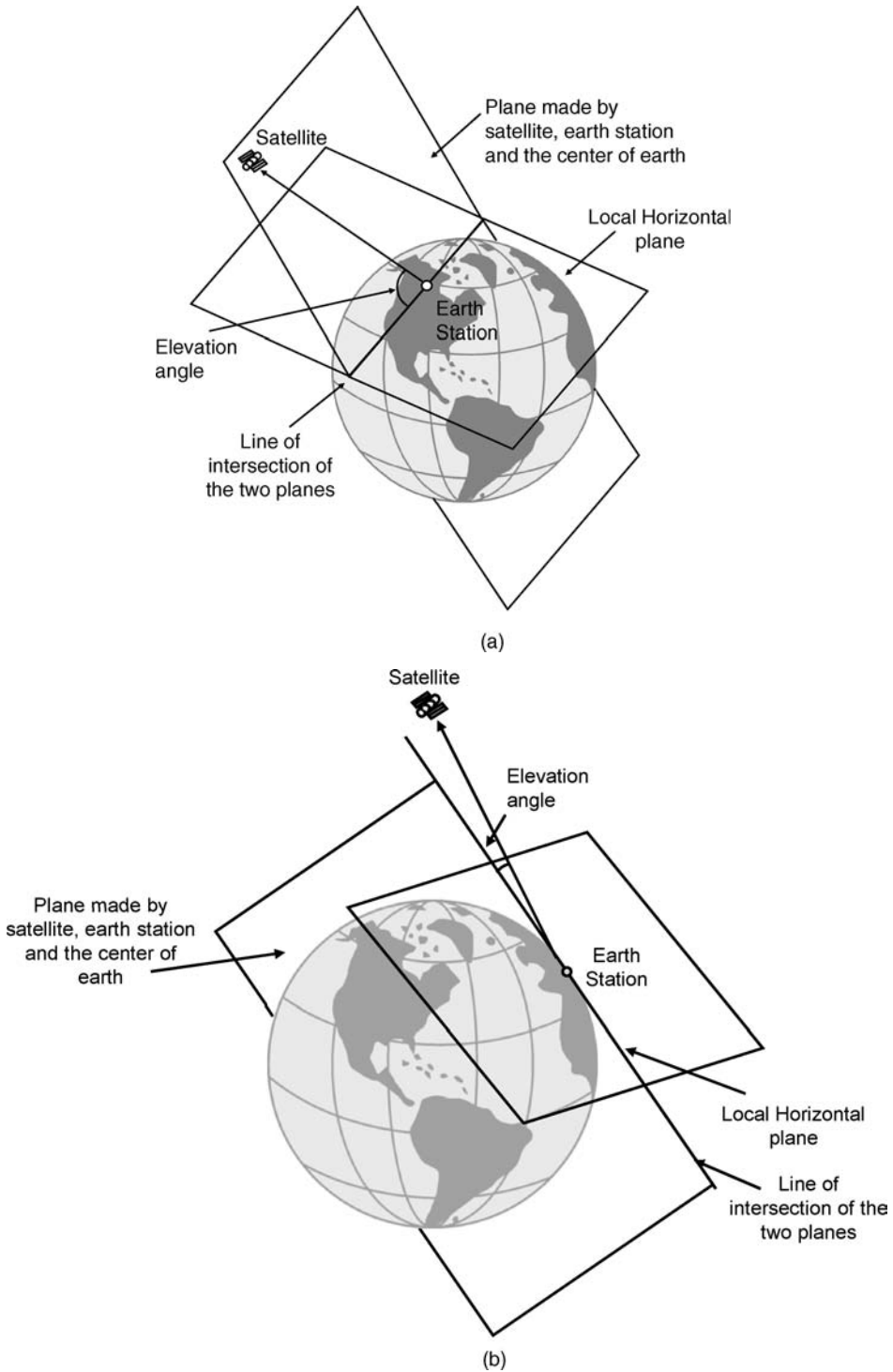


Figure 3.40 Earth station elevation angle

where

r = orbital radius, R = Earth's radius

θ_s = Satellite longitude, θ_L = Earth station longitude, θ_l = Earth station latitude

3.7.3 Computing the Slant Range

Slant range of a satellite is defined as the range or the distance of the satellite from the Earth station. The elevation angle E , as mentioned earlier, has a direct bearing on the slant range. The smaller the elevation angle of the Earth station, the larger is the slant range and the coverage angle. Refer to Figure 3.41.

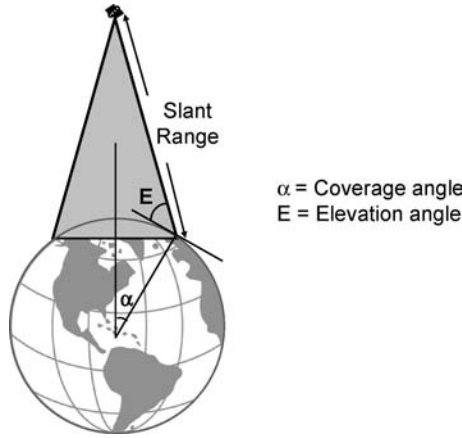


Figure 3.41 Elevation angle, slant range and coverage angle

The slant range can be computed from

$$\text{Slant range } D = \sqrt{R^2 + (R + H)^2 - 2R(R + H) \sin \left[E + \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E \right\} \right]} \quad (3.25)$$

$$\text{Coverage angle } \alpha = \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E \right\} \quad (3.26)$$

Where

R = radius of the Earth

E = angle of elevation

H = height of the satellite above the surface of the Earth

It is evident from the above expression that a zero angle of elevation leads to the maximum coverage angle. A larger slant range means a longer propagation delay time and a greater impairment of signal quality, as the signal has to travel a greater distance through the Earth's atmosphere.

3.7.4 Computing the Line-of-Sight Distance between Two Satellites

Refer to Figure 3.42. The line-of-sight distance between two satellites placed in the same circular orbit can be computed from triangle ABC formed by the points of location of two satellites and the centre of the Earth. The line-of-sight distance AB in this case is given by

$$AB = \sqrt{(AC^2 + BC^2 - 2 AC BC \cos \theta)} \quad (3.27)$$

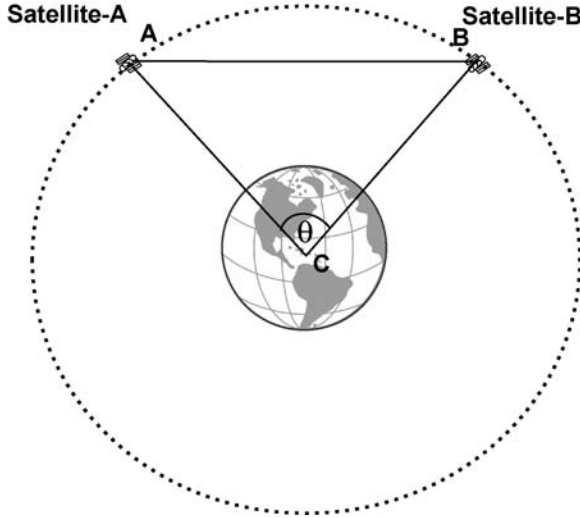


Figure 3.42 Line-of-sight distance between two satellites

Note also that angle θ will be the angular separation of the longitudes of the two satellites. For example, if the two satellites are located at 30°E and 60°E , θ would be equal to 30° . If the two locations are 30°W and 60°E , then in that case θ would be 90° . The maximum line-of-sight distance between these two satellites occurs when the satellites are placed so that the line joining the two becomes tangent to the Earth's surface, as shown in Figure 3.43.

In this the case, the maximum line-of-sight distance (AB) equals $OA + OB$, which further equals $2OA$ or $2OB$ as $OA = OB$. If R is the radius of the Earth and H is the height of satellites above the surface of the Earth, then

$$OA = AC \sin \beta = (R + H) \sin \beta \quad (3.28)$$

Now

$$\beta = \cos^{-1} \left(\frac{R}{R + H} \right) \quad (3.29)$$

Therefore

$$OA = (R + H) \sin \left[\cos^{-1} \left(\frac{R}{R + H} \right) \right] \quad (3.30)$$

and

$$\text{Maximum line-of-sight distance} = 2(R + H) \sin \left[\cos^{-1} \left(\frac{R}{R + H} \right) \right] \quad (3.31)$$

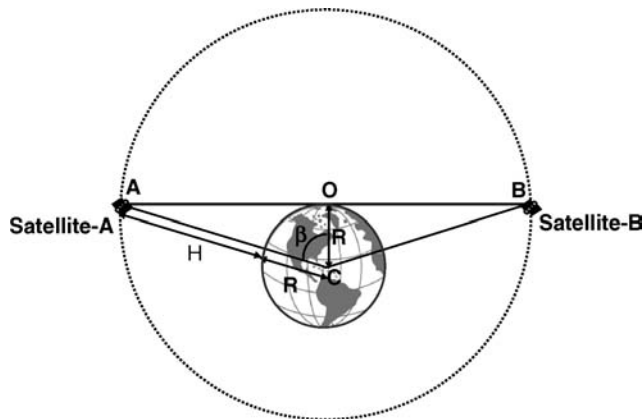


Figure 3.43 Maximum line-of-sight distance between two satellites

Problem 3.7

Determine the maximum possible line-of-sight distance between two geostationary satellites orbiting the Earth at a height of 36 000 km above the surface of the Earth. Assume the radius of the Earth to be 6370 km.

Solution:

Maximum line-of-sight distance can be computed from

$$2(R + H) \sin \left[\cos^{-1} \left\{ \frac{R}{(R + H)} \right\} \right]$$

where

R = Radius of Earth

H = Height of satellite above surface of Earth

Therefore, Maximum line-of-sight distance is given by:

$$\begin{aligned} & 2(6370 + 36\,000) \sin[\cos^{-1}\{6370/(6370 + 36\,000)\}] \\ &= 2(42\,370) \sin[\cos^{-1}\{6370/42\,370\}] = 84\,740 \times 0.989 = 83\,807.86 \text{ km} \end{aligned}$$

Problem 3.8

A satellite in the Intelsat-VI series is located at 37°W and another belonging to the Intelsat-VII series is located at 74°E (Figure 3.44). If both these satellites are in a circular equatorial geostationary orbit with an orbital radius of 42 164 km, determine the inter-satellite distance.

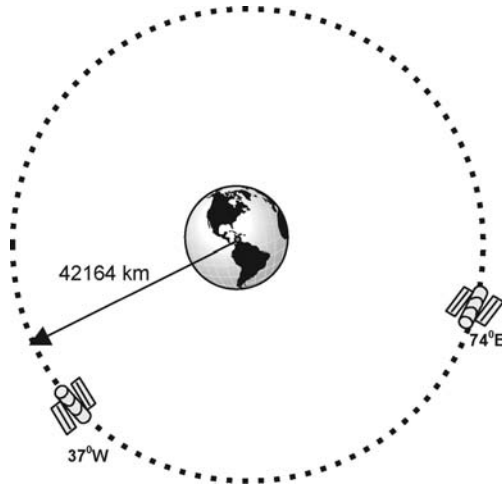


Figure 3.44 Figure for Problem 3.8

Solution: The inter-satellite distance can be computed from

$$\sqrt{(D_1^2 + D_2^2 - 2D_1D_2 \cos \theta)}$$

where

D_1 = orbital radius of the first satellite

D_2 = orbital radius of the second satellite

θ = angle formed by two radii

$$D_1 = D_2 = 42\,164 \text{ km and } \theta = 37^\circ + 74^\circ = 111^\circ$$

$$\begin{aligned} \text{Inter-satellite distance} &= \sqrt{[(42\,164)^2 + (42\,164)^2 - (2 \times 42\,164 \times 42\,164 \times \cos 111^\circ)]} \\ &= \sqrt{[2 \times (42\,164)^2 (1 - \cos 111^\circ)]} \\ &= 69\,486.27 \text{ km} \end{aligned}$$

Problem 3.9

An Earth station is located at 30°W longitude and 60°N latitude. Determine the Earth station's azimuth and elevation angles with respect to a geostationary satellite located at 50°W longitude. The orbital radius is 42 164 km. (Assume the radius of the Earth to be 6378 km.)

Solution: Since the Earth station is in the northern hemisphere and is located towards east of the satellite, the azimuth angle A is given by $(180^\circ + A')$, where A' can be computed

from

$$A' = \tan^{-1} \left(\frac{\tan |\theta_s - \theta_L|}{\sin \theta_l} \right)$$

where

θ_s = satellite longitude = 50°W

θ_L = Earth station longitude = 30°W

θ_l = Earth station latitude = 60°N

Therefore

$$A' = \tan^{-1} \left(\frac{\tan 20^\circ}{\sin 60^\circ} \right) = \tan^{-1} \left(\frac{0.364}{0.866} \right) = \tan^{-1}(0.42) = 22.8^\circ$$

and

$$A = 180^\circ + 22.8^\circ = 202.8^\circ$$

The Earth station elevation angle is given by

$$E = \tan^{-1} \left[\frac{r - R \cos \theta_l \cos |\theta_s - \theta_L|}{R \sin \{\cos^{-1}(\cos \theta_l \cos |\theta_s - \theta_L|)\}} \right] - \cos^{-1}(\cos \theta_l \cos |\theta_s - \theta_L|)$$

where

r = Satellite orbital radius

R = Earth's radius

Substituting the values of various parameters gives

$$\begin{aligned} E &= \tan^{-1} \left[\frac{42\,164 - 6378 \cos 60^\circ \cos 20^\circ}{6378 \sin \{\cos^{-1}(\cos 60^\circ \cos 20^\circ)\}} \right] - \cos^{-1}(\cos 60^\circ \cos 20^\circ) \\ &= \tan^{-1} \left[\frac{42\,164 - 2998}{6378 \sin(\cos^{-1} 0.47)} \right] - \cos^{-1} 0.47 \\ &= \tan^{-1} \left(\frac{39\,166}{5631} \right) - 62^\circ \\ &= 81.8^\circ - 62^\circ = 19.8^\circ \end{aligned}$$

Therefore,

$$\text{Azimuth} = 202.8^\circ \text{ and Elevation} = 19.8^\circ$$

Problem 3.10

Consider two Earth stations, X and Y, with longitudes at 60°W and 90°W respectively and latitudes at 30°N and 45°N respectively. They are communicating with each other via a geostationary satellite located at 105°W . Find the total delay in sending 500 kbs of information if the transmission speed is 10 Mbps. Assume the orbital radius to be 42 164 km and the radius of the Earth to be 6378 km.

Solution: In the first step, the elevation angles of the two Earth stations are determined:

Earth station X latitude, $\theta_{IX} = 30^\circ \text{ N}$

Earth station X longitude, $\theta_{LX} = 60^\circ \text{ W}$

Satellite longitude, $\theta_s = 105^\circ \text{ W}$

The elevation angle E_X of the Earth station X is given by

$$E_X = \tan^{-1} \left[\frac{r - R \cos \theta_{IX} \cos |\theta_s - \theta_{LX}|}{R \sin \{\cos^{-1}(\cos \theta_{IX} \cos |\theta_s - \theta_{LX}|)\}} \right] - \cos^{-1}(\cos \theta_{IX} \cos |\theta_s - \theta_{LX}|)$$

where

r = Satellite orbital radius

R = Earth's radius

$$\begin{aligned} E_X &= \tan^{-1} \left[\frac{42\,164 - 6378 \cos 30^\circ \cos 45^\circ}{6378 \sin \{\cos^{-1}(\cos 30^\circ \cos 45^\circ)\}} \right] - \cos^{-1}(\cos 30^\circ \cos 45^\circ) \\ &= \tan^{-1} \left(\frac{42\,164 - 6378 \times 0.612}{6378 \sin 52.3^\circ} \right) - \cos^{-1}(0.612) \\ &= \tan^{-1} \left(\frac{38260.66}{5044.998} \right) - 52.3^\circ = 82.6^\circ - 52.3^\circ = 30.3^\circ \end{aligned}$$

Earth station Y latitude, $\theta_{IY} = 45^\circ \text{ N}$

Earth station Y longitude, $\theta_{LY} = 90^\circ \text{ W}$

Satellite longitude, $\theta_s = 105^\circ \text{ W}$

The elevation angle E_Y of the Earth station Y is given by

$$\begin{aligned} E_Y &= \tan^{-1} \left(\frac{42\,164 - 6378 \cos 45^\circ \cos 15^\circ}{6378 \sin \{\cos^{-1}(\cos 45^\circ \cos 15^\circ)\}} \right) - \cos^{-1}(\cos 45^\circ \cos 15^\circ) \\ &= \tan^{-1} \left(\frac{37507.826}{4656.174} \right) - \cos^{-1} 0.683 \\ &= 82.92^\circ - 46.92^\circ = 36^\circ \end{aligned}$$

In the next step, the slant range of the two Earth stations will be determined. Refer to Figure 3.45. The slant range (d_X) of the Earth station X can be computed from

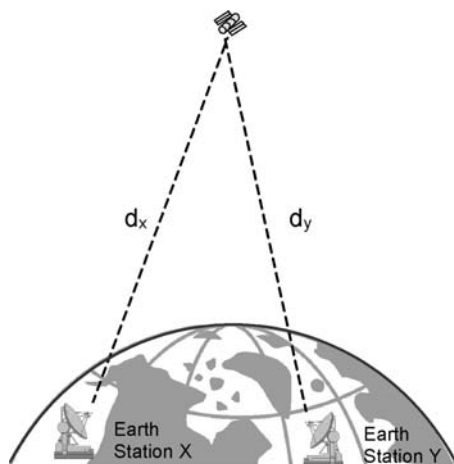


Figure 3.45 Figure for Problem 3.10

$$\begin{aligned}
 (d_x)^2 &= R^2 + (R + H)^2 - 2R(R + H) \sin \left[E + \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E_x \right\} \right] \\
 &= (6378 \times 10^3)^2 + (42164 \times 10^3)^2 - 2 \times 6378 \times 10^3 \times 42164 \times 10^3 \\
 &\quad \times \sin \left[30.3^\circ + \sin^{-1} \left\{ \left(\frac{6378 \times 10^3}{42164 \times 10^3} \right) \cos 30.3^\circ \right\} \right] \\
 &= 40678884 \times 10^6 + 1777802896 \times 10^6 - 537843984 \times 10^6 \times \sin 37.8^\circ \\
 &= 1488783418 \times 10^6
 \end{aligned}$$

This gives

$$d_x = \sqrt{1488783418 \times 10^6} = 38584.76 \text{ km}$$

The Slant range d_y for the Earth station Y can computed from

$$\begin{aligned}
 (d_y)^2 &= R^2 + (R + H)^2 - 2R(R + H) \sin \left[E + \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E_y \right\} \right] \\
 &= (6378 \times 10^3)^2 + (42164 \times 10^3)^2 - 2 \times 6378 \times 10^3 \times 42164 \times 10^3 \\
 &\quad \times \sin \left[36^\circ + \sin^{-1} \left\{ \left(\frac{6378 \times 10^3}{42164 \times 10^3} \right) \cos 36^\circ \right\} \right] \\
 &= 40678884 \times 10^6 + 1777802896 \times 10^6 - 537843984 \times 10^6 \times \sin 43^\circ \\
 &= 1451672183 \times 10^6
 \end{aligned}$$

This gives

$$d_Y = \sqrt{1\,451\,672\,183 \times 10^6} = 38\,100.8 \text{ km}$$

$$\text{Total range to be covered} = 38\,584.76 + 38\,100.8 = 76\,685.56 \text{ km}$$

$$\text{Propagation delay} = (76\,685.56 \times 10^3 / 3 \times 10^8) = 255.62 \text{ ms}$$

The time required to transmit 500 kbps of information at a transmission speed of 10 Mbps is given by

$$\frac{500\,000}{10^7} = 50 \text{ ms}$$

$$\text{Total delay} = 255.62 + 50 = 305.62 \text{ ms}$$

Problem 3.11

Two geostationary satellites, A and B, moving in an orbit of 42 164 km radius are stationed at 85°W and 25°W longitudes, as shown in Figure 3.46. The two satellites have slant ranges of 38 000 km and 36 000 km from a common Earth station. Determine the angular separation of the two satellites as viewed by the Earth station. Also find the intersatellite separation in orbit.

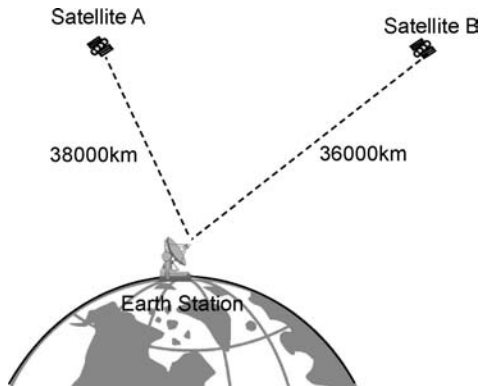


Figure 3.46 Figure for Problem 3.11

Solution: If d is the inter-satellite distance between the two satellites, then

$$d^2 = d_A^2 + d_B^2 - 2d_A d_B \cos \theta$$

where

θ = Angular separation between the two satellites

d_A = Slant range of Satellite A

d_B = Slant range of Satellite B

Also,

$$d^2 = r^2 + r^2 - 2r^2 \cos \beta = 2r^2(1 - \cos \beta)$$

where

r = Satellite orbital radius

β = Longitudinal difference between the two satellites

This gives

$$d_A^2 + d_B^2 - 2d_A d_B \cos \theta = 2r^2(1 - \cos \beta)$$

or

$$\theta = \cos^{-1} \left[\frac{d_A^2 + d_B^2 - 2r^2(1 - \cos \beta)}{2d_A d_B} \right]$$

Substituting these values gives

$$\begin{aligned} \theta &= \cos^{-1} \left[\frac{(38\,000)^2 + (36\,000)^2 - 2 \times (42\,164)^2(1 - \cos 60^\circ)}{2 \times 38\,000 \times 36\,000} \right] \\ &= \cos^{-1} \left(\frac{1444 \times 10^6 + 1296 \times 10^6 - 1777.8 \times 10^6}{2736 \times 10^6} \right) \\ &= \cos^{-1} \left(\frac{962.2}{2736} \right) = 69.4^\circ \end{aligned}$$

The intersatellite distance can be computed from

$$d = \sqrt{[2r^2(1 - \cos \beta)]} = 1.414 \times r \sqrt{(1 - \cos 60^\circ)} = 1.414 \times 42\,164 \sqrt{0.5} = 42\,164 \text{ km}$$

Angular separation = 69.4°

Inter-satellite distance = 42 164 km

3.8 Earth Coverage and Ground Tracks

Earth coverage, also known as the ‘footprint’, is the surface area of the Earth that can possibly be covered by a given satellite. In the discussion to follow, the effect of satellite altitude on Earth coverage provided by the satellite will be examined. Ground track is the path followed by the sub-satellite point. The effects of altitude of the satellite and its latitude on ground track are also discussed.

3.8.1 Satellite Altitude and the Earth Coverage Area

Refer to Figure 3.47. It is evident that the coverage area increases with the height of the satellite above the surface of the Earth. It varies from something like 1.5 % of the Earth’s surface area for a low Earth satellite orbit at 200 km to about 43 % of the Earth’s surface area for a satellite at a geostationary height of 36 000 km. Table 3.1 shows the variation of coverage area as a

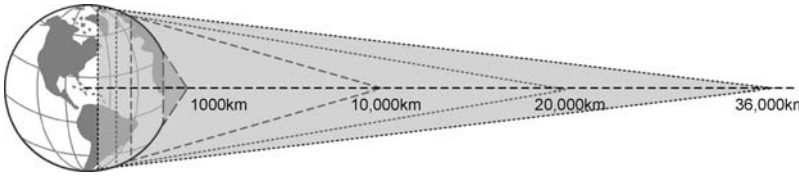


Figure 3.47 Satellite altitude and Earth coverage area

Table 3.1 Variation of the coverage area as a function of the satellite altitude

Satellite altitude (km)	Coverage area (% of Earth's surface area)
200	1.5
300	2.0
400	2.5
500	3.0
600	3.5
700	4.5
800	5.5
900	6.0
1 000	7.0
2 000	12.0
4 000	18.5
5 000	21.5
6 000	24.0
7 000	26.0
8 000	27.5
9 000	29.0
10 000	30.0
15 000	35.0
20 000	37.5
25 000	40.0
30 000	41.5
36 000	43.0

function of the satellite altitude. It can be seen from the table that the increase in coverage area with an increase in altitude is steeper in the beginning than it is as the altitude increases beyond 10 000 km. The coverage angle, from Equation (3.26), can be computed from

$$\text{Coverage angle } \alpha = \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E \right\}$$

For maximum possible coverage, $E = 0^\circ$. The expression reduces to

$$\text{Coverage angle } \alpha = \sin^{-1} \left(\frac{R}{R + H} \right) \quad (3.32)$$

The full coverage angle α can be computed to be approximately 150° for a satellite at 200 km and 17° for a satellite at a geostationary height of 36 000 km. Table 3.2 shows the variation of the full coverage angle as a function of the satellite altitude

Table 3.2 Variation of the full coverage angle as a function of the satellite altitude

Satellite altitude (km)	Full Coverage angle (deg)
200	150
300	144
400	138
500	134
600	130
700	126
800	124
900	120
1 000	118
2 000	100
4 000	76
5 000	68
6 000	60
7 000	56
8 000	52
9 000	48
10 000	44
15 000	32
20 000	26
25 000	22
30 000	18
36 000	17

3.8.2 *Satellite Ground Tracks*

The ground track of an orbiting satellite is the path followed by the sub-satellite point, i.e. the point formed by the projection of the line joining the orbiting satellite with the centre of the Earth on the surface of the Earth (Figure 3.48). If the Earth were not rotating, the ground track would simply be the circumference of the great circle formed by the bisection of the Earth with the orbital plane of the satellite. In reality, however, the Earth does rotate, with the result that the ground track gets modified from what it would be in the hypothetical case of a non-rotating Earth.

Two factors that influence the ground track due to Earth's rotation include the altitude of the satellite, which in turn determines the satellite's angular velocity, and the latitude at which the satellite is located, which determines the component of the Earth's rotation applicable at that point. In simple words, if we know the ground track that would have been there had the Earth been static, modification to this track at any given point in the satellite orbit would depend on the satellite altitude at that point and also on the latitude of that point.

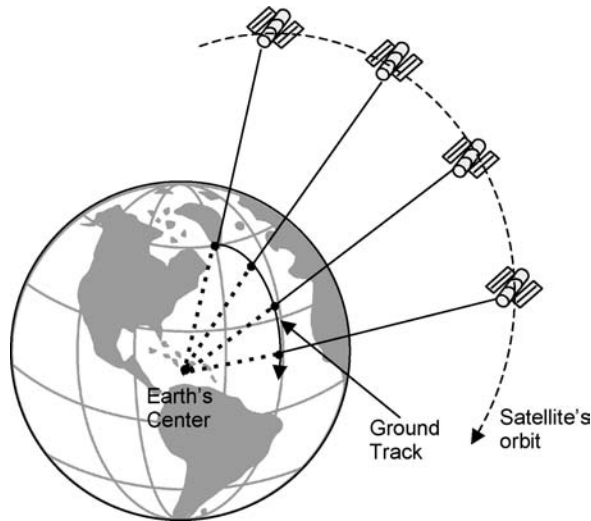


Figure 3.48 Satellite ground tracks

3.8.2.1 Effect of Altitude

The higher the altitude of the satellite, the smaller is the angular velocity and the greater will be the displacement of the ground track towards the west due to the Earth's rotation (Figure 3.49). In the case of circular orbits, the ground track of the satellite at a higher altitude will shift more than that of the satellite in a lower altitude orbit. In the case of eccentric orbits, the shift in the ground track is much less around the perigee point for the same reasons.

3.8.2.2 Effect of Latitude

The Earth's relative rotation rate decreases with an increase in latitude, becoming zero at the poles. As a consequence, the shift in the ground track reduces as the latitudes over which the satellite moves increase. In fact, the shift in the ground track is zero at the poles. Another point worth mentioning here is that in the case of a satellite in a prograde orbit, the ground track intersects increasingly westerly meridians. In the case of satellites in a prograde orbit, the ground track intersects increasingly westerly meridians when the satellite's angular speed is less than the rotational rate of Earth and increasingly easterly meridians when it is more than the Earth's rotation rate.

3.8.3 Orbit Inclination and Latitude Coverage

The northern and southern latitudes of the terrestrial segment covered by the satellite's ground track depend upon the satellite orbit inclination. The zone from the extreme northern latitude to the extreme southern latitude, which is symmetrical about the equator, is called the latitude coverage. Figure 3.50 illustrates the extent of latitude coverage for different inclinations.

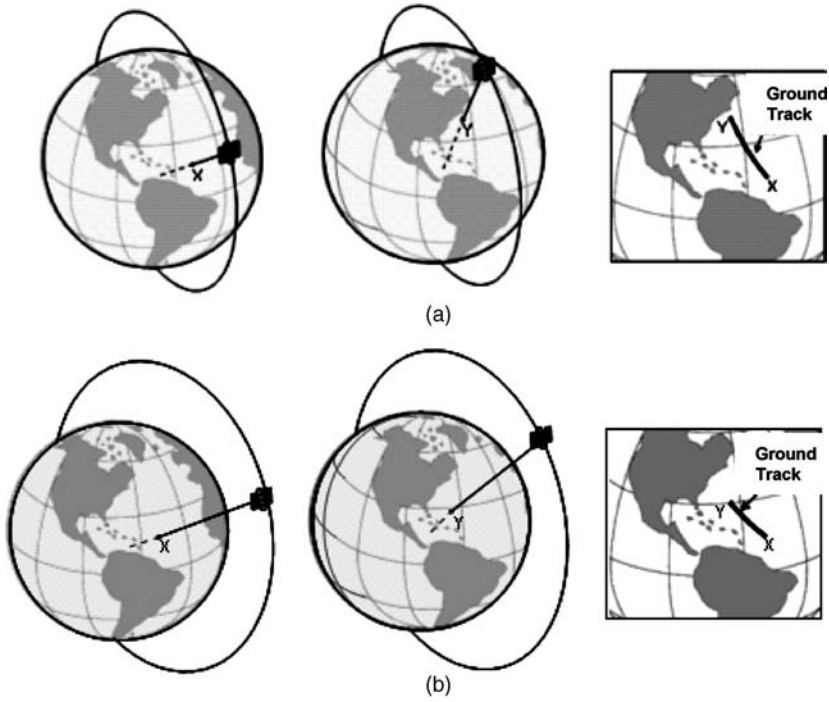


Figure 3.49 Effect of satellite altitude on the ground track

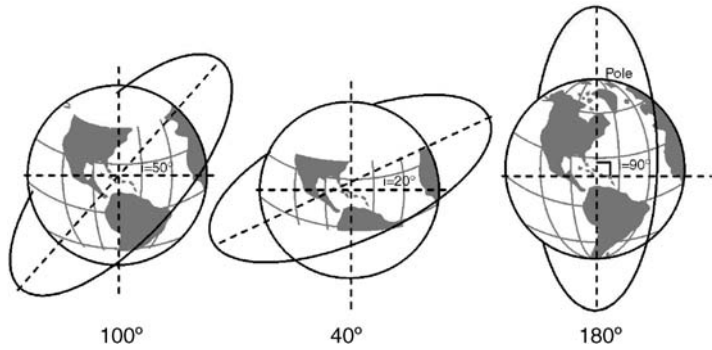


Figure 3.50 Effect of satellite orbital inclination on the latitude coverage

It can be seen that the latitude coverage is 100 % only in the case of polar orbits. The higher the orbit inclination, the greater is the latitude coverage. This also explains why an equatorial orbit is not useful for higher latitude regions and also why a highly inclined Molniya orbit is more suitable for the territories of Russia and other republics of the former USSR.

Problem 3.12

Determine the theoretical maximum area of the Earth's surface that would be in view from a geostationary satellite orbiting at a height of 35 786 km from the surface of the Earth. Also determine the area in view for a minimum elevation angle of 10° . (Assume that the radius of the Earth is 6378 km.)

Solution: Refer to Figure 3.51. For the maximum possible coverage angle, the elevation angle E must be 0° . In that case the coverage angle α is given by

$$\text{Coverage angle } \alpha = \sin^{-1} \left[\left(\frac{R}{R+H} \right) \cos E \right] = \sin^{-1} \left(\frac{R}{R+H} \right)$$

where

R = Earth's radius

H = Height of the satellite above the Earth's surface

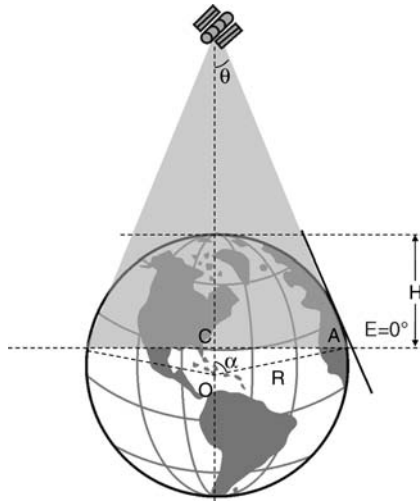


Figure 3.51 Figure for Problem 3.12

Thus

$$\alpha = \sin^{-1} \left(\frac{6378}{42164} \right) = 8.7^\circ$$

This gives

$$\theta = 90^\circ - \alpha - E = 90^\circ - 8.7^\circ = 81.3^\circ$$

In the right-angled triangle OAC, $OC = OA \times \sin 8.7^\circ$, (Angle OAC = 8.7°) and $OC = 6378 \times 0.151 = 963.1$ km. From the geometry, the covered surface area is given by

$$2\pi R(6378 - 963.1) = 2\pi \times 6378 \times 5414.9 = 216\,997\,546.7 \text{ km}^2$$

For $E = 10^\circ$,

$$\alpha = \sin^{-1} \left[\left(\frac{R}{R+H} \right) \cos E \right] = \sin^{-1} \left[\left(\frac{6378}{42\,164} \right) \cos 10^\circ \right] = 8.57^\circ$$

This gives

$$\theta = 90^\circ - 8.57^\circ - 10^\circ = 71.43^\circ$$

The new value of OC is

$$6378 \sin 18.57^\circ = 2028.2 \text{ km}$$

$$\text{Covered area} = 2\pi R(6378 - 2028.2) = 2\pi \times 6378 \times 4349.8 = 174\,314\,563.3 \text{ km}^2$$

Problem 3.13

A satellite is orbiting the Earth in a low Earth orbit inclined at 30° to the equatorial plane. Determine the extreme northern and southern latitudes swept by its ground track.

Solution:

The extreme latitudes covered in northern and southern hemispheres are the same as orbit inclination.

Therefore, extreme northern latitude covered = 30°N

Extreme southern latitude covered = 30°S

In fact, the ground track would sweep all latitudes between 30°N and 30°S .

Further Reading

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